UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington D.C. 20549
Form 10-K

(Mark One)
☑ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the fiscal year ended December 31, 2023
OR
☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the transition period from_____ to _____

Commission File Number: 1-12162

BorgWarner Inc.

(Exact name of registrant as specified in its charter)

Delaware	13-3404508
(State or other jurisdiction of Incorporation or organization)	(I.R.S. Employer Identification No.)

3850 Hamlin Road,
Auburn Hills, Michigan 48326
(Address of principal executive offices) (Zip Code)

Registrant's telephone number, including area code: (248) 754-9200

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock, par value $0.01 per share	BWA	New York Stock Exchange
1.00% Senior Notes due 2031	BWA31	New York Stock Exchange

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☑ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☑

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☑ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☑ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☑	Accelerated filer	☐	Non-accelerated filer	☐	Smaller reporting company	☐
Emerging growth company	☐						

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☑

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☑

The aggregate market value of the voting common stock of the registrant held by stockholders (not including voting common stock held by directors and executive officers of the registrant) on June 30, 2023 (the last business day of the most recently completed second fiscal quarter) was approximately $10 billion.

As of February 2, 2024, the registrant had 229,780,053 shares of voting common stock outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the following documents are incorporated herein by reference into the Part of the Form 10-K indicated.

Document	Part of Form 10-K into which incorporated
Portions of the BorgWarner Inc. Proxy Statement for the 2024 Annual Meeting of Stockholders	*Part III*

BORGWARNER INC.

FORM 10-K

YEAR ENDED DECEMBER 31, 2023

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CAUTIONARY STATEMENTS FOR FORWARD-LOOKING INFORMATION

Statements contained in this Annual Report on Form 10-K ("Form 10-K") (including Management's Discussion and Analysis of Financial Condition and Results of Operations) may contain forward-looking statements as contemplated by the 1995 Private Securities Litigation Reform Act (the "Act") that are based on management's current outlook, expectations, estimates and projections. Words such as "anticipates," "believes," "continues," "could," "designed," "effect," "estimates," "evaluates," "expects," "forecasts," "goal," "initiative," "intends," "may," "outlook," "plans," "potential," "predicts," "project," "pursue," "seek," "should," "target," "when," "will," "would," and variations of such words and similar expressions are intended to identify such forward-looking statements. Further, all statements, other than statements of historical fact contained or incorporated by reference in this Form 10-K, that we expect or anticipate will or may occur in the future regarding our financial position, business strategy and measures to implement that strategy, including changes to operations, competitive strengths, goals, expansion and growth of our business and operations, plans, references to future success and other such matters, are forward-looking statements. Accounting estimates, such as those described under the heading "Critical Accounting Policies and Estimates" in Item 7 of this Annual Report on Form 10-K, are inherently forward-looking. All forward-looking statements are based on assumptions and analyses made by us in light of our experience and our perception of historical trends, current conditions and expected future developments, as well as other factors we believe are appropriate in the circumstances. Forward-looking statements are not guarantees of performance and the Company's actual results may differ materially from those expressed, projected or implied in or by the forward-looking statements.

You should not place undue reliance on these forward-looking statements, which speak only as of the date of this Annual Report. Forward-looking statements are subject to risks and uncertainties, many of which are difficult to predict and generally beyond our control, that could cause actual results to differ materially from those expressed, projected or implied in or by the forward-looking statements. These risks and uncertainties, among others, include supply disruptions impacting us or our customers, such as the current shortage of semiconductor chips that has impacted original equipment manufacturer ("OEM") customers and their suppliers, including us; commodity availability and pricing, and an inability to achieve expected levels of recoverability in commercial negotiations with customers concerning these costs; competitive challenges from existing and new competitors, including OEM customers; the challenges associated with rapidly changing technologies, particularly as they relate to electric vehicles, and our ability to innovate in response; the difficulty in forecasting demand for electric vehicles and our electric vehicles revenue growth; potential disruptions in the global economy caused by Russia's invasion of Ukraine; the ability to identify targets and consummate acquisitions on acceptable terms; failure to realize the expected benefits of acquisitions on a timely basis; the possibility that our recently-completed tax-free spin-off of our former Fuel Systems and Aftermarket segments into a separate publicly traded company will not achieve its intended benefits; the failure to promptly and effectively integrate acquired businesses; the potential for unknown or inestimable liabilities relating to the acquired businesses; our dependence on automotive and truck production and is highly cyclical and subject to disruptions; our reliance on major OEM customers; impacts of any future strikes involving some of our OEM customers and any actions such OEM customers take in response; fluctuations in interest rates and foreign currency exchange rates; our dependence on information systems; the uncertainty of the global economic environment; the outcome of existing or any future legal proceedings, including litigation with respect to various claims, or governmental investigations, including related litigation; future changes in laws and regulations, including, by way of example, taxes and tariffs, in the countries in which we operate; impacts from any potential future acquisition or disposition transactions; and the other risks, noted in reports that we file with the Securities and Exchange Commission, including Item 1A, "Risk Factors" in our most recently-filed Form 10-K. We do not undertake any obligation to update or announce publicly any updates to or revisions to any of the forward-looking statements in this Form 10-Q to reflect any change in our expectations or any change in events, conditions, circumstances, or assumptions underlying the statements.

This section and the discussions contained in Item 1A, “Risk Factors,” and in Item 7, subheading “Critical Accounting Policies and Estimates” in this report, are intended to provide meaningful cautionary statements for purposes of the safe harbor provisions of the Act. This should not be construed as a complete list of all of the economic, competitive, governmental, technological and other factors that could adversely affect our expected consolidated financial position, results of operations or liquidity. Additional risks and uncertainties, including without limitation those not currently known to us or that we currently believe are immaterial, also may impair our business, operations, liquidity, financial condition and prospects.

Use of Non-GAAP Financial Measures

In addition to results presented in accordance with accounting principles generally accepted in the United States of America (“GAAP”), this report includes non-GAAP financial measures. The Company believes these non-GAAP financial measures provide additional information that is useful to investors in understanding the underlying performance and trends of the Company. Readers should be aware that non-GAAP financial measures have inherent limitations and should be cautious with respect to the use of such measures. To compensate for these limitations, we use non-GAAP measures as comparative tools, together with GAAP measures, to assist in the evaluation of our operating performance or financial condition. We calculate these measures using the appropriate GAAP components in their entirety and compute them in a manner intended to facilitate consistent period-to-period comparisons. The Company’s method of calculating these non-GAAP measures may differ from methods used by other companies. These non-GAAP measures should not be considered in isolation or as a substitute for those financial measures prepared in accordance with GAAP. Where non-GAAP financial measures are used, the most directly comparable GAAP financial measure, as well as the reconciliation to the most directly comparable GAAP financial measure, can be found in this report.

PART I

Item 1. Business

BorgWarner Inc. (together with its Consolidated Subsidiaries, the "Company" or "BorgWarner") is a Delaware corporation incorporated in 1987. The Company is a global product leader in clean and efficient technology solutions for combustion, hybrid and electric vehicles. Its products help improve vehicle performance, propulsion efficiency, stability and air quality. The Company manufactures and sells these products worldwide, primarily to original equipment manufacturers ("OEMs") of light vehicles (passenger cars, sport-utility vehicles ("SUVs"), vans and light trucks). The Company's products are also sold to OEMs of commercial vehicles (medium-duty trucks, heavy-duty trucks and buses) and off-highway vehicles (agricultural and construction machinery and marine applications). The Company also manufactures and sells its products to certain tier one vehicle systems suppliers and into the aftermarket for light, commercial and off-highway vehicles. The Company operates manufacturing facilities serving customers in Europe, the Americas and Asia and is an original equipment supplier to nearly every major automotive OEM in the world.

Charging Forward - Electrification Portfolio Strategy

In 2021, the Company announced its strategy to aggressively grow its eProducts over time through organic investments and technology-focused acquisitions. eProducts include all products utilized on or for electric vehicles ("EVs") plus those same products and components that are included in hybrid powertrains whose underlying technologies are adaptable or applicable to those used in or for EVs. The Company believes it is well positioned for the industry's anticipated migration to EVs.

In June 2023, the Company announced the next phase of its *Charging Forward* strategy, which focuses on profitably growing eProducts while maximizing the value of the Company's Foundational products portfolio. Foundational products include all products utilized on internal combustion engines plus those same products and components that are also included in hybrid powertrains. As a result of executing its strategy, the Company expects that by 2027, it will achieve over $10 billion in annual eProduct sales, deliver eProduct adjusted operating margin of approximately 7% and maintain its double-digit adjusted operating margin for its Foundational products portfolio. During the year ended December 31, 2023, the Company's eProduct revenue was approximately $2.0 billion, or 14% of its total revenue.

On July 3, 2023, BorgWarner completed the previously announced spin-off ("Spin-Off") of its Fuel Systems and Aftermarket segments in a transaction intended to qualify as tax free to the Company's stockholders for U.S. federal income tax purposes, which was accomplished by the distribution of 100% of the outstanding common stock of PHINIA, Inc. ("PHINIA") to holders of record of common stock of the Company on a pro-rata basis. Each holder of record of common stock of the Company received one share of PHINIA common stock for every five shares of common stock of the Company held on June 23, 2023, the record date for the distribution ("Distribution Date"). In lieu of fractional shares of PHINIA, shareholders of the Company received cash. PHINIA is an independent public company trading under the symbol "PHIN" on the New York Stock Exchange.

The historical results of operations and the financial position of PHINIA for periods prior to the Spin-Off are presented as discontinued operations in the accompanying Consolidated Financial Statements.

Recent Acquisitions

Acquisitions are an integral component of the Company's growth and value creation strategy. Below are summaries of recent acquisitions. Refer to Note 2, "Acquisitions and Dispositions," to the Consolidated Financial Statements in Item 8 of this report for more information.

Eldor Corporation's Electric Hybrid Systems Business

On December 1, 2023, the Company completed its acquisition of the electric hybrid systems business segment of Eldor Corporation ("Eldor"), which is headquartered in Italy. The acquisition is expected to complement its existing ePropulsion product portfolio by enhancing the Company's engineering capabilities in power electronics. The Company paid €72 million ($78 million) at closing, with up to €175 million ($191 million) in earn-out payments that could be paid over the two years following closing. The Company's current estimates indicate that the minimum threshold for the earn-out target will not be achieved, thus no amount of the earn-out payment has been included in the purchase consideration.

Hubei Surpass Sun Electric Charging Business

On March 1, 2023, the Company completed its acquisition of the electric vehicle solution, smart grid and smart energy businesses of Hubei Surpass Sun Electric, pursuant to an Equity Transfer Agreement. The acquisition is expected to complement the Company's existing European and North American charging footprint by adding a presence in China. The total consideration was ¥288 million ($42 million), including ¥268 million ($39 million) of base purchase price and ¥20 million ($3 million) of estimated earn-out payments. The Company paid ¥217 million ($31 million) of the base purchase price in the year ended December 31, 2023. The remaining ¥51 million ($8 million) of base purchase price is payable in two installments with the last payment due before April 30, 2025. In addition, pursuant to the agreement, the Company could be obligated to remit up to ¥103 million ($15 million), in the form of contingent payments over approximately two years following the closing.

Drivetek AG

On December 1, 2022, the Company acquired Drivetek AG, an engineering and product development company located in Switzerland. This acquisition is expected to strengthen the Company's power electronics capabilities in auxiliary inverters to accelerate the growth of the High Voltage eFan business. The total consideration was ₣27 million ($29 million) of base purchase price paid at closing, and ₣10 million ($10 million) of estimated earn-out payments that could be paid in the form of contingent payments over the three years following closing.

Rhombus Energy Solutions

On July 29, 2022, the Company acquired Rhombus Energy Solutions, a provider of charging solutions in the North American market. The acquisition is expected to complement the Company's existing European charging footprint to accelerate organic growth and adds North American regional presence to its charging business. The Company paid $131 million at closing, and up to $30 million could be paid in the form of contingent payments over the three years following closing. The Company's current estimates indicate that the minimum thresholds for these earn-out targets will not be achieved, thus no amount for the earn-out payments has been included in the purchase consideration.

Santroll Automotive Components

On March 31, 2022, the Company acquired Santroll Automotive Components, a carve-out of Santroll Electric Auto's eMotor business. The acquisition is expected to strengthen the Company's vertical integration, scale and portfolio breadth in light vehicle eMotors while allowing for increased speed to market. The total final consideration was $192 million, including approximately ¥1.0 billion ($152 million) of base purchase price and ¥0.25 billion ($40 million) of originally estimated earn-out payments. The Company paid approximately ¥1.0 billion ($157 million) of base purchase price in the year ended December 31, 2022 and no longer expects to recapture a previously anticipated $5 million of post-closing adjustments, which has been recorded in Other operating expense, net. Pursuant to the Equity Transfer Agreement for the acquisition, the obligation of the Company to remit up to ¥0.3 billion (approximately

$47 million) of earn-out payments was contingent upon achievement of certain sales volume targets and certain estimated future volume targets associated with newly awarded business. During the year ended December 31, 2023, the Company paid approximately ¥0.2 billion ($24 million) to settle the remaining earn-out liability and related adjustments.

AKASOL AG

On June 4, 2021, the Company completed a voluntary public takeover offer for shares of AKASOL AG ("AKASOL"), resulting in ownership of 89% of AKASOL's outstanding shares. The Company paid approximately €648 million ($788 million) to settle the offer. During 2021, the Company increased its ownership to 93% through the subsequent purchase of additional shares. On February 10, 2022, the Company completed a merger squeeze-out process to obtain the remaining shares, resulting in 100% ownership. The acquisition is expected to further strengthen BorgWarner's commercial vehicle and industrial electrification capabilities, which positions the Company to capitalize on what it believes to be a fast-growing battery module and pack market.

Financial Information About Reportable Segments

Refer to Note 24, "Reportable Segments and Related Information," to the Consolidated Financial Statements in Item 8 of this report for financial information about the Company's reportable segments.

Narrative Description of Reportable Segments

The Company reports its results under three reportable segments: Air Management, Drivetrain & Battery Systems and ePropulsion. In previous years, the Company presented its results under four reportable segments: Air Management, ePropulsion & Drivetrain, Fuel Systems and Aftermarket. In the first quarter of 2023, the Company elected to disaggregate Air Management and ePropulsion & Drivetrain segments into Air Management, Drivetrain & Battery Systems and ePropulsion and reported its results in a total of five reportable segments: Air Management, Drivetrain & Battery Systems, ePropulsion, Fuel Systems and Aftermarket. As a result of the Spin-Off, Fuel Systems and Aftermarket are no longer reportable segments.

Net sales by reportable segment were as follows:

	Year Ended December 31,		
(in millions)	2023	2022	2021
Air Management	$ 7,833	$ 7,137	$ 6,867
Drivetrain & Battery Systems	4,348	3,735	3,660
ePropulsion	2,166	1,906	1,427
Inter-segment eliminations	(149)	(143)	(151)
Net sales	$ 14,198	$ 12,635	$ 11,803

The sales information presented above does not include the sales by the Company's unconsolidated joint ventures (see sub-heading "Joint Ventures" below). Such unconsolidated sales totaled approximately $732 million, $734 million, and $852 million for the years ended December 31, 2023, 2022 and 2021, respectively.

Air Management

The Air Management segment develops and manufactures products to improve fuel economy, reduce emissions and enhance performance. The Air Management segment's technologies include turbochargers, eBoosters, eTurbos, timing systems, emissions systems, thermal systems, gasoline ignition technology, smart remote actuators, powertrain sensors, cabin heaters, battery heaters, battery charging and direct current charging stations.

The Air Management segment's emissions, thermal and turbocharger systems provide several benefits, including increased power for a given engine size, improved fuel economy, reduced emissions and optimized temperatures in propulsion systems and vehicle cabins. Sales of turbochargers for light vehicles represented approximately 21%, 26% and 24% of the Company's net sales for the years ended December 31, 2023, 2022 and 2021, respectively. No other single product line accounted for more than 10% of consolidated net sales in any of the years presented.

The Air Management segment's timing systems enable precise control of air and exhaust flow through the engine, improving fuel economy and emissions. The Air Management segment is a leading manufacturer of timing systems for OEMs around the world.

The Air Management segment's powertrain products include an array of highly engineered products that complement and enhance the efficiency improvements delivered by many other air management technologies.

Drivetrain & Battery Systems

The Company's Drivetrain & Battery Systems segment's technologies include battery modules and systems, control modules, friction and mechanical clutch products for automatic transmissions, torque-management products and rear-wheel drive ("RWD") and all-wheel drive ("AWD") transfer case systems and coupling systems.

The Drivetrain & Battery Systems segment's battery products include high-performance lithium-ion battery systems for electrified bus, truck and off-highway applications. The battery products provide commercial vehicle performance with a high-power output for safe, reliable and durable operation. The ultra-high energy battery system for electric drivetrains offers up to 4,000 cycles, significantly enhancing electric commercial vehicle range and is compatible with hyper-charging infrastructure. The battery management system provides safety and efficiency, with liquid cooling preventing overheating.

The Drivetrain & Battery Systems segment's friction and mechanical products for automatic transmissions include dual clutch modules, friction clutch modules, friction and separator plates, transmission bands, torque converter clutches, one-way clutches and torsional vibration dampers. Controls products for automatic transmissions feature electro-hydraulic solenoids for standard and high-pressure hydraulic systems, transmission solenoid modules and dual clutch control modules.

The Drivetrain & Battery Systems segment's torque management products include rear-wheel drive ("RWD") and all-wheel drive ("AWD") transfer case systems, front-wheel drive ("FWD")-AWD coupling systems and cross-axle coupling systems. The segment is developing electronically controlled torque management devices and systems that will benefit vehicle energy efficiency and vehicle dynamics.

ePropulsion

The Company's ePropulsion segment's products and technologies provide industry-leading performance and efficiency with quick-to-market solutions powering current and next-generation electric and hybrid vehicles.

The ePropulsion segment's technologies include power electronics such as inverters, onboard chargers, DC/DC converters and combination boxes (multiple combined power electronics components). Rotating electric machines are also part of the ePropulsion portfolio, including eMotors and generators as well as fully integrated drive modules ("iDM") consisting of inverter, eMotor and gear reducer. Additionally, the segment's products include electronic controls such as engine control units, transmission control units, battery management systems, propulsion controllers and domain controllers.

The ePropulsion segment's inverter products power many of the global leading electric and hybrid vehicles.

Additionally, the ePropulsion segment's iDMs combine all the benefits of our inverters, eMotors and gear reducers in a single package optimized for cost, performance, noise vibration and harshness and packaging. iDMs contain full software that offers functional safety and cybersecurity. This capability comes from deep experience of over 40 years in the field of automotive software. Applications of iDMs include a wide range of electric and hybrid vehicles globally.

Joint Ventures

As of December 31, 2023, the Company had eight joint ventures in which it had a less-than-100% ownership interest. Results from the six joint ventures in which the Company is the majority owner and has a controlling financial interest are consolidated as part of the Company's results. Results from the two joint ventures in which the Company exercises significant influence but does not have a controlling financial interest, were reported by the Company using the equity method of accounting pursuant to which the Company records its proportionate share of each joint venture's income or loss each period.

Management of the unconsolidated joint ventures is shared with the Company's respective joint venture partners. Certain information concerning the Company's joint ventures is set forth below:

Joint venture	Products	Year organized	Percentage owned by the Company	Location of operation	Joint venture partner
Unconsolidated:					
NSK-Warner K.K.	Transmission components	1964	50 %	Japan/China	NSK Ltd.
Turbo Energy Private Limited	Turbochargers	1987	32.6 %	India	Sundaram Finance Limited; Brakes India Limited
Consolidated:					
BuradaWarner LLC	Valvetrain and fuel injection equipment	1977	70 %	Korea	BU RA DA Company Limited
BorgWarner Transmission Systems Korea Ltd.[1]	Transmission components	1987	60 %	Korea	NSK-Warner
Beijing Delphi Wan Yuan Engine Management Systems Co. Ltd.	Engine management systems	1999	51 %	China	Beijing Wan Yuan Industry Corporation
BorgWarner Shenglong (Ningbo) Co. Ltd.	Fans and fan drives	1999	70 %	China	Ningbo Shenglong Automotive Powertrain Systems Co., Ltd.
BorgWarner TorqTransfer Systems Beijing Co. Ltd.	Transfer cases	2000	80 %	China	Beijing Hainachuan Automotive Parts Holding Co., Ltd.
BorgWarner United Transmission Systems Co. Ltd.	Transmission components	2009	66 %	China	China Automobile Development United Investment Co., Ltd.

[1] BorgWarner Inc. owns 50% of NSK-Warner, which has a 40% interest in BorgWarner Transmission Systems Korea Ltd. This ownership gives the Company an additional indirect effective ownership percentage of 20% in BorgWarner Transmission Systems Korea Ltd., resulting in a total effective ownership interest of 80%.

Financial Information About Geographic Areas

The Company has a global presence. During the year ended December 31, 2023, approximately 16% of the Company's net sales were generated in the United States, and 84% were generated outside the United States. Refer to Note 24, "Reportable Segments and Related Information," to the Consolidated Financial Statements in Item 8 of this report for additional financial information about geographic areas.

Product Lines and Customers

During the year ended December 31, 2023, approximately 82% of the Company's net sales were for light-vehicle applications; approximately 10% were for commercial-vehicle applications; approximately 6% were for off-highway vehicle applications; and approximately 2% were to distributors of aftermarket replacement parts.

The Company's worldwide net sales to the following customers (including their subsidiaries) were approximately as follows:

	Year Ended December 31,		
Customer	2023	2022	2021
Ford	14 %	15 %	13 %
Volkswagen	11 %	9 %	9 %

No other single customer accounted for more than 10% of the Company's consolidated net sales in any of the years presented. Sales to the Company's top ten customers represented 68% of sales for the year ended December 31, 2023.

The Company's automotive products are generally sold directly to OEMs, substantially pursuant to negotiated annual contracts, long-term supply agreements or terms and conditions as may be modified by the parties. Deliveries are subject to periodic authorizations based upon OEM production schedules. The Company typically ships its products directly from its plants to the OEMs.

Sales and Marketing

Each of the Company's businesses within its reportable segments has its own sales function. Account executives for each of the Company's businesses are assigned to serve specific customers for one or more businesses' products. Account executives spend the majority of their time in direct contact with customers' purchasing and engineering employees and are responsible for servicing existing business and for identifying and obtaining new business. Because of their close relationship with customers, account executives are able to identify and meet customers' needs based upon their knowledge of the Company's product design and manufacturing capabilities. Upon securing a new order, account executives participate in product launch team activities and serve as a key interface with customers. In addition, sales and marketing employees of the Company's reportable segments often work together to explore cross-development opportunities where appropriate.

Seasonality

The Company's operations are directly related to the automotive and commercial-vehicle industry. Consequently, the Company's segments may experience seasonal fluctuations to the extent vehicle production slows at certain times of the year. For example, model changeovers and vacations during the summer months have generally resulted in lower sales volume in the Company's third quarter, and in China, the Company typically experiences lower sales in the first quarter due to the Chinese New Year.

Research and Development

The Company conducts advanced propulsion research. This advanced engineering function seeks to leverage know-how and expertise across product lines to create new electrified propulsion systems and modules that can be commercialized. This function oversees the Company's investments in certain venture capital funds that provide seed money for start-up businesses developing new technologies pertinent to the automotive industry and the Company's propulsion strategies.

In addition, each of the Company's businesses within its Air Management, Drivetrain & Battery Systems and ePropulsion reportable segments has its own research and development (“R&D”) organization, including engineers and technicians, engaged in R&D activities at facilities worldwide. The Company also operates testing facilities such as prototype, measurement and calibration, life-cycle testing and dynamometer laboratories.

By working closely with OEMs and anticipating their future product needs, the Company’s R&D personnel conceive, design, develop and manufacture new proprietary components and systems. R&D personnel also work to improve current products and production processes. The Company believes its commitment to R&D will allow it to continue to obtain new orders from its OEM customers.

The Company’s net R&D expenditures are primarily included in selling, general and administrative expenses of the Consolidated Statements of Operations. Customer reimbursements are netted against gross R&D expenditures as they are considered a recovery of cost. Customer reimbursements for prototypes are recorded net of prototype costs based on customer contracts, typically either when the prototype is shipped or when it is accepted by the customer. Customer reimbursements for engineering services are recorded when performance obligations are satisfied in accordance with the contract. Financial risks and rewards transfer upon shipment, acceptance of a prototype component by the customer or upon completion of the performance obligation as stated in the respective customer agreement.

	Year Ended December 31,		
(in millions)	2023	2022	2021
Gross R&D expenditures	$ 856	$ 787	$ 694
Customer reimbursements	(139)	(86)	(108)
Net R&D expenditures	$ 717	$ 701	$ 586

Net R&D expenditures as a percentage of net sales were 5.1%, 5.5% and 5.0% for the years ended December 31, 2023, 2022 and 2021, respectively.

Intellectual Property

The Company has approximately 5,700 active domestic and foreign patents and patent applications pending or under preparation and receives royalties from licensing patent rights to others. While it considers its patents on the whole to be important, the Company does not consider any single patent, any group of related patents or any single license essential to its operations in the aggregate or to the operations of any of the Company’s business groups individually. The expiration of the patents individually and in the aggregate is not expected to have a material effect on the Company’s financial position or future operating results. The Company owns numerous trademarks, some of which are valuable, but none of which are essential to its business in the aggregate.

The Company owns the “BorgWarner” trade name and numerous trademarks which are material to the Company's business.

Competition

The Company's reportable segments compete worldwide with a number of other manufacturers and distributors that produce and sell similar products. Many of these competitors are larger and have greater resources than the Company. Technological innovation, application engineering development, quality, price, delivery and program launch support are the primary methods of competition.

The Company's major non-OEM competitors are Robert Bosch GmbH, Denso Corporation, Garrett Motion, Hitachi, Ltd., Magna Powertrain (an operating unit of Magna International Inc.), Valeo, Schaeffler Group and Vitesco Technologies. The Company also competes with certain start-ups in electrification.

In addition, a number of the Company's major OEM customers manufacture, for their own use and for others, products that compete with the Company's products. Other current OEM customers could elect to manufacture products to meet their own requirements or to compete with the Company. There is no assurance that the Company's business will not be adversely affected by increased competition in the markets in which it operates.

For many of its products, the Company's competitors include suppliers in parts of the world that enjoy economic advantages such as lower labor costs, lower health care costs, lower tax rates and, in some cases, export subsidies and/or raw materials subsidies. Also, see Item 1A, "Risk Factors."

Human Capital Management

The Company's ability to sustain and grow its business requires it to hire, retain and develop a highly skilled and diverse management team and workforce worldwide. The Company believes the skills, experience, and industry knowledge of its employees significantly benefit its operations and performance.

The Compensation Committee of the Board of Directors oversees human capital management and assesses whether efforts to promote and advance environmental, social and governance ("ESG") initiatives, practices, and objectives, as appropriate, are effectively reflected in executive compensation. The full Board of Directors oversees talent reviews and succession planning for the Company.

As of December 31, 2023, the Company had a salaried and hourly workforce as follows:

Americas	11,800
Asia	12,200
Europe	15,900
Total workforce	39,900
Salaried	13,000
Hourly	26,900
Total workforce	39,900

The Company uses an array of practices to attract, develop and retain highly qualified talent, including the following:

- *Diversity, Equity & Inclusion ("DE&I")*. Ultimate responsibility for DE&I at BorgWarner lies with the Company's CEO, while the Board of Directors monitors initiatives and performance. The Company cultivates a culture where employees are treated with respect and their differences are valued. The Company is continually reviewing its policies, programs and processes to ensure alignment with its DE&I strategy. The Company undertakes a variety of recruitment and retention initiatives that serve as a strategic opportunity to build a diverse talent and leadership pipeline.

In 2022, the Company set goals to advance DE&I and support its commitment to creating an inclusive and sustainable workforce. Those goals include:

- 35% of global workforce are women by 2026
- 30% of U.S. workforce are racially/ethnically diverse by 2026
- Pay parity for all by 2026
- 80% or above on the BorgWarner Beliefs index score from the Company's employee engagement survey by 2026

As of December 31, 2023:

- Women composed 30.1% of the Company's global workforce.[1]
- Racial/ethnic minorities composed 29.5% of the Company's total U.S. workforce.[1]
- The Company's latest pay equity analysis identified that, on average, women received compensation equal to 99.0% of that received by men across the Company's global workforce for substantially similar work. In the U.S., racial/ethnic minorities received compensation of 100.9% compared to compensation received by non-minorities for substantially similar work.[2]
- The Company's 2023 employee engagement survey achieved a 76% score on the BorgWarner Beliefs index.

In addition to the Company's DE&I goals progress, as of December 31, 2023:

- Five of eight board members (63%) were women and/or racially/ethnically diverse.
- Four of 10 executive management team members (40%) were women and/or racially/ethnically diverse.
- Women composed 17.4% of the Company's leadership (those who participate in the management incentive plan), 25.0% of the Company's salaried workforce, 32.9% of the Company's hourly workforce, and 36.3% of the Company's new hires in 2023.[1]
- Racial/ethnic minorities composed 20.6% of the Company's U.S. leadership, 23.6% of the Company's U.S. salaried workforce, 34.9% of the Company's U.S. hourly workforce, and 44.9% of the Company's U.S. new hires in 2023.[1]

- *Engagement & Retention*. The Company utilizes various strategies to attract, engage and retain the brightest and best talent. It recognizes and rewards employee contributions with competitive pay and benefits. The Company closely monitors employee turnover as part of its efforts to improve retention and to spot any potential opportunities for improvement. In the year ended December 31, 2023, annual voluntary employee turnover was 11.0%, annual voluntary turnover for our salaried workforce was 9.7%, and annual voluntary turnover for our hourly workforce was 11.7%.[1]

- *Education & Development.* The Company provides formal development opportunities at all levels and stages of the career journey of its employees. These opportunities are delivered in a variety of formats to make its portfolio of solutions flexible, accessible, scalable and translatable to meet the needs of our evolving workplace and workforce. The Company is also committed to preparing its workforce for the transition from combustion to electrification. In 2023, the Company delivered training programs created in partnership with elite universities to increase the knowledge and skills of its engineers to enable them to work in an electrification environment.

- *Health & Safety*. The safety of the Company's employees is vitally important, and the Company is dedicated to continuously improving safety performance. The Company's safety performance is rooted in robust safety management systems consisting of leading safety indicators, integrating detailed metrics into safety scorecards, engaging employees at every level, training and

[1] Data excludes employees acquired through the Eldor acquisition.
[2] The Company's most recent pay equity study was conducted in 2023 based on compensation and employees as of December 31, 2022. The analysis included employees from salaried early-in-career through vice president roles.

prevention initiatives, performing risk assessments and inspections, sharing best practices, hosting safety conferences, and sponsoring recognition programs.

In 2023, the Company set goals for the safety of its workforce. Those goals include:

- Perform in the top quartile for Total Recordable Incident Rate ("TRIR") and Lost Time Incident Rate ("LTIR").[3]
- Implement and then maintain ISO 45001 certification at 100% of its manufacturing sites.[4]

In the year ended December 31, 2023:

- The Company's global workforce accident TRIR was 0.36, which was within top quartile performance. The top quartile for motor vehicle parts manufacturing was lower than or equal to 1.2 according to the BLS.[3]
- The Company's global workforce accident LTIR was 0.21. The top quartile for motor vehicle parts manufacturing was lower than or equal to 0.2 according to the BLS.[3]
- 93% of the Company's manufacturing sites were ISO 45001 certified.[4]

Approximately 12.5% of the Company's U.S. workforce is unionized. These employees, located at one facility in the state of New York, are covered by a collective bargaining agreement that expires in September 2024. Employees at certain international facilities are also unionized. The Company believes the current relations with its workforce to be satisfactory. The Company recognizes that, in many of the locations where it operates, employees have freedom of association rights with third-party organizations such as labor unions. The Company respects and supports those rights, including the right to collective bargaining, in accordance with local laws.

Raw Materials

The Company uses a variety of raw materials in the production of its products, including aluminum, copper, nickel, plastic resins, steel, certain alloy elements and semiconductor chips. Manufacturing operations for each of the Company's operating segments are dependent upon natural gas, fuel oil and electricity.

The Company uses a variety of tactics in an attempt to limit the impact of supply shortages and inflationary pressures. The Company's global procurement organization works to accelerate cost reductions, purchase from lower cost regions, optimize the supply base, mitigate risk and collaborate on its buying activities. In addition, the Company uses long-term contracts, cost sharing arrangements, design changes, customer buy programs and limited financial instruments to help control costs. The Company intends to use similar measures in 2024 and beyond. Refer to Note 17, "Financial Instruments," to the Consolidated Financial Statements in Item 8 of this report for information related to the Company's hedging activities.

For 2024, the Company believes there will be continued inflationary pressures in certain raw materials, labor and energy. While the Company sees inflation decreasing in some areas it does not expect to see "deflation," which means that it expects supplier costs to remain elevated relative to prior years. Supplies of raw materials are adequate and available from multiple sources to support the Company's manufacturing requirements.

[3] Based on U.S. Bureau of Labor Statistics (the "BLS"), Survey of Occupational Injuries and Illnesses Data, motor vehicle parts manufacturing (NAICS 336300). TRIR and LTIR exclude safety performance for Eldor, which was acquired in the fourth quarter of 2023.
[4] Certified to ISO 45001:2018. The scope of this performance indicator is for manufacturing locations that supply directly to original equipment manufacturers, excluding locations during their first 18 months of production and newly acquired sites during their first 18 months with the Company.

Regulations

The Company operates in a constantly evolving global regulatory environment and is subject to numerous and varying regulatory requirements for its product performance and material content. The Company's practice is to identify potential regulatory and quality risks early in the design and development process and proactively manage them throughout the product lifecycle through the use of routine assessments, protocols, standards, performance measures and audits. New regulations and changes to existing regulations are managed in collaboration with the Company's OEM customers and implemented through its global systems and procedures designed to ensure compliance with existing laws and regulations. The Company demonstrates material content compliance through the International Material Data System ("IMDS"), which is the automotive industry material data system. In the IMDS, all materials used for automobile manufacturing are archived and maintained to meet the obligations placed on the automobile manufacturers, and thus on their suppliers, by national and international standards, laws and regulations.

The Company works collaboratively with a number of stakeholder groups, including government agencies, such as the National Highway Traffic Safety Administration, its customers and its suppliers to proactively engage in federal, state and international public policy processes.

Refer to Item 7, "Management's Discussion and Analysis of Financial Condition and Results of Operations," for a discussion of the impact of environmental regulations on the Company's business. Also, see Item 1A, "Risk Factors."

Available Information

Through its Internet website (www.borgwarner.com), the Company makes available, free of charge, its Annual Report on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, all amendments to those reports, and other filings with the Securities and Exchange Commission as soon as reasonably practicable after they are filed or furnished. The Company also makes the following documents available on its Internet website: the Audit Committee Charter; the Compensation Committee Charter; the Corporate Governance Committee Charter; the Company's Corporate Governance Guidelines; the Company's Code of Ethical Conduct; and the Company's Code of Ethics for CEO and Senior Financial Officers. You may also obtain a copy of any of the foregoing documents, free of charge, if you submit a written request to Investor Relations, 3850 Hamlin Road, Auburn Hills, Michigan 48326. You can also find the Company's public filings at a website maintained by the SEC, http://www.sec.gov, which contains reports, proxy and information statements, and other information regarding issuers that file electronically with the SEC.

Information About Executive Officers of the Company

Set forth below are the names, ages, positions and certain other information concerning the executive officers of the Company as of February 8, 2024.[5]

Name (Age)	Present Position (Effective Date)	Positions Held During the Past Five Years (Effective Date)
Frederic B. Lissalde (56)	President and Chief Executive Officer (2018)	• Autoliv, Inc., Member of Board of Directors (2020 – Present)
Kevin A. Nowlan (52)	Executive Vice President, Chief Financial Officer (2019)	• Meritor Inc., Senior Vice President, President, Trailer, Components and Chief Financial Officer (2018 – 2019) • Federal Reserve Bank of Chicago – Detroit Branch, Member of Board of Directors (2022 – Present)
Tonit M. Calaway (56)	Executive Vice President, Chief Administrative Officer, General Counsel and Secretary (2020)	• Executive Vice President, Chief Legal Officer and Secretary (2018 - 2020) • Air Products & Chemicals, Inc., Member of Board of Directors (2022 – Present) • W.P. Carey Inc., Member of Board of Directors (2020 – Present)
Tania Wingfield (57)	Executive Vice President, Chief Human Resources Officer (2022)	• Vice President and General Manager, North America Aftermarket (2021 – 2022) • Vice President and Integration Champion (2020 – 2021) • Vice President, Engineering, PowerDrive Systems (2017 – 2020)
Craig D. Aaron (46)	Vice President and Controller (2022)	• Vice President and Treasurer (2019 – 2022) • Vice President, Finance, BorgWarner Morse Systems (2016 – 2019)
Stefan Demmerle (59)	Vice President and President and General Manager, PowerDrive Systems (2015)	• Vice President of the Company and President and General Manager of BorgWarner PowerDrive Systems (2015 – Present)
Joseph F. Fadool (57)	Vice President and President and General Manager, Emissions, Thermal and Turbo Systems (2019)	• Vice President of the Company and President and General Manager, Turbo Systems LLC (2019) • Vice President of the Company and President and General Manager, BorgWarner Emissions Systems LLC and BorgWarner Thermal Systems Inc. (2017 – 2019)
Paul A. Farrell (57)	Executive Vice President and Chief Strategy Officer (2020)	• Delphi Technologies PLC, Senior Vice President Strategy, Sales and Corporate Development (2020) • Delphi Technologies PLC, Senior Vice President Strategy and Corporate Development (2019 – 2020) • Delphi Technologies PLC, Senior Vice President Strategic Planning and Product Marketing (2017 – 2019)
Isabelle McKenzie (54)	Vice President and President and General Manager, Morse Systems (2023)	• Vice President & General Manager, Americas, Power-Drive Systems (2020 - 2023) • Vice President Global Engineering, Transmission Systems (2014 - 2020)
Volker Weng (53)	Vice President and President and General Manager, Drivetrain & Battery Systems (formerly known as Drivetrain Systems) (2019)	• Vice President of the Company and President and General Manager, BorgWarner Emissions Systems LLC and BorgWarner Thermal Systems Inc. (2019) • Vice President and General Manager, Europe, BorgWarner Emissions Systems LLC and BorgWarner Thermal Systems Inc. (2017 – 2019)

[5] On November 30, 2023, Kevin A. Nowlan, Executive Vice President and Chief Financial Officer of the Company, notified the Company of his intention to retire as Executive Vice President and Chief Financial Officer effective March 1, 2024. On December 5, 2023, the Company announced the appointment of Craig D. Aaron, currently the Company's Vice President and Controller, to the role of Executive Vice President and Chief Financial Officer, effective March 1, 2024, succeeding Mr. Nowlan in that role.

Item 1A. Risk Factors

The following risk factors and other information included in this Annual Report on Form 10-K should be considered. The risks and uncertainties described below are not the only ones we face. Additional risks and uncertainties not presently known to us or that we currently deem immaterial also may impact our business operations. If any of the following risks occur, our business, including our financial performance, financial condition, operating results and cash flows, could be adversely affected.

Risks related to our strategy

Our Charging Forward strategy may prove unsuccessful.

In 2021, we announced our strategy to aggressively grow our eProduct portfolio over time through organic investments and technology-focused acquisitions. We believe we are well positioned for the industry's anticipated migration to EV. In June 2023, we announced the next phase of our *Charging Forward* strategy which focuses on profitably growing eProducts while maximizing the value of our Foundational product portfolio. As a result of executing this strategy, we expect that by 2027, we will achieve over $10 billion in annual eProduct sales, deliver eProduct adjusted operating margin of approximately 7% and maintain double-digit adjusted operating margins for our Foundational products portfolio.

We may not meet our goals due to many factors, including any of the risks identified in the paragraph that follows, failure to develop new products that our customers will purchase, technology changes that could render our products obsolete, or the introduction of new technology to which we do not have access, among other things.

We expect to continue to pursue business ventures, acquisitions, and strategic alliances that leverage our technology capabilities and enhance our customer base, geographic representation, and scale to complement our current businesses. We regularly evaluate potential growth opportunities, some of which could be material. While we believe that such transactions are an integral part of our long-term strategy, there are risks and uncertainties related to these activities. Assessing a potential growth opportunity involves extensive due diligence. However, the amount of information we can obtain about a potential growth opportunity can be limited, and we can give no assurance that past or future business ventures, acquisitions, and strategic alliances will positively affect our financial performance or will perform as planned. Assessing a price for potential transactions is inexact. We may not be able to successfully assimilate or integrate companies that we have acquired or will acquire in the future, including their personnel, financial systems, distribution, operations and general operating procedures. Failure to execute our growth strategy could adversely affect our business.

The failure to realize the expected benefits of acquisitions and other risks associated with acquisitions could adversely affect our business.

The success of our acquisitions is dependent, in part, on our ability to realize the expected benefits from combining our businesses and businesses that we acquire. To realize these anticipated benefits, both companies must be successfully combined, which is subject to our ability to consolidate operations, corporate cultures and systems and to eliminate redundancies and costs. If we are unsuccessful in combining companies, the anticipated benefits of the acquisitions may not be realized fully or at all or may take longer to realize than expected. Further, there is potential for unknown or inestimable liabilities relating to the acquired businesses. In addition, the actual integration may result in additional and unforeseen expenses, which could reduce the anticipated benefits of the acquisitions.

The combination of independent businesses is a complex, costly and time-consuming process that requires significant management attention and resources. It is possible that the integration process could result in the loss of key employees, the disruption of our operations, the inability to maintain or increase

our competitive presence, inconsistencies in standards, controls, procedures and policies, difficulties in achieving anticipated cost savings, synergies, business opportunities and growth prospects from the acquisition, the diversion of management's attention to integration matters and/or difficulties in the assimilation of employees and corporate cultures. Any or all of these factors could adversely affect our ability to maintain relationships with customers and employees or to achieve the anticipated benefits of the acquisition and could have an adverse effect on the combined company. In addition, many of these factors are outside of our control, and any one of these factors could result in increased costs, decreases in the amount of expected revenues and additional diversion of management's time and energy, which could materially adversely impact our business, financial condition and results of operations.

We may not be able to execute dispositions of assets or businesses successfully.

When we decide to dispose of assets or a business, we may have difficulty finding buyers or alternative exit strategies on acceptable terms in a timely manner, which could delay our ability to achieve our strategic objectives. We may also dispose of a business at a price or on terms that are less desirable than we had anticipated. Buyers of the assets or business may from time to time agree to indemnify us for operations of such businesses after the closing. We cannot be assured that any of these indemnification provisions will fully protect us, and as a result may face unexpected liabilities that adversely affect our business, financial condition and results of operations. In addition, we may experience fewer synergies than expected or even negative synergies from separating a business, and the impact of the disposition on our financial results may be larger than projected.

After reaching an agreement for the disposition of a business, we are subject to satisfaction of pre-closing conditions as well as necessary regulatory and governmental approvals on acceptable terms, which, if not satisfied or obtained, may prevent us from completing the transaction. Such regulatory and governmental approvals may be required in jurisdictions around the world, and any delays in the timing of such approvals could materially delay or prevent the transaction.

Goodwill and indefinite-lived intangible assets, which are subject to periodic impairment evaluations, represent a significant portion of our total assets. An impairment charge on these assets could have a material adverse impact on our financial condition and results of operations.

We have recorded goodwill and indefinite-lived intangible assets related to acquisitions. We periodically assess these assets to determine if they are impaired. Significant negative industry or macroeconomic trends, disruptions to our business, inability to effectively integrate acquired businesses, unexpected significant changes or planned changes in use of the assets, dispositions and market capitalization declines may impair these assets.

We review goodwill and indefinite-lived intangible assets for impairment either annually or whenever changes in circumstances indicate that the carrying value may not be recoverable. The risk of impairment to goodwill and indefinite-lived intangible assets is higher during the early years following an acquisition. This is because the fair values of these assets align very closely with what was paid to acquire the reporting units to which these assets are assigned. As a result, the difference between the carrying value of the reporting unit and its fair value (typically referred to as "headroom") is smaller at the time of acquisition. Until this headroom grows over time, due to business growth or lower carrying value of the reporting unit, a relatively small decrease in reporting unit fair value can trigger impairment charges. When impairment charges are triggered, they tend to be material due to the size of the assets involved. Future acquisitions could present similar risks. Any charges relating to such impairments could adversely affect our results of operations in the periods recognized.

Risks related to the Spin-Off of PHINIA Inc.

The Spin-Off may not achieve the anticipated benefits and may expose us to additional risks.

We may not realize the anticipated strategic, financial, operational or other benefits of the Spin-Off. We cannot predict with certainty when the benefits expected from the Spin-Off will occur or the extent to which they will be achieved. There is no assurance that following the Spin-Off each separate company will be successful. We may face material challenges in connection with the Spin-Off, including but not limited to, the diversion of management time on matters relating to the Spin-Off; the impact of having to operate under the terms of transition service agreements; the impact on our ability to retain talent; and potential impacts on our relationships with customers, suppliers, employees and other counterparties. In addition, we have incurred one-time costs and may incur ongoing costs in connection with, or as a result of, the spin-off, including costs of operating as independent, publicly-traded companies that the separate businesses will no longer be able to share. Those costs may exceed our estimates or could negate some of the benefits we expect to realize. Further, while it is intended that the transaction will be tax-free to the Company's stockholders for U.S. federal income tax purposes, there is no assurance that the transaction will qualify for this treatment. If the Spin-Off is ultimately determined to be taxable, either the Company, PHINIA, or the Company's stockholders could incur income tax liabilities that could be significant. If we do not realize the anticipated benefits of the Spin-Off, it could adversely affect our business, results of operations, cash flows and financial condition.

Potential indemnification liabilities pursuant to the spin-off of PHINIA could materially and adversely affect our business.

In connection with the Spin-Off, we entered into a separation and distribution agreement and related agreements with PHINIA to govern the Spin-Off and the relationship between the two companies following the completion of the Spin-Off. These agreements provide for specific indemnity and liability obligations of each party and could lead to disputes between us. If we are required to indemnify the other parties under the circumstances set forth in these agreements, we may be subject to future liabilities. In addition, with respect to the liabilities for which the other parties have agreed to indemnify us under these agreements, there can be no assurance that the indemnity rights we have against such other parties will be sufficient to protect us against the full amount of the liabilities or that such other parties will be able to fully satisfy its indemnification obligations. It is also possible that a court could disregard the allocation of assets and liabilities agreed to between the Company and such other parties and require the Company to assume responsibility for obligations allocated to such other parties. Each of these risks could negatively affect our business and financial statements.

Risks related to our industry

Conditions in the automotive industry may adversely affect our business.

Our financial performance depends on conditions in the global automotive industry. Automotive and truck production and sales are cyclical and sensitive to general economic conditions and other factors, including interest rates, consumer credit, and consumer spending and preferences. Economic declines that result in significant reduction in automotive or truck production would have an adverse effect on our sales to OEMs.

We face strong competition.

We compete globally with a number of other manufacturers and distributors that produce and sell similar products. Price, quality, delivery, technological innovation, engineering development and program launch support are the primary elements of competition. Our competitors include vertically integrated units of our major OEM customers, as well as a large number of independent domestic and international suppliers. Additionally, our competitors include start-ups that may be well funded, with the result that they could have more operational and financial flexibility than we have. A number of our competitors are larger than we are, and some competitors have greater financial and other resources than we do. Although many OEMs have indicated that they will continue to rely on outside suppliers, a number of major OEM customers have indicated their intent to insource certain components that we produce, and many do manufacture products for their own uses that directly compete with our products. These OEMs could elect to manufacture such products for their own uses in place of the products we currently supply. Our traditional OEM customers, faced with intense international competition, have continued to expand their global sourcing of components. As a result, we have experienced competition from suppliers in other parts of the world that enjoy economic advantages, such as lower labor costs, lower health care costs, lower tax rates and, in some cases, export or raw materials subsidies. Increased competition could adversely affect our business. In addition, any of our competitors may foresee the course of market development more accurately than we do, develop products that are superior to our products, produce similar products at a cost that is lower than our cost, or adapt more quickly than we do to new technologies or evolving customer requirements. As a result, our products may not be able to compete successfully with our competitors' products, and we may not be able to meet the growing demands of customers. These trends may adversely affect our sales as well as the profit margins on our products.

If we do not respond appropriately, the evolution of the automotive industry could adversely affect our business.

The automotive industry is increasingly focused on improved vehicle efficiency and reduced emissions, including the development of hybrid and electric vehicles, largely as a result of changing consumer preferences and increasingly stringent global regulatory requirements related to climate change, and of advanced driver-assistance technologies, with the goal of developing and introducing a commercially viable, fully automated driving experience. There has also been an increase in consumer preferences for mobility-on-demand services, such as car and ride sharing, as opposed to automobile ownership, which may result in a long-term reduction in the number of vehicles per capita. In addition, some industry participants are exploring transportation through alternatives to automobiles. These evolving areas have also attracted increased competition from entrants outside the traditional automotive industry. While we are focused on driving growth through our ability to capitalize on certain potential trends, such as the move toward hybrid and electric vehicles, some of the focuses and trends are not part of our product line or strategy, which could have an adverse impact on our results of operations. If we do not continue to innovate and develop, or acquire, new and compelling products that capitalize upon new technologies in response to OEM and consumer preferences, this could have an adverse impact on our results of operations.

We may be subject to potential governmental investigations and related proceedings relating to vehicle emissions standards.

In recent years, within the automotive industry, there have been governmental investigations and related proceedings relating to alleged or actual violations of vehicle emissions standards. Alleged violations by BorgWarner of existing or future emissions standards could result in government investigations and other legal proceedings, the recall of one or more of our products, negotiated remedial actions, fines,

disgorgement of profits, restricted product offerings, reputational harm or a combination of any of those items. Any of these actions could have a material adverse effect on our business and financial results.

Risks related to our business

We are under substantial pressure from OEMs to reduce the prices of our products.

There is substantial and continuing pressure on OEMs to reduce costs, including costs of products we supply. OEM customers expect annual price reductions in our business. To maintain our profit margins, we seek price reductions from our suppliers, improved production processes to increase manufacturing efficiency, and streamlined product designs to reduce costs, and we attempt to develop new products, the benefits of which support stable or increased prices. Our ability to pass through increased raw material or other inflationary costs to our OEM customers is limited, with cost recovery often less than 100% and often on a delayed basis. Inability to reduce costs in an amount equal to annual price reductions, increases in raw material costs, and increases in employee wages and benefits could have an adverse effect on us.

We continue to face volatile costs of commodities used in the production of our products and elevated levels of inflation.

We use a variety of commodities (including aluminum, copper, nickel, plastic resins, steel, other raw materials and energy) and materials purchased in various forms such as castings, powder metal, forgings, stampings and bar stock. Beginning in 2021, we have experienced price increases for base metals (e.g., steel, aluminum and nickel), precious metals (e.g., palladium) and raw materials that are primarily used in batteries for electric vehicles (e.g., lithium and cobalt). Increasing commodity costs negatively impact our operating margins and results. We have sought to alleviate the impact of increasing costs by including material pass-through provisions in our customer contracts wherever possible and by selectively hedging certain commodity exposures. The discontinuation or lessening of our ability to pass through or hedge increasing commodity costs could adversely affect our business.

From time to time, commodity prices may also fall rapidly. If this happens, suppliers may withdraw capacity from the market until prices improve, which may cause periodic supply interruptions. The same may be true of transportation carriers and energy providers. If these supply interruptions occur, it could adversely affect our business.

In addition, during 2022 and 2023, many global economies, including the United States, experienced elevated levels of inflation more generally, which drove an increase in input costs. Following non-contractual negotiations, we reached cost-recovery agreements with various customers in 2022 and 2023, but these agreements did not enable us to recover 100 percent of our increased costs, and as a result, our operating margins were negatively impacted. While we will continue to negotiate the pass through and recovery of higher costs with our customers, continued increasing levels of inflation could adversely affect our business.

Changes in U.S. administrative policy, including changes to existing trade agreements and any resulting changes in international trade relations, may have an adverse effect on us.

The United States has maintained tariffs on certain imported steel, aluminum and items originating from China. These tariffs have increased the cost of raw materials and components we purchase. The imposition of tariffs by the United States has resulted in retaliatory tariffs from a number of countries, including China, which increase the cost of products we sell. If the U.S. or other countries impose additional tariffs, that will have a further adverse impact on us.

We use important intellectual property in our business. If we are unable to protect our intellectual property or if a third party makes assertions against us or our customers relating to intellectual property rights, our business could be adversely affected.

We own important intellectual property, including patents, trademarks, copyrights, and trade secrets and are involved in numerous licensing arrangements. Our intellectual property plays an important role in maintaining our competitive position in a number of the markets that we serve. Our competitors may develop technologies that are similar or superior to our proprietary technologies or design around the patents we own or license. Further, as we expand our operations in jurisdictions where the enforcement of intellectual property rights is less robust, the risk of others duplicating our proprietary technologies increases, despite efforts we undertake to protect them. Our inability to protect or enforce our intellectual property rights or claims that we are infringing intellectual property rights of others could adversely affect our business and our competitive position.

We are subject to business continuity risks associated with increasing centralization of our information technology ("IT") systems.

To improve efficiency and reduce costs, we have regionally centralized the information systems that support our business processes such as invoicing, payroll, and general management operations. If the centralized systems are disrupted or disabled, key business processes could be interrupted, which could adversely affect our business.

A failure of or disruption in our information technology infrastructure, including a disruption related to cybersecurity, could adversely impact our business and operations.

We rely on the capacity, reliability and security of our IT systems and infrastructure. IT systems are vulnerable to disruptions, including those resulting from natural disasters, cyber-attacks or failures in third-party provided services. Disruptions and attacks on our IT systems pose a risk to the security of our systems and our ability to protect our networks and the confidentiality, availability and integrity of information and data and that of third parties, including our employees. Some cyber-attacks depend on human error or manipulation, including phishing attacks or schemes that use social engineering to gain access to systems or carry out disbursement of funds or other frauds, which raise the risks from such events and the costs associated with protecting against such attacks. Although we have implemented security policies, processes, and layers of defense designed to help identify and protect against intentional and unintentional misappropriation or corruption of our systems and information, and disruptions of our operations, we have been, and likely will continue to be, subjected to such attacks or disruptions. Future attacks or disruptions could potentially lead to the inappropriate disclosure of confidential information, including our intellectual property or employee data, improper use of our systems and networks, access to and manipulation and destruction of our or third-party data, production downtimes, lost revenues, inappropriate disbursement of funds and both internal and external supply shortages. In addition, we may be required to incur significant costs to protect against damage caused by such attacks or disruptions in the future. These consequences could cause significant damage to our reputation, affect our relationships with our customers and suppliers, lead to claims against us and ultimately adversely affect our business.

Our business success depends on attracting and retaining qualified personnel.

Our ability to sustain and grow our business requires us to hire, retain and develop a highly skilled and diverse management team and workforce worldwide. In particular, any unplanned turnover or inability to attract and retain key employees and employees with engineering, technical and software capabilities in numbers sufficient for our needs could adversely affect our business.

Our profitability and results of operations may be adversely affected by program launch difficulties.

The launch of new business is a complex process, the success of which depends on a wide range of factors, including the production readiness of our manufacturing facilities and manufacturing processes and those of our suppliers, as well as factors related to tooling, equipment, employees, initial product quality and other factors. Our failure to successfully launch new business, or our inability to accurately estimate the cost to design, develop and launch new business, could have an adverse effect on our profitability and results of operations.

To the extent we are not able to successfully launch new business, vehicle production at our customers could be significantly delayed or shut down. Such situations could result in significant financial penalties to us or a diversion of personnel and financial resources to improving launches rather than investment in continuous process improvement or other growth initiatives and could result in our customers shifting work away from us to a competitor, all of which could result in loss of revenue or loss of market share and could have an adverse effect on our profitability and cash flows.

Part of our workforce is unionized, which could subject us to work stoppages.

As of December 31, 2023, approximately 12.5% of our U.S. workforce was unionized. We have a domestic collective bargaining agreement for one facility in New York, which expires in September 2024. The workforce at certain of our international facilities is also unionized. A prolonged dispute with our employees could have an adverse effect on our business.

Work stoppages, production shutdowns and similar events could significantly disrupt our business.

Because the automotive industry relies heavily on just-in-time delivery of components during the assembly and manufacture of vehicles, a work stoppage or production shutdown at one or more of our manufacturing and assembly facilities could have adverse effects on our business. Similarly, if one or more of our customers were to experience a work stoppage or production shutdown, that customer would likely halt or limit purchases of our products, which could result in the shutdown of the related manufacturing facilities. A significant disruption in the supply of a key component due to supply constraints, such as the constraints experienced in 2021 and 2022 related to semiconductor chips, or due to a work stoppage or production shutdown at one of our suppliers or any other supplier could have the same consequences and, accordingly, have an adverse effect on our financial results.

Changes in interest rates and asset returns could increase our pension funding obligations and reduce our profitability.

We have unfunded obligations under certain of our defined benefit pension and other postemployment benefit plans. The valuation of our future payment obligations under the plans and the related plan assets is subject to significant adverse changes if the credit and capital markets cause interest rates and projected rates of return to decline. Such declines could also require us to make significant additional contributions to our pension plans in the future. Additionally, a material deterioration in the funded status of the plans could significantly increase our pension expenses and reduce profitability in the future.

We also sponsor post-employment medical benefit plans in the U.S. that are unfunded. If medical costs continue to increase or actuarial assumptions are modified, this could have an adverse effect on our business.

We are subject to extensive environmental regulations that are subject to change and involve significant risks.

Our operations are subject to laws governing, among other things, emissions to air, discharges to waters, and the generation, management, transportation and disposal of waste and other materials. The operation of automotive parts manufacturing plants entails risks in these areas, and we cannot assure that we will not incur material costs or liabilities as a result. Through various acquisitions over the years, we have acquired a number of manufacturing facilities, and we cannot assure that we will not incur material costs and liabilities relating to activities that predate our ownership. In addition, potentially significant expenditures could be required to comply with evolving interpretations of existing environmental, health and safety laws and regulations or any new such laws and regulations (including concerns about global climate change and its impact) that may be adopted in the future. Costs associated with failure to comply with such laws and regulations could have an adverse effect on our business.

Our operations may be affected by greenhouse emissions and climate change and related regulations.

Climate change is receiving increasing attention worldwide, which has led to significant legislative and regulatory efforts to limit greenhouse gas emissions. Our manufacturing plants use energy, including electricity and natural gas, and certain of our plants emit amounts of greenhouse gas that may be affected by these legislative and regulatory efforts. Greenhouse gas regulation could increase the price of the electricity we purchase, increase costs for use of natural gas, potentially restrict access to or the use of natural gas, require us to purchase allowances to offset our own emissions or result in an overall increase in costs of raw materials, any one of which could increase our costs, reduce competitiveness in a global economy or otherwise negatively affect our financial condition, results of operations and reputation. Many of our suppliers face similar circumstances. Supply disruptions would raise market rates and jeopardize the continuity of production and could have an adverse effect on our financial results.

Climate changes could also disrupt our operations by impacting the availability and cost of materials within our supply chain, and could also increase insurance and other operating costs. These factors may impact our decisions to construct new facilities.

We have liabilities related to environmental, product warranties, litigation and other claims.

We and certain of our current and former direct and indirect corporate predecessors, subsidiaries and divisions have been identified by the United States Environmental Protection Agency and certain state environmental agencies and private parties as potentially responsible parties at various hazardous waste disposal sites under the Comprehensive Environmental Response, Compensation and Liability Act and equivalent state laws, and, as such, may be liable for the cost of clean-up and other remedial activities at such sites. While responsibility for clean-up and other remedial activities at such sites is typically shared among potentially responsible parties based on an allocation formula, we could have greater liability under applicable statutes. Refer to Note 21, “Contingencies,” to the Consolidated Financial Statements in item 8 of this report for further discussion.

We provide product warranties to our customers for some of our products. Under these product warranties, we may be required to bear costs and expenses for the repair or replacement of these products. As suppliers become more integrally involved in the vehicle design process and assume more of the vehicle assembly functions, auto manufacturers are increasingly looking to their suppliers for contribution when faced with recalls and product warranty claims. A recall claim brought against us, or a product warranty claim brought against us, could adversely impact our results of operations. In addition, a recall claim could require us to review our entire product portfolio to assess whether similar issues are present in other product lines, which could result in significant disruption to our business and could have

an adverse impact on our results of operations. We cannot assure that costs and expenses associated with these product warranties will not be material or that those costs will not exceed any amounts accrued for such product warranties in our financial statements.

We are currently, and may in the future become, subject to legal proceedings and commercial or contractual disputes. These claims typically arise in the normal course of business and may include, but not be limited to, commercial or contractual disputes with our customers and suppliers, intellectual property matters, personal injury, product liability, environmental and employment claims. There is a possibility that such claims may have an adverse impact on our business that is greater than we anticipate. While we maintain insurance for certain risks, the amount of insurance may not be adequate to cover all insured claims and liabilities. The incurrence of significant liabilities for which there is no, or insufficient, insurance coverage could adversely affect our business.

Compliance with and changes in laws could be costly and could affect our operating results.

We have operations in multiple countries that can be impacted by expected and unexpected changes in the legal and business environments in which we operate. Compliance-related issues in certain countries associated with laws such as the Foreign Corrupt Practices Act and other anti-corruption laws could adversely affect our business. We have internal policies and procedures relating to compliance with such laws; however, there is a risk that such policies and procedures will not always protect us from the improper acts of employees, agents, business partners, joint venture partners, or representatives, particularly in the case of recently acquired operations that may not have significant training in applicable compliance policies and procedures. Violations of these laws, which are complex, may result in criminal penalties, sanctions and/or fines that could have an adverse effect on our business, financial condition, and results of operations and reputation.

Changes that could impact the legal environment include new legislation, new regulations, new policies, investigations and legal proceedings, and new interpretations of existing legal rules and regulations, in particular, changes in import and export control laws or exchange control laws, additional restrictions on doing business in countries subject to sanctions, additional limitations on greenhouse gas emissions or other matters related to climate change and other changes in laws in countries where we operate or intend to operate.

Changes in tax laws or tax rates taken by taxing authorities and tax audits could adversely affect our business.

Changes in tax laws or tax rates, the resolution of tax assessments or audits by various tax authorities, and the inability to fully utilize our tax loss carryforwards and tax credits could adversely affect our operating results. In addition, we may periodically restructure our legal entity organization. If taxing authorities were to disagree with our tax positions in connection with any such restructurings, our effective tax rate could be materially affected.

Our tax filings for various periods are subject to audit by the tax authorities in most jurisdictions where we conduct business. We have received tax assessments from various taxing authorities and are currently at varying stages of appeals and/or litigation regarding these matters. These audits may result in assessment of additional taxes that are resolved with the authorities or through the courts. We believe these assessments may occasionally be based on erroneous and even arbitrary interpretations of local tax law. Resolution of any tax matters involves uncertainties, and there are no assurances that the outcomes will be favorable.

We are subject to risks related to our international operations.

We have manufacturing and technical facilities in many regions, including Europe, Asia, and the Americas. For 2023, approximately 84% of our consolidated net sales were outside the U.S. Consequently, our results could be affected by changes in trade, monetary and fiscal policies, trade restrictions or prohibitions, import or other charges or taxes, fluctuations in foreign currency exchange rates, limitations on the repatriation of funds, changing economic conditions, unreliable intellectual property protection and legal systems, insufficient infrastructures, social unrest, political instability and disputes, international terrorism and other factors that may be discrete to a particular country or geography. Compliance with multiple and potentially conflicting laws and regulations of various countries is challenging, burdensome and expensive.

The financial statements of foreign subsidiaries are translated to U.S. Dollars using the period-end exchange rate for assets and liabilities and an average exchange rate for each period for revenues, expenses and capital expenditures. The local currency is typically the functional currency for our foreign subsidiaries. Significant foreign currency fluctuations and the associated translation of those foreign currencies to U.S. Dollars could adversely affect our business. Additionally, significant changes in currency exchange rates, particularly the Euro, Korean Won and Chinese Renminbi, could cause fluctuations in the reported results of our businesses' operations that could negatively affect our results of operations.

Because we are a U.S. holding company, one significant source of our funds is distributions from our non-U.S. subsidiaries. Certain countries in which we operate have adopted or could institute currency exchange controls that limit or prohibit our non-U.S. subsidiaries' ability to convert local currency into U.S. Dollars or to make payments outside the country. This could subject us to the risks of local currency devaluation and business disruption.

Our business in China is subject to aggressive competition and is sensitive to economic, political, and market conditions.

Maintaining a strong position in the Chinese market is a key component of our global growth strategy. The automotive supply market in China is highly competitive, with competition from many of the largest global manufacturers and numerous smaller domestic manufacturers. As the Chinese market evolves, we anticipate that market participants will act aggressively to increase or maintain their market share. Increased competition may result in price reductions, reduced margins and our inability to gain or hold market share. In addition, our business in China is sensitive to economic, political, social and market conditions that drive sales volumes in China. If we are unable to maintain our position in the Chinese market or if vehicle sales in China decrease, our business and financial results could be adversely affected.

A downgrade in the ratings of our debt could restrict our ability to access the debt capital markets.

Changes in the ratings that rating agencies assign to our debt may ultimately impact our access to the debt capital markets and the costs we incur to borrow funds. If ratings for our debt fall below investment grade, our access to the debt capital markets could become restricted and our cost of borrowing or the interest rate for any subsequently issued debt would likely increase.

Our revolving credit agreement includes an increase in interest rates if the ratings for our debt are downgraded. The interest cost on our revolving credit agreement is based on a rating grid. Further, an increase in the level of our indebtedness and related interest costs may increase our vulnerability to adverse general economic and industry conditions and may affect our ability to obtain additional financing.

We could incur additional restructuring charges as we continue to execute actions in an effort to improve future profitability and competitiveness and to optimize our product portfolio and may not achieve the anticipated savings and benefits from these actions.

We have initiated and may continue to initiate restructuring actions designed to improve the competitiveness of our business and sustain our margin profile, optimize our product portfolio or create an optimal legal entity structure. We may not realize anticipated savings or benefits from past or future actions in full or in part or within the time periods we expect. We are also subject to the risks of labor unrest, negative publicity and business disruption in connection with our actions. Failure to realize anticipated savings or benefits from our actions could have an adverse effect on our business.

Risks related to our customers

We rely on sales to major customers.

We rely on sales to OEMs around the world of varying credit quality and manufacturing demands. Supply to several of these customers requires significant investment by us. We base our growth projections, in part, on commitments made by our customers. These commitments generally renew yearly during a program life cycle. Among other things, the level of production orders we receive is dependent on the ability of our OEM customers to design and sell products that consumers desire to purchase. If actual production orders from our customers do not approximate such commitments due to a variety of factors, including non-renewal of purchase orders, a customer's financial hardship or other unforeseen reasons, it could adversely affect our business.

Some of our sales are concentrated. Our worldwide sales in 2023 to Ford and Volkswagen constituted approximately 14% and 11% of our 2023 consolidated net sales, respectively. Sales to the Company’s top ten customers represented 68% of sales for the year ended December 31, 2023.

We are sensitive to the effects of our major customers’ labor relations.

All three of our primary North American customers, Ford, Stellantis, and General Motors, have major union contracts with the International Union, United Automobile, Aerospace and Agricultural Implement Workers of America (“UAW”). Additionally, there is effort from the UAW to unionize other North American OEM plants, the outcome of which is difficult to predict. Because of domestic OEMs’ dependence on a single union, we are affected by labor difficulties and work stoppages at OEMs’ facilities, such as the UAW strikes that occurred in 2023. Such stoppages at OEMs’ facilities could halt our businesses with those facilities and an increase in the number of OEMs facilities with union contracts with the UAW could increase the negative impact to our business. Similarly, a majority of our global customers’ operations outside of North America are also represented by various unions. Any extended work stoppage at one or more of our customers could have an adverse effect on our business.

Risks related to our suppliers

We could be adversely affected by supply shortages of components from our suppliers.

In an effort to manage and reduce the cost of purchased goods and services, we have been rationalizing our supply base. As a result, we remain dependent on fewer sources of supply for certain components used in the manufacture of our products. We select suppliers based on total value (including total landed price, quality, delivery, and technology), taking into consideration their production capacities and financial condition. We expect that they will deliver to our stated written expectations.

However, there can be no assurance that capacity limitations, industry shortages, labor or social unrest, weather emergencies, commercial disputes, government actions, riots, wars, such as Russia's invasion of Ukraine in 2022, sabotage, cyber-attacks, non-conforming parts, acts of terrorism, "Acts of God," or other problems that our suppliers experience will not result in occasional shortages or delays in their supply of components to us. During 2021, and to a lesser extent in 2022, trailing impacts of the shutdowns and production declines related, in part, to COVID-19, created supply constraints of certain components, particularly semiconductor chips. These supply constraints have had significant impacts on global industry production levels. If we experience a prolonged shortage of critical components from any of our suppliers and cannot procure the components from other sources, we may be unable to meet the production schedules for some of our key products and could miss customer delivery expectations. In addition, with fewer sources of supply for certain components, each supplier may perceive that it has greater leverage and, therefore, some ability to seek higher prices from us at a time that we face substantial pressure from OEMs to reduce the prices of our products, which could adversely affect our customer relations and business.

Suppliers' economic distress could result in the disruption of our operations and could adversely affect our business.

Rapidly changing industry conditions such as volatile production volumes; our need to seek price reductions from our suppliers as a result of the substantial pressure we face from OEMs to reduce the prices of our products; credit tightness; changes in foreign currency exchange rates; raw material, commodity, tariffs, transportation, and energy price escalation; drastic changes in consumer preferences; and other factors could adversely affect our supply chain, and sometimes with little advance notice. These conditions could also result in increased commercial disputes and supply interruption risks. In certain instances, it would be difficult and expensive for us to change suppliers that are critical to our business. On occasion, we must provide financial support to distressed suppliers or take other measures to protect our supply lines. We cannot predict with certainty the potential adverse effects these costs might have on our business.

We are subject to possible insolvency of financial counterparties.

We engage in numerous financial transactions and contracts, including insurance policies, letters of credit, credit line agreements, financial derivatives, and investment management agreements involving various counterparties. We are subject to the risk that one or more of these counterparties may become insolvent and, therefore, be unable to meet its obligations under such contracts.

Other risks

A variety of other factors could adversely affect our business.

Any of the following could materially and adversely affect our business: the loss of or changes in supply contracts or sourcing strategies of our major customers or suppliers; start-up expenses associated with new vehicle programs or delays or cancellation of such programs; low levels of utilization of our manufacturing facilities, which can be dependent on a single product line or customer; inability to recover engineering and tooling costs; market and financial consequences of recalls that may be required on products we supplied; delays or difficulties in new product development; the possible introduction of similar or superior technologies by others; global excess capacity and vehicle platform proliferation; and the impact of fire, flood, or other natural disasters, including pandemics and quarantines.

Item 1B. Unresolved Staff Comments

The Company has received no written comments regarding its periodic or current reports from the staff of the Securities and Exchange Commission that were issued 180 days or more preceding the end of its 2023 fiscal year that remain unresolved.

Item 1C. Cybersecurity

BorgWarner's Board of Directors acknowledges the importance of upholding the trust and confidence of its customers, business partners, employees and other stakeholders. The Board, in conjunction with the Audit Committee, is involved in the oversight of the Company's risk management program, including its Cybersecurity Program. The Cybersecurity Program is managed by the Chief Information Officer ("CIO"), whose information technology ("IT") team is responsible for enterprise-wide information technology, including cybersecurity strategy, policy, standards, architecture and processes. The Cybersecurity Program, including its standards, processes and practices, is benchmarked against recognized cybersecurity frameworks. The Cybersecurity Program continually enhances the enterprise security structure and contingency plans with the goal of preventing cybersecurity incidents to the extent feasible, while simultaneously increasing the organization system resilience to minimize the business impact should an incident occur.

Risk Management and Strategy

Collaborative Approach: The Company has implemented a comprehensive, cross-functional approach to identifying, preventing and mitigating cybersecurity threats and incidents. The Cybersecurity Program has various tools and programs in place to monitor and address potential threats and incidents impacting the Company's operations and to determine the materiality of and ensure timely public disclosure of such threat or incident, if appropriate.

Technical Safeguards: The Company deploys technical safeguards designed to protect the Company's information systems from cybersecurity threats. The Company deploys tools in an effort to detect vulnerabilities, and when a weakness is identified, the Company seeks to assess the significance of the impact and mitigate before the weakness is exploited by an unauthorized actor.

Incident Response and Recovery Plan: The Company has an incident response and recovery plan, which details the steps to be taken from the initial internal reporting of a potential cybersecurity incident.

Third-Party Risk Management: The Company is developing processes and procedures to identify and oversee cybersecurity risks presented by third parties, including service providers, vendors and other users of the Company's systems.

Education and Awareness: The Company provides regular, mandatory training for applicable personnel on cybersecurity threats to help them identify, avoid and address cybersecurity threats and to communicate the Company's Cybersecurity Program, including applicable policies, standards, processes and practices.

Governance

The Board and the Audit Committee actively discuss cybersecurity risks with management and among themselves. The CIO reports on the Company's Cybersecurity Program and the Company's approach to cybersecurity risk management to the Audit Committee of the Board of Directors two times a year and to the full Board periodically, as appropriate. These reports include updates on the Company's cybersecurity risks and threats, the status of projects to strengthen the Company's information security systems, assessments of the information security program, recent developments, evolving standards, vulnerability assessments, third-party and independent reviews, the emerging threat landscape, technological trends and information security considerations arising with respect to the Company's peers and third parties.

The Audit Committee and Board receive prompt and timely information regarding cybersecurity threats and incidents that meet specified thresholds, as well as ongoing updates regarding any such threats or incidents until they have been addressed.

The Cybersecurity Program and related initiatives are managed by the CIO, and the Company's IT team is responsible for enterprise-wide informational technology, coordinating with various functions and business groups to ensure they are following best practices. The current CIO has over two decades of experience in various roles in information technology and information security. The CIO and the IT team work with the business to implement the Cybersecurity Program, which is designed to protect the Company's information systems from cybersecurity threats and to promptly respond to any cybersecurity incidents in accordance with the Company's Cybersecurity Incident Response Plan. The CIO and the IT team use detection tools to monitor for cybersecurity threats and incidents in real time, apply mitigation and remediation steps and then report such threats to the Audit Committee and the Board, as appropriate.

The Company's efforts include a wide range of actions, including audits, assessments, tabletop exercises, threat modeling, vulnerability testing and other exercises focused on evaluating and improving the effectiveness of the Company's cybersecurity measures and planning. The Company engages in periodic assessment and testing of the Cybersecurity Program and may periodically engage a third-party expert to conduct the assessment, audits and testing. The results of such assessments, audits and testing are reported to the CIO and the Audit Committee or full Board, as applicable, and the Company makes adjustments as appropriate.

As of the date of this report, the Company is not aware of any material risks from cybersecurity threats that have materially affected or are reasonably likely to materially affect the Company, including its business strategy, results of operations, or financial condition. Despite the extensive approach we take to cybersecurity, we may not be successful in preventing or mitigating a cybersecurity incident that could have a material adverse effect on the Company or its stakeholders. See Item 1A. "Risk Factors" for a discussion of cybersecurity risks.

Item 2. Properties

As of December 31, 2023, the Company had 82 manufacturing, assembly and technical locations worldwide. The Company's worldwide headquarters are located in a leased facility in Auburn Hills, Michigan. In general, the Company believes its facilities to be suitable and adequate to meet its current and reasonably anticipated needs.

The following is additional information concerning principal manufacturing, assembly and technical facilities operated by the Company, its subsidiaries, and affiliates.

Segments	Americas	Europe	Asia	Total
Air Management	14	13	16	43
Drivetrain & Battery Systems	4	5	7	16
ePropulsion	8	6	9	23

The table above excludes unconsolidated joint ventures as of December 31, 2023 and administrative offices. Of the facilities noted above, 35 have leased land rights or a leased facility.

Item 3. Legal Proceedings

The Company is subject to a number of claims and judicial and administrative proceedings (some of which involve substantial amounts) arising out of the Company's business or relating to matters for which the Company may have a contractual indemnity obligation. See Note 21, "Contingencies," to the Consolidated Financial Statements in Item 8 of this report for a discussion of environmental, product liability, derivative and other litigation, which is incorporated herein by reference.

Item 4. Mine Safety Disclosures

Not applicable.

PART II

Item 5. Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities

The Company's common stock is listed for trading on the New York Stock Exchange under the symbol BWA. As of February 2, 2024, there were 1,464 holders of record of common stock.

While the Company currently expects that quarterly cash dividends will continue to be paid in the future at levels comparable to recent historical levels, the dividend policy is subject to review and change at the discretion of the Board of Directors.

The line graph below compares the cumulative total shareholder return on the Company's Common Stock with the cumulative total return of companies on the Standard & Poor's (S&P's) 500 Stock Index, and companies within Standard Industrial Code ("SIC") 3714 - Motor Vehicle Parts.

COMPARISON OF 5 YEAR CUMULATIVE TOTAL RETURN[1]
Among BorgWarner Inc., the S&P 500 Index, and SIC 3714 Motor Vehicle Parts



[1]$100 invested on 12/31/2018 in stock or index, including reinvestment of dividends. Fiscal year ending December 31.
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BWA and S&P 500 data are from Capital IQ; SIC Code Index data is from Research Data Group

	December 31,					
	2018	2019	2020	2021	2022	2023
BorgWarner Inc.[1]	$ 100.00	$ 127.16	$ 115.46	$ 136.71	$ 124.16	$ 127.65
S&P 500[2]	$ 100.00	$ 131.49	$ 155.68	$ 200.37	$ 164.08	$ 207.21
SIC Code Index[3]	$ 100.00	$ 128.74	$ 152.65	$ 162.23	$ 117.41	$ 116.08

[1] BorgWarner Inc.
[2] S&P 500 — Standard & Poor's 500 Total Return Index
[3] Standard Industrial Code ("SIC") 3714-Motor Vehicle Parts

Purchase of Equity Securities

In January 2020, the Company's Board of Directors authorized the purchase of up to $1 billion of the Company's common stock. This share repurchase authorization expired in January 2023 with approximately $544 million remaining for repurchase under that authorization. In November 2023, the Company's Board of Directors authorized the purchase of up to $544 million of the Company's common stock, which replaced the previous repurchase authorization. This share repurchase authorization does not expire. As of December 31, 2023, the Company had repurchased $177 million of common stock under this repurchase authorization. Shares purchased under this authorization may be repurchased in the open market at prevailing prices and at times and in amounts to be determined by management as market conditions and the Company's capital position warrant. The Company may use Rule 10b5-1 and 10b-18 plans to facilitate share repurchases. Repurchased shares will be deemed common stock held in treasury and may subsequently be reissued.

Employee transactions include restricted stock withheld to offset statutory minimum tax withholding that occurs upon vesting of restricted stock. The BorgWarner Inc. 2023 Stock Incentive Plan provides that the withholding obligations be settled by the Company retaining stock that is part of the award. Withheld shares will be deemed common stock held in treasury and may subsequently be reissued for general corporate purposes.

The following table provides information about the Company's purchases of its equity securities that are registered pursuant to Section 12 of the Securities Exchange Act of 1934, as amended (the "Exchange Act") during the quarter ended December 31, 2023:

Issuer Purchases of Equity Securities

Period	Total number of shares purchased	Average price per share	Total number of shares purchased as part of publicly announced plans or programs	Approximate dollar value of shares that may yet be purchased under plans or programs (in millions)
October 1, 2023 - October 31, 2023				
Common Stock Repurchase Program	—	$ —	—	$ 544
Employee transactions	—	$ —	—	
November 1, 2023 - November 30, 2023				
Common Stock Repurchase Program	3,000,000	$ 33.65	3,000,000	$ 443
Employee transactions	4,573	$ 34.15	—	
December 1, 2023 - December 31, 2023				
Common Stock Repurchase Program	2,264,923	$ 33.52	2,264,923	$ 367
Employee transactions	194	$ 33.95	—	

Equity Compensation Plan Information

As of December 31, 2023, the number of shares of options, warrants and rights outstanding under the Company's equity compensation plans, the weighted average exercise price of outstanding options, restricted common stock, warrants and rights and the number of securities remaining available for issuance were as follows:

Plan category	Number of securities to be issued upon exercise of outstanding options, warrants and rights (a)	Weighted average exercise price of outstanding options, warrants and rights (b)	Number of securities remaining available for future issuance under equity compensation plans (excluding securities reflected in column (a)) (c)
Equity compensation plans approved by security holders	1,817,795	$ 47.48	9,126,458
Equity compensation plans not approved by security holders	—	$ —	—
Total	1,817,795	$ 47.48	9,126,458

Item 6. [Reserved]

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

INTRODUCTION

BorgWarner Inc. and Consolidated Subsidiaries (the "Company" or "BorgWarner") is a global product leader in clean and efficient technology solutions for combustion, hybrid and electric vehicles. BorgWarner's products help improve vehicle performance, propulsion efficiency, stability and air quality. These products are manufactured and sold worldwide, primarily to original equipment manufacturers ("OEMs") of light vehicles (passenger cars, sport-utility vehicles ("SUVs"), vans and light trucks). The Company's products are also sold to other OEMs of commercial vehicles (medium-duty trucks, heavy-duty trucks and buses) and off-highway vehicles (agricultural and construction machinery and marine applications). The Company also manufactures and sells its products to certain tier one vehicle systems suppliers and into the aftermarket for light, commercial and off-highway vehicles. The Company operates manufacturing facilities serving customers in Europe, the Americas and Asia and is an original equipment supplier to nearly every major automotive OEM in the world.

Charging Forward - Electrification Portfolio Strategy

In 2021, the Company announced its strategy to aggressively grow its eProducts over time through organic investments and technology-focused acquisitions. eProducts include all products utilized on or for electric vehicles ("EVs") plus those same products and components that are included in hybrid powertrains whose underlying technologies are adaptable or applicable to those used in or for EVs. The Company believes it is well positioned for the industry's anticipated migration to EVs.

In June 2023, the Company announced the next phase of its *Charging Forward* strategy, which focuses on profitably growing eProducts while maximizing the value of the Company's Foundational products portfolio. Foundational products include all products utilized on internal combustion engines plus those same products and components that are also included in hybrid powertrains. As a result of executing its strategy, the Company expects that by 2027, it will achieve over $10 billion in annual eProduct sales, deliver eProduct adjusted operating margin of approximately 7% and maintain its double-digit adjusted operating margin for its Foundational products portfolio. During the year ended December 31, 2023, the Company's eProduct revenue was approximately $2.0 billion, or 14% of its total revenue.

On July 3, 2023, BorgWarner completed the previously announced spin-off ("Spin-Off") of its Fuel Systems and Aftermarket segments in a transaction intended to qualify as tax free to the Company's stockholders for U.S. federal income tax purposes, which was accomplished by the distribution of 100% of the outstanding common stock of PHINIA, Inc. ("PHINIA") to holders of record of common stock of the Company on a pro-rata basis. Each holder of record of common stock of the Company received one share of PHINIA common stock for every five shares of common stock of the Company held on June 23, 2023, the record date for the distribution ("Distribution Date"). In lieu of fractional shares of PHINIA, shareholders of the Company received cash. PHINIA is an independent public company trading under the symbol "PHIN" on the New York Stock Exchange.

The historical results of operations and the financial position of PHINIA for periods prior to the Spin-Off are presented as discontinued operations in the accompanying Consolidated Financial Statements.

Acquisitions

Eldor Corporation's Electric Hybrid Systems Business

On December 1, 2023, the Company completed its acquisition of the electric hybrid systems business segment of Eldor Corporation ("Eldor"), which is headquartered in Italy. The acquisition is expected to complement its existing ePropulsion product portfolio by enhancing the Company's engineering capabilities in power electronics. The Company paid €72 million ($78 million) at closing, with up to €175 million ($191 million) in earn-out payments that could be paid over the two years following closing. The Company's current estimates indicate that the minimum threshold for the earn-out target will not be achieved, thus no amount of the earn-out payment has been included in the purchase consideration.

Hubei Surpass Sun Electric Charging Business

On March 1, 2023, the Company completed its acquisition of the electric vehicle solution, smart grid and smart energy businesses of Hubei Surpass Sun Electric, pursuant to an Equity Transfer Agreement. The acquisition is expected to complement the Company's existing European and North American charging footprint by adding a presence in China. The total consideration was ¥288 million ($42 million), including ¥268 million ($39 million) of base purchase price and ¥20 million ($3 million) of estimated earn-out payments. The Company paid ¥217 million ($31 million) of the base purchase price in the year ended December 31, 2023. The remaining ¥51 million ($8 million) of base purchase price is payable in two installments with the last payment due before April 30, 2025. In addition, pursuant to the agreement, the Company could be obligated to remit up to ¥103 million ($15 million), in the form of contingent payments over approximately two years following the closing.

Drivetek AG

On December 1, 2022, the Company acquired Drivetek AG, an engineering and product development company located in Switzerland. This acquisition is expected to strengthen the Company's power electronics capabilities in auxiliary inverters to accelerate the growth of the High Voltage eFan business. The total consideration was ₣27 million ($29 million) of base purchase price paid at closing, and ₣10 million ($10 million) of estimated earn-out payments that could be paid in the form of contingent payments over the three years following closing.

Rhombus Energy Solutions

On July 29, 2022, the Company acquired Rhombus Energy Solutions, a provider of charging solutions in the North American market. The acquisition is expected to complement the Company's existing European charging footprint to accelerate organic growth and adds North American regional presence to its charging business. The Company paid $131 million at closing, and up to $30 million could be paid in the

form of contingent payments over the three years following closing. The Company's current estimates indicate that the minimum thresholds for these earn-out targets will not be achieved, thus no amount for the earn-out payments has been included in the purchase consideration.

Santroll Automotive Components

On March 31, 2022, the Company acquired Santroll Automotive Components, a carve-out of Santroll Electric Auto's eMotor business. The acquisition is expected to strengthen the Company's vertical integration, scale and portfolio breadth in light vehicle eMotors while allowing for increased speed to market. The total final consideration was $192 million, including approximately ¥1.0 billion ($152 million) of base purchase price and ¥0.25 billion ($40 million) of originally estimated earn-out payments. The Company paid approximately ¥1.0 billion ($157 million) of base purchase price in the year ended December 31, 2022 and no longer expects to recapture a previously anticipated $5 million of post-closing adjustments, which has been recorded in Other operating expense, net. Pursuant to the Equity Transfer Agreement for the acquisition, the obligation of the Company to remit up to ¥0.3 billion (approximately $47 million) of earn-out payments was contingent upon achievement of certain sales volume targets and certain estimated future volume targets associated with newly awarded business. During the year ended December 31, 2023, the Company paid approximately ¥0.2 billion ($24 million) to settle the remaining earn-out liability and related adjustments.

AKASOL AG

On June 4, 2021, the Company completed a voluntary public takeover offer for shares of AKASOL AG ("AKASOL"), resulting in ownership of 89% of AKASOL's outstanding shares. The Company paid approximately €648 million ($788 million) to settle the offer. During 2021, the Company increased its ownership to 93% through the subsequent purchase of additional shares. On February 10, 2022, the Company completed a merger squeeze-out process to obtain the remaining shares, resulting in 100% ownership. The acquisition is expected to further strengthen BorgWarner's commercial vehicle and industrial electrification capabilities, which positions the Company to capitalize on what it believes to be a fast-growing battery module and pack market.

Refer to Note 2, "Acquisitions and Dispositions," to the Consolidated Financial Statements in Item 8 of this report for more information. Results of operations for these acquisitions are included in the Company's financial information following their respective dates of acquisition.

Key Trends and Economic Factors

Commodities and Other Inflationary Impacts. Prices for commodities remain volatile, and since the beginning of 2021, the Company has experienced price increases for base metals (e.g., steel, aluminum and nickel), precious metals (e.g., palladium), silicon carbide, and raw materials that are primarily used in batteries for electric vehicles (e.g., lithium and cobalt). In addition, many global economies, including the United States, are experiencing elevated levels of inflation more generally, which is driving an increase in other input costs. As a result, the Company has experienced, and is continuing to experience, higher costs.

In 2022 and 2023, following non-contractual negotiations, the Company reached agreement for the pass through and recovery of higher costs with various customers. These agreements did not enable the Company to recover 100 percent of its increased costs, and as a result, the Company's operating margins were negatively impacted.

Foreign Currency Impacts. The rapid strengthening of the U.S. Dollar in 2022, which continued in 2023, albeit to a lesser extent, relative to major foreign currencies, including the Euro, Korean Won and Chinese Renminbi, and related translation of these currencies to the U.S. Dollar, unfavorably impacted

the Company's net sales, earnings and cash flows. Continued significant fluctuations of foreign currencies against the U.S. Dollar may further negatively impact the Company's financial results.

Outlook

The Company expects global industry production to be flat or to decrease modestly year over year in 2024. However, the Company expects net new business-related sales growth, due to the increased penetration of BorgWarner products, including eProducts, to drive a sales increase in excess of the change in industry production outlook. As a result, the Company expects increased revenue in 2024, excluding the impact of foreign currencies. The Company expects the earnings benefit of this revenue growth to be partially offset by the negative earnings impact of the acquisition of Eldor.

The Company maintains a positive long-term outlook for its global business and is committed to new product development and strategic investments to enhance its product leadership strategy. There are several trends that are driving the Company's long-term growth that management expects to continue, including adoption of product offerings for electrified vehicles and increasingly stringent global emissions standards that support demand for the Company's products that drive vehicle efficiency.

RESULTS OF OPERATIONS

A detailed comparison of the Company's 2021 operating results to its 2022 operating results can be found in the Management's Discussion and Analysis of Financial Condition and Results of Operations section in the Company's 2022 Annual Report on Form 10-K filed February 9, 2023.

The following table presents a summary of the Company's operatingresults:

	Year Ended December 31,			
(in millions, except per share data)	2023		2022	
Net sales		% of net sales		% of net sales
Air Management	$ 7,833	55.2 %	$ 7,137	56.5 %
Drivetrain & Battery Systems	4,348	30.6	3,735	29.6
ePropulsion	2,166	15.3	1,906	15.1
Inter-segment eliminations	(149)	(1.0)	(143)	(1.1)
Total net sales	14,198	100.0	12,635	100.0
Cost of sales	11,630	81.9	10,266	81.3
Gross profit	2,568	18.1	2,369	18.7
Selling, general and administrative expenses - R&D, net	717	5.1	701	5.5
Selling, general and administrative expenses - Other	599	4.2	589	4.7
Restructuring expense	79	0.6	48	0.4
Other operating expense, net	13	0.1	22	0.2
Operating income	1,160	8.2	1,009	8.0
Equity in affiliates' earnings, net of tax	(30)	(0.2)	(28)	(0.2)
Realized and unrealized loss on debt and equity securities	174	1.2	73	0.6
Interest expense, net	10	0.1	51	0.4
Other postretirement expense	15	0.1	—	—
Earnings from continuing operations before income taxes and noncontrolling interest	991	7.0	913	7.2
Provision for income taxes	289	2.0	195	1.5
Net earnings from continuing operations	702	4.9	718	5.7
Net (loss) earnings from discontinued operations	(7)	—	308	2.4
Net earnings	695	4.9	1,026	8.1
Net earnings from continuing operations attributable to the noncontrolling interest, net of tax	70	0.5	82	0.6
Net earnings attributable to BorgWarner Inc.	$ 625	4.4 %	$ 944	7.5 %
Earnings per share from continuing operations — diluted	$ 2.70		$ 2.69	

Net sales

Net sales for the year ended December 31, 2023 totaled $14,198 million, an increase of $1,563 million, or 12%, from the year ended December 31, 2022. The change in net sales for the year ended December 31, 2023 was primarily driven by the following:

- Favorable volume, mix and net new business increased sales approximately $1,418 million, or 10%. This increase was primarily driven by higher weighted average market production as estimated by the Company, which was up approximately 11% from the year ended December 31, 2022. The remaining increase primarily reflects the sales growth above market production, which the Company believes reflects higher demand for its products. Weighted average market production reflects light and commercial vehicle production as reported by IHS weighted for the Company's geographic exposure, as estimated by the Company.
- Fluctuations in foreign currencies resulted in a year-over-year decrease in sales of approximately $66 million, primarily due to the weakening of the Chinese Renminbi and Korean Won, partially offset by the strengthening of the Euro, in each case relative to the U.S. Dollar.

- Recoveries from the Company's customers of material cost inflation arising from non-contractual commercial negotiations with those customers and normal contractual customer commodity pass-through arrangements increased net sales by approximately $158 million.
- Acquisitions contributed $53 million in additional sales during the year ended December 31, 2023.

Cost of sales and gross profit
Cost of sales and cost of sales as a percentage of net sales were $11,630 million and 81.9%, respectively, during the year ended December 31, 2023, compared to $10,266 million and 81.3%, respectively, during the year ended December 31, 2022. The change in cost of sales for the year ended December 31, 2023 was primarily driven by the following:

- Higher sales volume, mix and net new business increased cost of sales by approximately $1,012 million.
- Fluctuations in foreign currencies resulted in a year-over-year decrease in cost of sales of approximately $30 million, primarily due to the weakening of the Chinese Renminbi and Korean Won, partially offset by the strengthening of the Euro, in each case relative to the U.S. Dollar.
- Cost of sales was also impacted by material cost inflation of approximately $170 million arising from non-contractual commercial negotiations and normal contractual supplier commodity pass-through arrangements with the Company's suppliers.

Gross profit and gross margin were $2,568 million and 18.1%, respectively, during the year ended December 31, 2023 compared to $2,369 million and 18.7%, respectively, during the year ended December 31, 2022. The decrease in gross margin was primarily due to the factors discussed above.

Selling, general and administrative expenses ("SG&A")
SG&A for the year ended December 31, 2023 was $1,316 million as compared to $1,290 million for the year ended December 31, 2022. SG&A as a percentage of net sales was 9.3% and 10.2% for the years ended December 31, 2023 and 2022, respectively. The change in SG&A was primarily attributable to:

- Research and development ("R&D") costs increased $16 million. R&D costs, net of customer reimbursements, were 5.1% of net sales in the year ended December 31, 2023, compared to 5.5% of net sales in the year ended December 31, 2022. The increase in R&D costs, net of customer reimbursements, was primarily due to increasing net investment related to the Company's eProduct portfolio. The Company will continue to invest in R&D programs, which are necessary to support short- and long-term growth.
- Increased administrative expenses of $19 million, primarily related to IT and travel.

Restructuring expense was $79 million and $48 million for the years ended December 31, 2023 and 2022, respectively, primarily related to employee benefit costs. Refer to Note 4 "Restructuring" to
the Consolidated Financial Statements in Item 8 of this report for more information.

In 2023, the Company announced a $130 million to $150 million restructuring plan to address structural costs in its Foundational products businesses. During the year ended December 31, 2023, the Company recorded $79 million of restructuring costs related to this plan.

Other operating expense, net was $13 million and $22 million for the years ended December 31, 2023 and 2022, respectively.

For the years ended December 31, 2023 and 2022, merger and acquisition expenses, net were $23 million and $9 million, respectively, primarily related to professional fees associated with specific acquisition initiatives.

During the year ended December 31, 2023, the Company recorded charges of $29 million, primarily related to the write down of a customer incentive asset, a service and lease agreement termination and impairment of certain property, plant and equipment.

During the year ended December 31, 2023, the Company recorded a $13 million gain, primarily related to the sale of a European manufacturing facility and other fixed assets. The sale of the facility was pursuant to a formal restructuring plan.

During the year ended December 31, 2022, the Company recorded a gain of $22 million in connection with the sale of its interest in BorgWarner Romeo Power LLC, in which the Company owned a 60% interest.

During the year ended December 31, 2022, the Company recorded an impairment charge of $30 million to remove the AKASOL indefinite-lived trade name as the Company no longer plans to utilize this trade name in the business.

Other operating expense, net is primarily comprised of items included within the subtitle “Non-comparable items impacting the Company’s earnings per diluted share and net earnings” below.

Equity in affiliates’ earnings, net of tax was $30 million and $28 million in the years ended December 31, 2023 and 2022, respectively. This line item is driven by the results of the Company’s unconsolidated joint ventures.

Realized and unrealized loss on debt and equity securities was $174 million and $73 million for the years ended December 31, 2023 and 2022, respectively. This line item reflects the net realized and unrealized gains or losses recognized due to valuing the Company’s investments at fair value. For the year ended December 31, 2023, this primarily related to losses recognized to adjust the Company’s investment in Wolfspeed Inc. (“Wolfspeed”) convertible debt securities to fair value. During the year ended December 31, 2023, the Company sold all of the $500 million in convertible debt securities.

Interest expense, net was $10 million and $51 million in the years ended December 31, 2023 and 2022, respectively. The decrease was primarily due to recognition of a $28 million gain on extinguishment of the Company’s 3.375% and 5.000% Senior Notes and higher interest rates on cash and cash equivalents balances and lower expense related to the Company’s cross-currency swaps.

Other postretirement expense was $15 million in the year ended December 31, 2023 compared to an immaterial amount in the year ended December 31, 2022. The increase in other postretirement expense for the year ended December 31, 2023, was primarily due to higher interest cost in 2023.

Provision for income taxes was $289 million for the year ended December 31, 2023 resulting in an effective tax rate of 29%. This compared to $195 million or 21% for the year ended December 31, 2022.

In 2023, the Company recognized a discrete tax benefit of approximately $19 million related to the resolution of tax audits and reductions in certain unrecognized tax benefits and accrued interest related to matters for which the statute of limitation had lapsed. In addition, the Company recognized a discrete tax benefit of approximately $50 million in relation to the Spin-Off, a discrete tax benefit of approximately $30 million in relation to various changes in filling positions for prior years, and a discrete tax expense of approximately $79 million in relation to changes in judgment related to the recovery of deferred tax assets, primarily due to the impact of the Spin-Off on the allocation of the Company’s profits across jurisdictions for tax purposes as well as various tax structuring actions and strategies.

In 2022, the Company recognized discrete tax benefits of $23 million, primarily related to a reduction in certain unrecognized tax benefits and accrued interest related to a matter for which the statute of limitations had lapsed and favorable provision-to-return adjustments.

For further details, see Note 7, “Income Taxes,” to the Consolidated Financial Statements in Item 8 of this report.

Net earnings attributable to the noncontrolling interest, net of tax of $70 million for the year ended December 31, 2023 decreased by $12 million compared to the year ended December 31, 2022. This decrease was primarily due to a decline in demand for certain of the Company’s Foundational products in China as well as the reduction arising from the Company’s 2023 purchase of the noncontrolling interest related to SeohanWarner Turbo Systems Ltd. in Korea.

Non-comparable items impacting the Company's earnings per diluted share and net earnings

The Company's earnings per diluted share were $2.70 and $2.69 for the years ended December 31, 2023 and 2022, respectively. The non-comparable items presented below are calculated after tax using the corresponding effective tax rate discrete to each item and the weighted average number of diluted shares for each of the years then ended. The Company believes the following table is useful in highlighting non-comparable items that impacted its earnings per diluted share:

	Year Ended December 31,	
Non-comparable items:	2023	2022
Restructuring expense	$ (0.24)	$ (0.15)
Merger and acquisition expense, net	(0.09)	(0.03)
Asset impairments and lease modifications	(0.10)	(0.13)
Gain on sales of businesses	0.02	0.04
Other non-comparable items	(0.07)	—
Realized and unrealized loss on debt and equity securities	(0.73)	(0.25)
Gain on sale of assets	0.04	—
Gain on debt extinguishment	0.09	—
Tax adjustments[1]	0.05	0.10
Total impact of non-comparable items per share — diluted:	$ (1.03)	$ (0.42)

1 In 2023, the Company recognized a discrete tax benefit of approximately $19 million related to the resolution of tax audits and reductions in certain unrecognized tax benefits and accrued interest related to matters for which the statute of limitations had lapsed. In addition, the Company recognized a discrete tax benefit of approximately $50 million related to the Spin-Off, a discrete tax benefit of approximately $30 million due to various changes in filling positions for prior years, and a discrete tax expense of approximately $79 million for changes in judgement on the recovery of certain deferred tax assets. During the year ended December 31, 2022, the Company recognized discrete tax benefits of $23 million, primarily related to a reduction in certain unrecognized tax benefits and accrued interest for a matter in which the statute of limitations had lapsed.

Results by Reportable Segment

The Company's business is aggregated into three reportable segments: Air Management, Drivetrain & Battery Systems and ePropulsion. These segments are strategic business groups, which are managed separately as each represents a specific grouping of related automotive components and systems. In previous years, the Company presented its results under four reportable segments: Air Management, ePropulsion & Drivetrain, Fuel Systems and Aftermarket. In the first quarter of 2023, the Company elected to disaggregate Air Management and ePropulsion & Drivetrain segments into Air Management, Drivetrain & Battery Systems and ePropulsion and reported its results in a total of five reportable segments: Air Management, Drivetrain & Battery Systems, ePropulsion, Fuel Systems and Aftermarket. As a result of the Spin-Off, Fuel Systems and Aftermarket are no longer reportable segments.

Segment Adjusted Operating Income (Loss) is the measure of segment income or loss used by the Company. Segment Adjusted Operating Income (Loss) is comprised of operating income adjusted for restructuring, merger, acquisition and divestiture expense, intangible asset amortization expense, impairment charges and other items not reflective of ongoing operating income or loss. The Company believes Segment Adjusted Operating Income (Loss) is most reflective of the operational profitability or loss of its reportable segments.

Segment Adjusted Operating Income excludes certain corporate costs, which primarily represent headquarters' expenses not directly attributable to the individual segments. Corporate expenses not allocated to Segment Adjusted Operating Income were $278 million and $282 million for the years ended December 31, 2023 and 2022, respectively. The decrease in corporate expenses in 2023 is primarily related to cost savings realized post PHINIA spin-off.

The following table presents net sales and Segment Adjusted Operating Income for the Company's reportable segments:

	Year ended December 31, 2023			Year ended December 31, 2022		
(in millions)	Net sales	Segment Adjusted Operating Income (Loss)	% margin	Net sales	Segment Adjusted Operating Income (Loss)	% margin
Air Management	$ 7,833	$ 1,171	14.9 %	$ 7,137	$ 1,073	15.0 %
Drivetrain & Battery Systems	4,348	545	12.5 %	3,735	449	12.0 %
ePropulsion	2,166	(90)	(4.2)%	1,906	(88)	(4.6)%
Inter-segment eliminations	(149)	—		(143)	—	
Totals	$ 14,198	$ 1,626		$ 12,635	$ 1,434	

The **Air Management** segment's net sales for the year ended December 31, 2023 increased $696 million, or 10%, and Segment Adjusted Operating Income increased $98 million from the year ended December 31, 2022. Foreign currencies resulted in a year-over-year decrease in sales of approximately $6 million, primarily due to the weakening of the Chinese Renminbi and Korean Won, partially offset by the strengthening of the Euro, in each case relative to the U.S. Dollar. Acquisitions contributed $25 million in additional sales during the year ended December 31, 2023. The increase excluding the impact of foreign currencies was primarily due to approximately $596 million of volume, mix and net new business driven by increased demand for the Company's products and higher weighted average market production compared to the prior year, non-contractual commercial negotiations and normal contractual customer commodity pass-through arrangements with the Company's customers. Segment Adjusted Operating margin was 14.9% for the year ended December 31, 2023, compared to 15.0% in the year ended December 31, 2022. The Segment Adjusted Operating margin decrease was primarily due to the impacts of charging business losses.

The **Drivetrain & Battery Systems** segment's net sales for the year ended December 31, 2023 increased $613 million, or 16%, and Segment Adjusted Operating Income increased $96 million from the year ended December 31, 2022. Foreign currencies resulted in a year-over-year decrease in sales of approximately $17 million, primarily due to the weakening of the Chinese Renminbi, partially offset by the strengthening of the Euro, in each case relative to the U.S. Dollar. The increase excluding these items was primarily due to approximately $604 million of volume, mix and net new business driven by increased demand for the Company's products and higher weighted average market production compared to the prior year, non-contractual commercial negotiations and normal contractual customer commodity pass-through arrangements with the Company's customers. Segment Adjusted Operating margin was 12.5% in the year ended December 31, 2023, compared to 12.0% in the year ended December 31, 2022. The Segment Adjusted Operating margin increase was primarily due to conversion on higher sales and customer recoveries, partially offset by higher costs due to inflation and battery systems business losses.

The **ePropulsion** segment's net sales for the year ended December 31, 2023 increased $260 million, or 14%, and Segment Adjusted Operating Loss increased $2 million from the year ended December 31, 2022. Foreign currencies resulted in a year-over-year decrease in sales of approximately $43 million, primarily due to the weakening of the Chinese Renminbi relative to the U.S. Dollar. Acquisitions contributed $28 million in additional sales during the year ended December 31, 2023. The increase excluding the impact of foreign currencies was primarily due to approximately $224 million of volume, mix and net new business driven by higher weighted average market production compared to the prior year, non-contractual commercial negotiations and normal contractual customer commodity pass-through arrangements with the Company's customers. Segment Adjusted Operating margin was (4.2)% in the year ended December 31, 2023, compared to (4.6)% in the year ended December 31, 2022. The Segment Adjusted Operating Loss was primarily due to investments in R&D for eProducts. The Segment Adjusted Operating margin was relatively flat as conversion on higher sales related to eProduct growth and customer recoveries were offset by higher R&D for eProducts, higher input costs due to inflation and additional depreciation expense.

LIQUIDITY AND CAPITAL RESOURCES

The Company maintains various liquidity sources, including cash and cash equivalents and the unused portion of its multi-currency revolving credit agreement. As of December 31, 2023, the Company had liquidity of $3,534 million, comprised of cash and cash equivalent balances of $1,534 million and an undrawn revolving credit facility of $2,000 million. The Company was in full compliance with its covenants under the revolving credit facility and had full access to its undrawn revolving credit facility. The total debt expected to mature through the end of 2024 is $73 million. Given the Company's strong liquidity position, management believes that it will have sufficient liquidity and will maintain compliance with all covenants through at least the next 12 months.

As of December 31, 2023, cash balances of $891 million were held by the Company's subsidiaries outside of the United States. Cash and cash equivalents held by these subsidiaries are used to fund foreign operational activities and future investments, including acquisitions. The majority of cash and cash equivalents held outside the United States is available for repatriation. The Company uses its U.S. liquidity primarily for various corporate purposes, including but not limited to debt service, share repurchases, dividend distributions, acquisitions and other corporate expenses.

The Company has a $2.0 billion multi-currency revolving credit facility, which includes a feature that allows the facility to be increased by $1.0 billion with bank group approval. This facility matures in September 2028. The credit facility agreement contains customary events of default and one key financial covenant, which is a debt-to-EBITDA (Earnings Before Interest, Taxes, Depreciation and Amortization) ratio. The Company was in compliance with the financial covenant at December 31, 2023. At December 31, 2023 and 2022, the Company had no outstanding borrowings under this facility.

The Company's commercial paper program allows the Company to issue $2.0 billion of short-term, unsecured commercial paper notes under the limits of its multi-currency revolving credit facility. Under this program, the Company may issue notes from time to time and use the proceeds for general corporate purposes. The Company had no outstanding borrowings under this program as of December 31, 2023 and 2022.

The total current combined borrowing capacity under the multi-currency revolving credit facility and commercial paper program cannot exceed $2.0 billion.

In addition to the revolving credit facility, the Company's universal shelf registration statement filed with the U.S. Securities and Exchange Commission provides the Company with the ability to issue various debt and equity securities subject to market conditions.

On February 8, 2023, April 26, 2023, July 26, 2023 and November 8, 2023, the Company's Board of Directors declared quarterly cash dividends of $0.17, $0.17, $0.11 and $0.11 per share of common stock, respectively. These dividends were paid on March 15, 2023, June 15, 2023, September 15, 2023 and December 15, 2023, respectively. The Company's third and fourth quarter cash dividend rate compared to the second quarter dividend rate reflects the impact of the Spin-Off.

From a credit quality perspective, the Company has a credit rating of BBB from Standard & Poor's, Baa1 from Moody's and BBB+ from Fitch Ratings. The current outlook from each of Fitch, Standard & Poor's and Moody's is stable. None of the Company's debt agreements require accelerated repayment in the event of a downgrade in credit ratings.

Cash Flows

Operating Activities

(in millions)	Year Ended December 31, 2023	2022
OPERATING ACTIVITIES OF CONTINUING OPERATIONS		
Net earnings from continuing operations	$ 702	$ 718
Adjustments to reconcile net earnings from continuing operations to net cash provided by operating activities from continuing operations:		
Depreciation and tooling amortization	515	483
Intangible asset amortization	67	69
Restructuring expense, net of cash paid	66	41
Stock-based compensation expense	58	64
(Gain) loss on sales of businesses	(5)	(16)
Gain on debt extinguishment	(28)	—
Realized and unrealized loss on debt and equity securities	174	73
Deferred income tax benefit	(44)	(76)
Other non-cash adjustments	4	(3)
Adjustments to reconcile net earnings from continuing operations to net cash provided by operating activities from continuing operations	1,509	1,353
Retirement plan contributions	(19)	(21)
Changes in assets and liabilities:		
Receivables	(482)	(409)
Inventories	(72)	(158)
Accounts payable and accrued expenses	375	433
Other assets and liabilities	86	(18)
Net cash provided by operating activities from continuing operations	$ 1,397	$ 1,180

Net cash provided by operating activities was $1,397 million and $1,180 million in the years ended December 31, 2023 and 2022, respectively. The increase for the year ended December 31, 2023, compared with the year ended December 31, 2022, was primarily due to higher net earnings adjusted for non-cash charges partially offset by changes in working capital.

Investing Activities

(in millions)	Year Ended December 31, 2023	2022
INVESTING ACTIVITIES OF CONTINUING OPERATIONS		
Capital expenditures, including tooling outlays	$ (832)	$ (622)
Payments for businesses acquired, net of cash and restricted cash acquired	(109)	(312)
Proceeds from sale of businesses, net of cash divested	9	27
Proceeds from settlement of net investment hedges, net	25	40
Proceeds from (payments for) investments in debt and equity securities, net	284	(473)
Proceeds from asset disposals and other, net	30	20
Net cash used in investing activities from continuing operations	$ (593)	$ (1,320)

Net cash used in investing activities was $593 million and $1,320 million in the years ended December 31, 2023 and 2022, respectively. In 2023, the Company acquired the electric vehicle solution, smart grid and smart energy businesses of Hubei Surpass Sun Electric and the electric hybrid systems business segment of Eldor Corporation. In 2022, the Company acquired Rhombus Energy Solutions, Santroll Automotive Components and Drivetek AG. Additionally, in 2022, the Company invested $500 million in Wolfspeed convertible debt securities, which were sold in 2023. These impacts were partially offset by proceeds related to the liquidation of the Company's investment in Romeo Power, Inc., the sale of the Company's 60% interest in BorgWarner Romeo Power LLC and proceeds from net investment hedges. As a percentage of sales, capital expenditures were 5.9% and 4.9% for the years ended December 31, 2023 and 2022, respectively.

Financing Activities

(in millions)	Year Ended December 31, 2023	2022
FINANCING ACTIVITIES OF CONTINUING OPERATIONS		
Additions to debt	$ 18	$ 5
Repayments of debt, including current portion	(451)	(13)
Payments for debt issuance costs	(3)	—
Payments for purchase of treasury stock	(177)	(240)
Payments for stock-based compensation items	(25)	(18)
Purchase of noncontrolling interest	(15)	(56)
Payments for contingent consideration	(23)	—
Net distribution from PHINIA	401	—
Dividends paid to BorgWarner stockholders	(130)	(161)
Dividends paid to noncontrolling stockholders	(116)	(81)
Net cash used in financing activities from continuing operations	$ (521)	$ (564)

Net cash used in financing activities was $521 million during the year ended December 31, 2023 compared to $564 million in the year ended December 31, 2022. Net cash used in financing activities during the year ended December 31, 2023 was primarily related to the $177 million of BorgWarner share repurchases, $130 million in dividends paid to the Company's stockholders, $116 million in dividends paid to noncontrolling stockholders of the Company's consolidated joint ventures and $23 million in contingent consideration payments. Additionally, during the year ended December 31, 2023, the Company used $15 million to purchase the noncontrolling interest related to SeohanWarner Turbo Systems Ltd. in Korea. Finally, during the year ended December 31, 2023, the Company executed the Spin-Off and received a net distribution, part of which was utilized to purchase and extinguish a portion of senior notes due in 2025.

Contractual Obligations

The Company's significant cash requirements for contractual obligations as of December 31, 2023 primarily consisted of the principal and interest payments on its notes payable and long-term debt, non-cancelable lease obligations, capital spending obligations and purchase obligations. The principal amount of notes payable and long-term debt was $3,785 million as of December 31, 2023. The projected interest payments over the terms of that debt were $761 million as of December 31, 2023. Refer to Note 14, "Notes Payable and Debt," to the Consolidated Financial Statements in Item 8 of this report for more information.

As of December 31, 2023, non-cancelable lease obligations were $225 million. Refer to Note 22, "Leases and Commitments," to the Consolidated Financial Statements in Item 8 of this report for more information. Capital spending obligations were $148 million as of December 31, 2023.

On November 16, 2022, the Company entered into a strategic partnership with Wolfspeed in which the Company invested $500 million in Wolfspeed's convertible debt securities and simultaneously entered into an agreement under which Wolfspeed agreed to provide a silicon carbide manufacturing capacity corridor to the Company. Under this agreement, beginning in 2024, the Company will purchase silicon carbide parts with an aggregate total price equal to or greater than the corridor amount totaling a minimum of $184 million, annually through 2029. On September 21, 2023, and November 15, 2023, the Company sold $100 million and the remaining $400 million, respectively, of the Wolfspeed convertible debt securities. The silicone carbine parts agreement is independent of the sale of the convertible debt securities and remains effective as of December 31, 2023.

Management believes that the combination of cash from operations, cash balances, available credit facilities, and the universal shelf registration capacity will be sufficient to satisfy the Company's cash needs for its current level of operations and its planned operations for the foreseeable future. Management will continue to balance the Company's needs for organic growth, inorganic growth, debt reduction, cash conservation and return of cash to shareholders.

Postretirement Defined Benefits

The Company's policy is to fund its defined benefit pension plans in accordance with applicable government regulations and to make additional contributions when appropriate. At December 31, 2023, all legal funding requirements had been met. The Company contributed $21 million, $22 million and $24 million to its defined benefit pension plans in the years ended December 31, 2023, 2022 and 2021, respectively.

The Company expects to contribute a total of $20 million to $30 million into its defined benefit pension plans during 2024. Of the $20 million to $30 million in projected 2024 contributions, $6 million are contractually obligated, while any remaining payments would be discretionary.

The funded status of all pension plans was a net unfunded position of $94 million and $173 million at December 31, 2023 and 2022, respectively. The decrease in the net unfunded position was a result of a lower projected benefit obligation, which was primarily due to discontinued operations and actuarial losses during the period. The main driver of these losses was the decrease of 0.70% in the weighted average discount rate for Non-U.S. plans. Of the total net unfunded amounts, $39 million and $34 million at December 31, 2023 and 2022, respectively, were related to plans in Germany, where there is no tax deduction allowed under the applicable regulations to fund the plans; hence, the common practice is to make contributions as benefit payments become due.

Other postemployment benefits primarily consist of health care benefits for certain former employees and retirees of the Company's U.S. operations. The Company funds these benefits as retiree claims are

incurred. Other postemployment benefits had an unfunded status of $33 million and $37 million at December 31, 2023 and 2022, respectively.

The Company believes it will be able to fund the requirements of these plans through cash generated from operations or other available sources of financing for the foreseeable future.

Refer to Note 18, “Retirement Benefit Plans,” to the Consolidated Financial Statements in Item 8 of this report for more information regarding costs and assumptions for employee retirement benefits.

OTHER MATTERS

Contingencies

In the normal course of business, the Company is party to various commercial and legal claims, actions and complaints, including matters involving warranty claims, intellectual property claims, governmental investigations and related proceedings, general liability and other risks. It is not possible to predict with certainty whether or not the Company will ultimately be successful in any of these commercial and legal matters or, if not, what the impact might be. The Company’s management does not believe that adverse outcomes in any of these commercial and legal claims, actions and complaints are reasonably likely to have a material adverse effect on the Company’s results of operations, financial position or cash flows. An adverse outcome could, nonetheless, be material to the results of operations or cash flows.

Environmental

The Company and certain of its current and former direct and indirect corporate predecessors, subsidiaries and divisions have been identified by the United States Environmental Protection Agency and certain state environmental agencies and private parties as potentially responsible parties (“PRPs”) at various hazardous waste disposal sites under the Comprehensive Environmental Response, Compensation and Liability Act (“Superfund”) and equivalent state laws and, as such, may be presently liable for the cost of clean-up and other remedial activities at 17 such sites. Responsibility for clean-up and other remedial activities at a Superfund site is typically shared among PRPs based on an allocation formula.

The Company believes that none of these matters, individually or in the aggregate, will have a material adverse effect on its results of operations, financial position or cash flows. Generally, this is because either the estimates of the maximum potential liability at a site are not material or the liability will be shared with other PRPs, although no assurance can be given with respect to the ultimate outcome of any such matter.

Refer to Note 21, “Contingencies,” to the Consolidated Financial Statements in Item 8 of this report for further details and information respecting the Company’s environmental liability.

CRITICAL ACCOUNTING POLICIES AND ESTIMATES

The consolidated financial statements are prepared in conformity with accounting principles generally accepted in the United States (“GAAP”). In preparing these financial statements, management has made its best estimates and judgments of certain amounts included in the financial statements, giving due consideration to materiality. Critical accounting policies are those that are most important to the portrayal of the Company’s financial condition and results of operations. Some of these policies require management's most difficult, subjective or complex judgments in the preparation of the financial statements and accompanying notes. Management makes estimates and assumptions about the effect of matters that are inherently uncertain, relating to the reporting of assets, liabilities, revenues, expenses and the disclosure of contingent assets and liabilities. The Company’s most critical accounting policies are discussed below.

Business combinations The Company allocates the cost of an acquired business to the assets acquired and liabilities assumed based on their estimated fair values at the date of acquisition. The excess value of the cost of an acquired business over the estimated fair value of the assets acquired and liabilities assumed is recognized as goodwill. The valuation of the acquired assets and liabilities will impact the determination of future operating results. The Company uses a variety of information sources to determine the value of acquired assets and liabilities, including third-party appraisers for the values and lives of property, identifiable intangibles and inventories, and actuaries for defined benefit retirement plans. Goodwill is assigned to reporting units as of the date of the related acquisition. If goodwill is assigned to more than one reporting unit, the Companyutilizes a method that is consistent with the manner in which the amount of goodwill in a business combination is determined. Costs related to the acquisition of a business are expensed as incurred.

Acquired intangible assets include customer relationships, developed technology and trade names. The Company estimates the fair value of acquired intangible assets using various valuation techniques. The primary valuation techniques used include forms of the income approach, specifically the relief-from-royalty and multi-period excess earnings valuation methods. Under these valuation approaches, the Company is required to make estimates and assumptions from a market participant perspective, which may include revenue growth rates, estimated earnings, royalty rates, obsolescence factors, contributory asset charges, customer attrition and discount rates. Under the multi-period excess earnings method, value is estimated as the present value of the benefits anticipated from ownership of the asset, in excess of the returns required on the investment in contributory assets that are necessary to realize those benefits. The intangible asset’s estimated earnings are determined as the residual earnings after quantifying estimated earnings from contributory assets. When the Company estimates fair value using the relief-from-royalty method, it calculates the cost savings associated with owning rather than licensing the assets. Assumed royalty rates are applied to projected revenue for the remaining useful lives of the assets to estimate the royalty savings.

While the Company uses its best estimates and assumptions, fair value estimates are inherently uncertain and subject to refinement. As a result, during the measurement period, which may be up to one year from the acquisition date, the Company may record adjustments to the assets acquired and liabilities assumed, with the corresponding offset to goodwill. Any adjustments required after the measurement period are recorded in the consolidated statement of earnings.

Future changes in the judgments, assumptions and estimates that are used in acquisition valuations and intangible asset and goodwill impairment testing, including discount rates or future operating results and related cash flow projections, could result in significantly different estimates of the fair values in the future. An increase in discount rates, a reduction in projected cash flows or a combination of the two could lead to a reduction in the estimated fair values, which may result in impairment charges that could materially affect the Company’s financial statements in any given year.

Impairment of long-lived assets, including definite-lived intangible assets The Company reviews the carrying value of its long-lived assets, whether held for use or disposal, including other amortizing intangible assets, when events and circumstances warrant such a review under ASC Topic 360. In assessing long-lived assets for an impairment loss, assets are grouped with other assets and liabilities at the lowest level for which identifiable cash flows are largely independent of the cash flows of other assets and liabilities. In assessing long-lived assets for impairment, management generally considers individual facilities to be the lowest level for which identifiable cash flows are largely independent. A recoverability review is performed using the undiscounted cash flows if there is a triggering event. If the undiscounted cash flow test for recoverability identifies a possible impairment, management will perform a fair value analysis. Management determines fair value under ASC Topic 820 using the appropriate valuation technique of market, income or cost approach. If the carrying value of a long-lived asset is considered impaired, an impairment charge is recorded for the amount by which the carrying value of the long-lived asset exceeds its fair value.

Management believes that the estimates of future cash flows and fair value assumptions are reasonable; however, changes in assumptions underlying these estimates could affect the valuations. Significant judgments and estimates used by management when evaluating long-lived assets for impairment include (i) an assessment as to whether an adverse event or circumstance has triggered the need for an impairment review; (ii) undiscounted future cash flows generated by the asset; and (iii) fair valuation of the asset. Events and conditions that could result in impairment in the value of long-lived assets include changes in the industries in which the Company operates, particularly the impact of a downturn in the global economy, as well as competition and advances in technology, adverse changes in the regulatory environment, or other factors leading to reduction in expected long-term sales or profitability.

Goodwill and other indefinite-lived intangible assets During the fourth quarter of each year, the Company tests goodwill for impairment by either performing a qualitative assessment or a quantitative analysis. The qualitative assessment evaluates various events and circumstances, such as macroeconomic conditions, industry and market conditions, cost factors, relevant events and financial trends, that may impact a reporting unit's fair value. Using this qualitative assessment, the Company determines whether it is more-likely-than-not the reporting unit's fair value exceeds its carrying value. If it is determined that it is not more-likely-than-not the reporting unit's fair value exceeds the carrying value, or upon consideration of other factors, including recent acquisition, restructuring or disposal activity or to refresh the fair values, the Company performs a quantitative goodwill impairment analysis. In addition, the Company may test goodwill in between annual test dates if an event occurs or circumstances change that could more-likely-than-not reduce the fair value of a reporting unit below its carrying value.

The Company performs a quantitative analysis on each reporting unit to refresh its respective fair value using a combined income and market approach. The market approach is based on market multiples (revenue and “EBITDA”, defined as earnings before interest, taxes, depreciation and amortization) and requires an estimate of appropriate multiples based on market data for comparable companies. The market valuation models and other financial ratios used by the Company require certain assumptions and estimates regarding the applicability of those models to the Company’s facts and circumstances.

The Company believes the assumptions and estimates used to determine the estimated fair value are reasonable. Different assumptions could materially affect the estimated fair value. The primary assumptions affecting the Company’s 2023 goodwill quantitative impairment review are as follows:

- **Discount rates:** The Company used a range of 12.5% to 14.5%weighted average cost of capital (“WACC”) as the discount rates for future cash flows. The WACC is intended to represent a rate of return that would be expected by a market participant.
- **Operating income margin:** The Company used historical and expected operating income margins, which may vary based on the projections of the reporting unit being evaluated.

- **Revenue growth rates:** The Company used a global automotive market industry growth rate forecast adjusted to estimate its own market participation for product lines.

In addition to the above primary assumptions, the Company notes the following risks to volume and operating income assumptions that could have an impact on the discounted cash flow models:

- The automotive industry is cyclical, and the Company's results of operations could be adversely affected by industry downturns.
- The automotive industry is evolving, and if the Company does not respond appropriately, its results of operations could be adversely affected.
- The Company is dependent on market segments that use its key products and could be affected by decreasing demand in those segments.
- The Company is subject to risks related to international operations.

Based on the assumptions outlined above, the impairment testing conducted in the fourth quarter of 2023 indicated the Company's goodwill assigned to the respective reporting units was not impaired. Future changes in the judgments, assumptions and estimates from those used in acquisition-related valuations and goodwill impairment testing, including discount rates or future operating results and related cash flow projections, could result in significantly different estimates of the fair values in the future. Due to the Company's recent acquisitions, there is less headroom (the difference between the carrying value and the fair value) associated with certain of the Company's reporting units. Based on the impairment testing conducted in 2023, the amounts by which the estimated fair values of the Company's goodwill reporting units exceeded their carrying values ranged from 22% to 139%. An increase in discount rates, a reduction in projected cash flows or a combination of the two could lead to a reduction in the estimated fair values, which may result in impairment charges that could materially affect the Company's financial statements in any given year.

Similar to goodwill, the Company can elect to perform the impairment test for indefinite-lived intangibles other than goodwill (primarily trade names) using a qualitative analysis, considering similar factors as outlined in the goodwill discussion, in order to determine if it is more-likely-than-not that the fair value of the trade names is less than the respective carrying values. If the Company elects to perform or is required to perform a quantitative analysis, the test consists of a comparison of the fair value of the indefinite-lived intangible asset to the carrying value of the asset as of the impairment testing date. The Company estimates the fair value of indefinite-lived intangibles using the relief-from-royalty method, which it believes is an appropriate and widely used valuation technique for such assets. The fair value derived from the relief-from-royalty method is measured as the discounted cash flow savings realized from owning such trade names and not being required to pay a royalty for their use.

Refer to Note 12, "Goodwill and Other Intangibles," to the Consolidated Financial Statements in Item 8 of this report for more information regarding goodwill.

Product warranties The Company provides warranties on some, but not all, of its products. The warranty terms are typically from one to three years. Provisions for estimated expenses related to product warranty are made at the time products are sold. These estimates are established using historical information about the nature, frequency and average cost of warranty claim settlements as well as product manufacturing and industry developments and recoveries from third parties. Management actively studies trends of warranty claims and takes action to improve product quality and minimize warranty claims. Costs of product recalls, which may include the cost of the product being replaced as well as the customer's cost of the recall, including labor to remove and replace the recalled part, are accrued as part of the Company's warranty accrual at the time an obligation becomes probable and can be reasonably estimated. Management believes that the warranty accrual is appropriate; however, if actual claims incurred differ from the original estimates or there are changes in our assumptions, it could materially affect the Company's financial statements.

At December 31, 2023, the total accrued warranty liability was $196 million. The accrual is represented as $91 million in Other current liabilities and $105 million in Other non-current liabilities on the Consolidated Balance Sheets.

Refer to Note 13, "Product Warranty," to the Consolidated Financial Statements in Item 8 of this report for more information regarding product warranties.

Postretirement defined benefits The Company provides postretirement defined benefits to a number of its current and former employees. Costs associated with postretirement defined benefits include pension and other postemployment health care expenses for former employees, retirees and surviving spouses and dependents.

The Company's defined benefit pension and other postemployment benefit plans are accounted for in accordance with ASC Topic 715. The determination of the Company's obligation and expense for its pension and other postemployment benefits, such as retiree health care, is dependent on certain assumptions used by actuaries in calculating such amounts. Certain assumptions, including the expected long-term rate of return on plan assets, discount rate, rates of increase in compensation and health care costs trends are described in Note 18, "Retirement Benefit Plans," to the Consolidated Financial Statements in Item 8 of this report. The effects of any modification to those assumptions, or actual results that differ from assumptions used, are either recognized immediately or amortized over future periods in accordance with GAAP.

The primary assumptions affecting the Company's accounting for employee benefits under ASC Topics 712 and 715 as of December 31, 2023 are as follows:

- **Expected long-term rate of return on plan assets**: The expected long-term rate of return is used in the calculation of net periodic benefit cost. The required use of the expected long-term rate of return on plan assets may result in recognized returns that are greater or less than the actual returns on those plan assets in any given year. Over time, however, the expected long-term rate of return on plan assets is designed to approximate actual earned long-term returns. The expected long-term rate of return for pension assets has been determined based on various inputs, including historical returns for the different asset classes held by the Company's trusts and its asset allocation, as well as inputs from internal and external sources regarding expected capital market return, inflation and other variables. The Company also considers the impact of active management of the plans' invested assets. In determining its pension expense for the year ended December 31, 2023, the Company used long-term rates of return on plan assets ranging from 1.3% to 12% outside of the U.S. and 5% in the U.S.

Actual returns on U.S. pension assets were 5.2 % and (17.9)% for the years ended December 31, 2023 and 2022, respectively, compared to the expected rate of return assumptions of 5% and 4.8%, respectively, for the same years ended.

Actual returns on U.K. pension assets were 3.2% and (34.8)% for the years ended December 31, 2023 and 2022, respectively, compared to the expected rate of return assumption of 5.3% and 4.1%, respectively, for the same years ended.

Actual returns on German pension assets were 9.9% and (19.7)% for the years ended December 31, 2023 and 2022, respectively, compared to the expected rate of return assumptions of 4.5% and 4.0%, respectively, for the same years ended.

- **Discount rate**: The discount rate is used to calculate pension and other postemployment benefit ("OPEB") obligations. In determining the discount rate, the Company utilizes a full-yield approach in the estimation of service and interest components by applying the specific spot rates along the yield curve used in the determination of the benefit obligation to the relevant projected cash flows. For its significant plans, the Company used discount rates ranging from 1.8% to 11.8% to determine its pension and other benefit obligations as of December 31, 2023, including weighted average discount rates of 5.1% in the U.S., 4.2% outside of the U.S. (including 4.6% in the U.K.) and 5.1% for U.S. other postemployment health care plans. The U.S. and U.K. discount rates reflect the fact that the U.S. and U.K. pension plans have been closed for new participants.

- **Health care cost trend**: For postemployment employee health care plan accounting, the Company reviews external data and Company-specific historical trends for health care cost to determine the health care cost trend rate assumptions. In determining the projected benefit obligation for postemployment health care plans as of December 31, 2023, the Company used health care cost trend rates of 6.3%, declining to an ultimate trend rate of 4.75% by the year 2026.

While the Company believes that these assumptions are appropriate, significant differences in actual experience or significant changes in these assumptions may materially affect the Company's pension and OPEB and its future expense.

The sensitivity to a 25 basis-point change in the assumptions for discount rate and expected return on assets related to 2024 pre-tax pension expense for Company sponsored U.S. and non-U.S. pension plans is expected to be negligible.

The following table illustrates the sensitivity to a change in discount rate for Company sponsored U.S. and non-U.S. pension plans on its pension obligations:

(in millions)	Impact on U.S. PBO	Impact on Non-U.S. PBO
25 basis point decrease in discount rate	$ 3	$ 16
25 basis point increase in discount rate	$ (3)	$ (16)

The sensitivity to a 25 basis-point change in the discount rate assumption and to the assumed health care cost trend related to the Company's OPEB obligation and service and interest cost is expected to be negligible.

Refer to Note 18, "Retirement Benefit Plans," to the Consolidated Financial Statements in Item 8 of this report for more information regarding the Company's retirement benefit plans.

Restructuring Restructuring costs may occur when the Company takes action to exit or significantly curtail a part of its operations or implements a reorganization that affects the nature and focus of

operations. A restructuring charge can consist of severance costs associated with reductions to the workforce, costs to terminate an operating lease or contract, professional fees and other costs incurred related to the implementation of restructuring activities.

The Company generally records costs associated with voluntary separations at the time of employee acceptance. Costs for involuntary separation programs are recorded when management has approved the plan for separation, the employees are identified and aware of the benefits they are entitled to and it is unlikely that the plan will change significantly. When a plan of separation requires approval by or consultation with the relevant labor organization or government, the costs are recorded upon agreement. Costs associated with benefits that are contingent on the employee continuing to provide service are expensed over the required service period.

Income taxes The Company accounts for income taxes in accordance with ASC Topic 740. Deferred tax assets and liabilities are recognized for the future tax consequences attributable to differences between financial statement carrying amounts of existing assets and liabilities and their respective tax bases and operating loss and tax credit carryforwards. Deferred tax assets and liabilities are measured using enacted tax rates expected to apply to taxable income in the years in which those temporary differences are expected to be recovered or settled. Deferred tax assets are reduced by a valuation allowance if, based on the weight of available evidence, it is more-likely-than-not that some portion or all of the deferred tax assets will not be realized.

Accounting for income taxes is complex, in part because the Company conducts business globally and, therefore, files income tax returns in numerous tax jurisdictions. Management judgment is required in determining the Company’s worldwide provision for income taxes and recording the related assets and liabilities, including accruals for unrecognized tax benefits and assessing the need for valuation allowances. In calculating the provision for income taxes on an interim basis, the Company uses an estimate of the annual effective tax rate based upon the facts and circumstances known at each interim period. In determining the need for a valuation allowance, the historical and projected financial performance of the operation recording the net deferred tax asset is considered along with any other pertinent information. Since future financial results may differ from previous estimates, periodic adjustments to the Company’s valuation allowance may be necessary.

The Company is subject to income taxes in the U.S. at the federal and state level and numerous non-U.S. jurisdictions. The determination of accruals for unrecognized tax benefits includes the application of complex tax laws in a multitude of jurisdictions across the Company's global operations. Management judgment is required in determining the accruals for unrecognized tax benefits. In the ordinary course of the Company’s business, there are many transactions and calculations where the ultimate tax determination is less than certain. Accruals for unrecognized tax benefits are established when, despite the belief that tax positions are supportable, there remain certain positions that do not meet the minimum probability threshold, which is a tax position that is more-likely-than-not to be sustained upon examination by the applicable taxing authority. The Company has certain U.S. state income tax returns and certain non-U.S. income tax returns that are currently under various stages of audit by applicable tax authorities. At December 31, 2023, the Company had a liability for tax positions the Company estimates are not more-likely-than-not to be sustained based on the technical merits, which is included in Other non-current liabilities. Nonetheless, the amounts ultimately paid, if any, upon resolution of the issues raised by the taxing authorities may differ materially from the amounts accrued for each year.

The Company records valuation allowances to reduce the carrying value of certain deferred tax assets to amounts that it expects are more-likely-than-not to be realized. Existing deferred tax assets, net operating losses and tax credits by jurisdiction and expectations of the ability to utilize these tax attributes are assessed through a review of past, current and estimated future taxable income and tax planning strategies.

Estimates of future taxable income, including income generated from prudent and feasible tax planning strategies resulting from actual or planned business and operational developments, could change in the near term, perhaps materially, which may require the Company to consider any potential impact to the assessment of the recoverability of the related deferred tax asset. Such potential impact could be material to the Company's consolidated financial condition or results of operations for an individual reporting period.

In future periods, the Company's effective tax rate and tax liability may be impacted due to changes in U.S. and non-U.S. tax laws and as a result of regulatory or legislative developments related to such laws. This could include U.S. and non-U.S. tax law developments related to changes to long-standing tax principles arising from proposals made by the Organization for Economic Co-operation and Development that seek to allocate greater taxing rights to countries where customers are located and establish a global minimum tax rate of at least 15%.

Refer to Note 7, "Income Taxes," to the Consolidated Financial Statements in Item 8 of this report for more information regarding income taxes.

New Accounting Pronouncements

Refer to Note 1, "Summary of Significant Accounting Policies," to the Consolidated Financial Statements in Item 8 of this report for more information regarding new applicable accounting pronouncements.

QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

The Company's primary market risks include fluctuations in interest rates and foreign currency exchange rates. The Company is also affected by changes in the prices of commodities used or consumed in its manufacturing operations. Some of its commodity purchase price risk is covered by supply agreements with customers and suppliers. Other commodity purchase price risk is occasionally addressed by hedging strategies, which include forward contracts. The Company enters into derivative instruments only with high credit quality counterparties and diversifies its positions across such counterparties in order to reduce its exposure to credit losses. The Company does not engage in any derivative instruments for purposes other than hedging specific operating risks.

The Company has established policies and procedures to manage sensitivity to interest rate, foreign currency exchange rate and commodity purchase price risk, which include monitoring the level of exposure to each market risk. For quantitative disclosures about market risk, refer to Note 17, "Financial Instruments," to the Consolidated Financial Statements in Item 8 of this report for information with respect to interest rate risk, foreign currency exchange rate risk and commodity purchase price risk.

Interest Rate Risk

Interest rate risk is the risk that the Company will incur economic losses due to adverse changes in interest rates. The Company manages its interest rate risk by monitoring its exposure to fixed and variable rates while attempting to optimize its interest costs. The Company selectively uses interest rate swaps to reduce market value risk associated with changes in interest rates (fair value hedges). At December 31, 2023, all of the Company's long-term debt had fixed interest rates.

Foreign Currency Exchange Rate Risk

Foreign currency exchange rate risk is the risk that the Company will incur economic losses due to adverse changes in foreign currency exchange rates. Currently, the Company's most significant currency exposures relate to the Brazilian Real, British Pound, Chinese Renminbi, Euro, Korean Won, Mexican Peso, Polish Zloty, Singapore Dollar and Thailand Baht. The Company mitigates its foreign currency exchange rate risk by establishing local production facilities and related supply chain participants in the markets it serves, by invoicing customers in the same currency as the source of the products and by funding some of its investments in foreign markets through local currency loans. The Company also monitors its foreign currency exposure in each country and implements strategies to respond to changing economic and political environments. In addition, the Company regularly enters into forward currency contracts, cross-currency swaps and foreign currency-denominated debt designated as net investment hedges to reduce exposure to translation exchange rate risk. As of December 31, 2023 and 2022, the Company recorded a deferred gain of $112 million and $196 million, respectively, both before taxes, for designated net investment hedges within accumulated other comprehensive income (loss).

The significant foreign currency translation adjustments, including the impact of the net investment hedges discussed above, during the years ended December 31, 2023 and 2022, are shown in the following table, which provides the percentage change in U.S. Dollars against the respective currencies and the approximate impacts of these changes recorded within other comprehensive income (loss) for the respective periods.

(in millions, except for percentages)	December 31, 2023		
Chinese Renminbi	(3)%	$	(61)
Korean Won	(3)%	$	(11)
Euro	3 %	$	9

(in millions, except for percentages)	December 31, 2022		
Chinese Renminbi	(8)%	$	(201)
Euro	(6)%	$	(46)
British Pound	(11)%	$	(40)
Korean Won	(6)%	$	(25)
India Rupee	(10)%	$	(11)

Commodity Price Risk

Commodity price risk is the possibility that the Company will incur economic losses due to adverse changes in the cost of raw materials used in the production of its products. Commodity forward and option contracts are occasionally executed to offset exposure to potential change in prices mainly for various non-ferrous metals and natural gas consumption used in the manufacturing of vehicle components. As of December 31, 2023 and 2022, the Company had no outstanding commodity swap contracts.

Disclosure Regarding Forward-Looking Statements

The matters discussed in this Item 7 include forward looking statements. See "Forward Looking Statements" at the beginning of this Annual Report on Form 10-K.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk

For quantitative and qualitative information regarding market risk, refer to the discussion in Item 7 of this report under the caption “Quantitative and Qualitative Disclosures about Market Risk.”

For information regarding interest rate risk, foreign currency exchange risk and commodity price risk, refer to Note 17, “Financial Instruments,” to the Consolidated Financial Statements in Item 8 of this report. For information regarding the levels of indebtedness subject to interest rate fluctuation, refer to Note 14, “Notes Payable and Debt,” to the Consolidated Financial Statements in Item 8 of this report. For information regarding the level of business outside the United States, which is subject to foreign currency exchange rate market risk, refer to Note 24, “Reportable Segments and Related Information,” to the Consolidated Financial Statements in Item 8 of this report.

Item 8. Financial Statements and Supplementary Data

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Report of Independent Registered Public Accounting Firm

To the Board of Directors and Stockholders of BorgWarner Inc.

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of BorgWarner Inc. and its subsidiaries (the "Company") as of December 31, 2023 and 2022, and the related consolidated statements of operations, of comprehensive income, of equity and of cash flows for each of the three years in the period ended December 31, 2023, including the related notes (collectively referred to as the "consolidated financial statements"). We also have audited the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2023 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the COSO.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in Management's Report on Internal Control Over Financial Reporting appearing under Item 9A. Our responsibility is to express opinions on the Company's consolidated financial statements and on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

As described in Management's Report on Internal Control Over Financial Reporting, management has excluded Hubei Surpass Sun Electric's charging business and Eldor Corporation's electric hybrid systems business from its assessment of internal control over financial reporting as of December 31, 2023 because they were acquired by the Company in purchase business combinations during 2023. We have also excluded Hubei Surpass Sun Electric's charging business and Eldor Corporation's electric hybrid systems business from our audit of internal control over financial reporting. Hubei Surpass Sun Electric's charging business and Eldor Corporation's electric hybrid systems business are wholly-owned subsidiaries whose total assets and total net sales excluded from management's assessment and our audit of internal control over financial reporting collectively represent less than 1% of the related consolidated financial statement amounts as of and for the year ended December 31, 2023.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that (i) relates to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Worldwide Provision for Income Taxes

As described in Notes 1 and 7 to the consolidated financial statements, the Company recorded income taxes from continuing operations of $289 million for the year ended December 31, 2023. Management judgment is required in determining the Company's worldwide provision for income taxes and recording the related assets and liabilities, including accruals for unrecognized tax benefits and assessing the need for valuation allowances. As disclosed by management, accounting for income taxes is complex, in part because the Company conducts business globally and, therefore, files income tax returns in numerous tax jurisdictions. The Company is subject to income taxes in the U.S. at the federal and state level and numerous non-U.S. jurisdictions. In the ordinary course of the Company's business, there are many transactions and calculations where the ultimate tax determination is less than certain. Accruals for unrecognized tax benefits are established when, despite the belief that tax positions are supportable, there remain certain positions that do not meet the minimum probability threshold, which is a tax position that is more-likely-than-not to be sustained upon examination by the applicable taxing authority. The determination of accruals for unrecognized tax benefits includes the application of complex tax laws in a multitude of jurisdictions across the Company's global operations. The Company records valuation allowances to reduce the carrying value of deferred tax assets to amounts that it expects are more-likely-than-not to be realized. The Company assesses existing deferred tax assets, net operating loss carryforwards, and tax credit carryforwards by jurisdiction and expectations of its ability to utilize these tax attributes through a review of past, current, and estimated future taxable income and tax planning strategies.

The principal considerations for our determination that performing procedures relating to management's worldwide provision for income taxes is a critical audit matter are (i) the significant judgment by management when developing the worldwide provision for income taxes, (ii) a high degree of auditor judgment, subjectivity and effort in performing procedures and evaluating management's worldwide provision for income taxes, including the accruals for unrecognized tax benefits and valuation allowances on deferred tax assets, and (iii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to management's worldwide provision for income taxes, including controls over accruals for

unrecognized tax benefits and valuation allowances on deferred tax assets. These procedures also included, among others, (i) testing the accuracy of the worldwide provision for income taxes, including the rate reconciliation and permanent and temporary differences, (ii) evaluating the completeness of the accruals for unrecognized tax benefits, (iii) evaluating the reasonableness of management's more-likely-than-not determination in consideration of the tax laws in relevant jurisdictions, and (iv) evaluating the reasonableness of management's assessment of the realizability of its deferred tax assets based on expectations of the ability to utilize its tax attributes through a review of historical and estimated future taxable income and tax planning strategies. Professionals with specialized skill and knowledge were used to assist in (i) testing the accuracy of the worldwide provision for income taxes, (ii) evaluating the completeness of the accruals for unrecognized tax benefits, (iii) evaluating the reasonableness of management's more-likely-than-not determination in consideration of the tax laws in relevant jurisdictions, and (iv) evaluating the reasonableness of management's assessment of the realizability of its deferred tax assets.

/s/ PricewaterhouseCoopers LLP

Detroit, Michigan
February 8, 2024

We have served as the Company's auditor since 2008.

BORGWARNER INC. AND CONSOLIDATED SUBSIDIARIES
CONSOLIDATED BALANCE SHEETS

(in millions, except share and per share amounts)	December 31, 2023	2022
ASSETS		
Cash, cash equivalents and restricted cash	$ 1,534	$ 1,083
Receivables, net	3,109	2,471
Inventories, net	1,313	1,217
Prepayments and other current assets	261	230
Current assets of discontinued operations	—	1,616
Total current assets	6,217	6,617
Property, plant and equipment, net	3,783	3,426
Investments and long-term receivables	364	819
Goodwill	3,013	2,978
Other intangible assets, net	564	619
Other non-current assets	512	489
Non-current assets of discontinued operations	—	2,046
Total assets	$ 14,453	$ 16,994
LIABILITIES AND EQUITY		
Notes payable and other short-term debt	$ 73	$ 60
Accounts payable	2,546	2,146
Other current liabilities	1,148	1,084
Current liabilities of discontinued operations	—	946
Total current liabilities	3,767	4,236
Long-term debt	3,707	4,140
Retirement-related liabilities	146	129
Other non-current liabilities	767	686
Non-current liabilities of discontinued operations	—	295
Total liabilities	8,387	9,486
Commitments and contingencies		
Capital stock:		
Preferred stock, $0.01 par value; authorized shares: 5,000,000; none issued and outstanding	—	—
Common stock, $0.01 par value; authorized shares: 390,000,000; issued shares: (2023 - 283,575,876; 2022 - 283,575,876); outstanding shares: (2023 - 229,783,795; 2022 - 234,122,211)	3	3
Non-voting common stock, $0.01 par value; authorized shares: 25,000,000; none issued and outstanding	—	—
Capital in excess of par value	2,689	2,675
Retained earnings	6,152	7,454
Accumulated other comprehensive loss	(828)	(876)
Common stock held in treasury, at cost: (2023 - 53,792,081 shares; 2022 - 49,453,665 shares)	(2,188)	(2,032)
Total BorgWarner Inc. stockholders' equity	5,828	7,224
Noncontrolling interest	238	284
Total equity	6,066	7,508
Total liabilities and equity	$ 14,453	$ 16,994

See Accompanying Notes to Consolidated Financial Statements.

BORGWARNER INC. AND CONSOLIDATED SUBSIDIARIES
CONSOLIDATED STATEMENTS OF OPERATIONS

(in millions, except per share amounts)	Year Ended December 31, 2023	2022	2021
Net sales	$ 14,198	$ 12,635	$ 11,803
Cost of sales	11,630	10,266	9,630
Gross profit	2,568	2,369	2,173
Selling, general and administrative expenses	1,316	1,290	1,085
Restructuring expense	79	48	108
Other operating expense, net	13	22	66
Operating income	1,160	1,009	914
Equity in affiliates' earnings, net of tax	(30)	(28)	(42)
Realized and unrealized loss on debt and equity securities	174	73	362
Interest expense, net	10	51	91
Other postretirement expense (income)	15	—	(7)
Earnings from continuing operations before income taxes and noncontrolling interest	991	913	510
Provision for income taxes	289	195	65
Net earnings from continuing operations	702	718	445
Net (loss) earnings from discontinued operations	(7)	308	194
Net earnings	695	1,026	639
Net earnings from continuing operations attributable to the noncontrolling interest, net of tax	70	82	102
Net earnings attributable to BorgWarner Inc.	$ 625	$ 944	$ 537
Amounts attributable to BorgWarner Inc.:			
Net earnings from continuing operations	$ 632	$ 636	$ 343
Net (loss) earnings from discontinued operations	(7)	308	194
Net earnings attributable to BorgWarner Inc.	$ 625	$ 944	$ 537
Earnings per share from continuing operations — basic	$ 2.71	$ 2.70	$ 1.44
Earnings per share from discontinued operations — basic	(0.03)	1.31	0.81
Earnings per share attributable to BorgWarner Inc. — basic	$ 2.68	$ 4.01	$ 2.25
Earnings per share from continuing operations — diluted	$ 2.70	$ 2.69	$ 1.43
Earnings per share from discontinued operations — diluted	(0.03)	1.30	0.81
Earnings per share attributable to BorgWarner Inc. — diluted	$ 2.67	$ 3.99	$ 2.24
Weighted average shares outstanding:			
Basic	232.8	235.5	238.1
Diluted	234.4	236.8	239.5

See Accompanying Notes to Consolidated Financial Statements.

BORGWARNER INC. AND CONSOLIDATED SUBSIDIARIES
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME

(in millions)	Year Ended December 31, 2023	2022	2021
Net earnings attributable to BorgWarner Inc.	$ 625	$ 944	$ 537
Other comprehensive (loss) income			
Foreign currency translation adjustments[1]	31	(327)	(102)
Hedge instruments[1]	24	4	—
Postretirement defined benefit plans[1]	(7)	(2)	202
Total other comprehensive (loss) income attributable to BorgWarner Inc.	48	(325)	100
Comprehensive income attributable to BorgWarner Inc.[1]	673	619	637
Net earnings attributable to noncontrolling interest, net of tax	70	82	102
Other comprehensive (loss) income attributable to the noncontrolling interest[1]	(8)	(24)	(6)
Comprehensive income	$ 735	$ 677	$ 733

[1] Net of income taxes.

See Accompanying Notes to Consolidated Financial Statements.

BORGWARNER INC. AND CONSOLIDATED SUBSIDIARIES
CONSOLIDATED STATEMENTS OF CASH FLOWS

(in millions)	Year Ended December 31, 2023	2022	2021
OPERATING ACTIVITIES OF CONTINUING OPERATIONS			
Net cash provided by operating activities (see Note 25)	$ 1,397	$ 1,180	$ 1,210
INVESTING ACTIVITIES OF CONTINUING OPERATIONS			
Capital expenditures, including tooling outlays	(832)	(622)	(514)
Capital expenditures for damage to property, plant and equipment	—	—	(2)
Insurance proceeds received for damage to property, plant and equipment	—	—	5
Payments for businesses acquired, net of cash and restricted cash acquired	(109)	(312)	(759)
Proceeds from sale of businesses, net of cash divested	9	27	22
Proceeds from settlement of net investment hedges, net	25	40	11
Proceeds from (payments for) investments in debt and equity securities, net	284	(473)	(20)
Proceeds from asset disposals and other, net	30	20	8
Net cash used in investing activities from continuing operations	(593)	(1,320)	(1,249)
FINANCING ACTIVITIES OF CONTINUING OPERATIONS			
Additions to debt	18	5	1,286
Repayments of debt, including current portion	(451)	(13)	(699)
Payments for debt issuance costs	(3)	—	(11)
Payments for purchase of treasury stock	(177)	(240)	—
Payments for stock-based compensation items	(25)	(18)	(15)
Purchase of noncontrolling interest	(15)	(56)	(33)
Payments for contingent consideration	(23)	—	—
Net distribution from PHINIA	401	—	—
Dividends paid to BorgWarner stockholders	(130)	(161)	(162)
Dividends paid to noncontrolling stockholders	(116)	(81)	(72)
Net cash (used in) provided by financing activities from continuing operations	(521)	(564)	294
CASH FLOWS FROM DISCONTINUED OPERATIONS			
Operating activities of discontinued operations	(85)	390	97
Investing activities of discontinued operations	(86)	(99)	(147)
Financing activities of discontinued operations	84	(3)	(8)
Net cash (used in) provided by discontinued operations	(87)	288	(58)
Effect of exchange rate changes on cash	—	(90)	(3)
Net increase (decrease) in cash, cash equivalents and restricted cash	196	(506)	194
Cash, cash equivalents and restricted cash at beginning of year	1,338	1,844	1,650
Cash, cash equivalents and restricted cash at end of year	$ 1,534	$ 1,338	$ 1,844
Less: Cash, cash equivalents and restricted cash of discontinued operations at end of year	$ —	$ 255	$ 295
Cash, cash equivalents and restricted cash of continuing operations at end of year	$ 1,534	$ 1,083	$ 1,549

See Accompanying Notes to Consolidated Financial Statements.

BORGWARNER INC. AND CONSOLIDATED SUBSIDIARIES
CONSOLIDATED STATEMENTS OF EQUITY

	Number of shares		BorgWarner Inc. stockholder's equity						
(in millions, except share data)	Issued common stock	Common stock held in treasury	Issued common stock	Capital in excess of par value	Treasury stock	Retained earnings	Accumulated other comprehensive income (loss)	Noncontrolling interests	Total
Balance, January 1, 2021	283,575,876	(44,645,173)	$ 3	$ 2,614	$ (1,834)	$ 6,296	$ (651)	$ 296	$ 6,724
Dividends declared ($0.68 per share)[1]	—	—	—	—	—	(162)	—	(84)	(246)
Net issuance for executive stock plan	—	89,787	—	19	2	—	—	—	21
Net issuance of restricted stock	—	756,402	—	5	20	—	—	—	25
Acquisition of AKASOL	—	—	—	—	—	—	—	96	96
Purchase and reclass of noncontrolling interest	—	—	—	(1)	—	—	—	(90)	(91)
Net earnings	—	—	—	—	—	537	—	102	639
Other comprehensive loss	—	—	—	—	—	—	100	(6)	94
Balance, December 31, 2021	283,575,876	(43,798,984)	$ 3	$ 2,637	$ (1,812)	$ 6,671	$ (551)	$ 314	$ 7,262
Dividends declared ($0.68 per share)[1]	—	—	—	—	—	(161)	—	(85)	(246)
Net issuance for executive stock plan	—	181,212	—	28	4	—	—	—	32
Net issuance of restricted stock	—	583,021	—	9	16	—	—	—	25
Purchase of treasury stock	—	(6,418,914)	—	—	(240)	—	—	—	(240)
Purchase/sale of noncontrolling interest	—	—	—	1	—	—	—	(3)	(2)
Net earnings	—	—	—	—	—	944	—	82	1,026
Other comprehensive loss	—	—	—	—	—	—	(325)	(24)	(349)
Balance, December 31, 2022	283,575,876	(49,453,665)	$ 3	$ 2,675	$ (2,032)	$ 7,454	$ (876)	$ 284	$ 7,508
Dividends declared ($0.56 per share)[1]	—	—	—	—	—	(130)	—	(95)	(225)
Net issuance for executive stock plan	—	238,708	—	9	5	—	—	—	14
Net issuance of restricted stock	—	687,799	—	7	16	—	—	—	23
Purchase of treasury stock	—	(5,264,923)	—	—	(177)	—	—	—	(177)
Purchase of noncontrolling interest	—	—	—	(2)	—	—	—	(13)	(15)
Net earnings	—	—	—	—	—	625	—	70	695
Other comprehensive income	—	—	—	—	—	—	64	(8)	56
Spin-Off of PHINIA	—	—	—	—	—	(1,797)	(16)	—	(1,813)
Balance, December 31, 2023	283,575,876	(53,792,081)	$ 3	$ 2,689	$ (2,188)	$ 6,152	$ (828)	$ 238	$ 6,066

[1] Dividends declared relate to BorgWarner common stock.

See Accompanying Notes to Consolidated Financial Statements.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

INTRODUCTION

BorgWarner Inc. (together with its Consolidated Subsidiaries, the “Company”) is a Delaware corporation incorporated in 1987. The Company is a global product leader in clean and efficient technology solutions for combustion, hybrid and electric vehicles. The Company’s products help improve vehicle performance, propulsion efficiency, stability and air quality. The Company manufactures and sells these products worldwide, primarily to original equipment manufacturers (“OEMs”) of light vehicles (passenger cars, sport-utility vehicles (“SUVs”), vans and light trucks). The Company's products are also sold to OEMs of commercial vehicles (medium-duty trucks, heavy-duty trucks and buses) and off-highway vehicles (agricultural and construction machinery and marine applications). The Company also manufactures and sells its products to certain tier one vehicle systems suppliers and into the aftermarket for light, commercial and off-highway vehicles. The Company operates manufacturing facilities serving customers in Europe, the Americas and Asia and is an original equipment supplier to nearly every major automotive OEM in the world.

On July 3, 2023, BorgWarner completed the previously announced spin-off (“Spin-Off”) of its Fuel Systems and Aftermarket segments by the distribution of 100% of the outstanding common stock of PHINIA, Inc. (“PHINIA”) to holders of record of common stock of the Company on a pro-rata basis. Each holder of record of common stock of the Company received one share of PHINIA common stock for every five shares of common stock of the Company held on June 23, 2023, the record date for the distribution (“Distribution Date”). In lieu of fractional shares of PHINIA, shareholders of the Company received cash. PHINIA is an independent public company trading under the symbol “PHIN” on the New York Stock Exchange.

In connection with the Spin-Off, the Company entered into several agreements with PHINIA on or prior to the Distribution Date that, among other things, provide a framework for the Company’s relationship with PHINIA after the Spin-Off, including a separation and distribution agreement, an employee matters agreement, a tax matters agreement, an intellectual property cross-license agreement and a transition services agreement through which the Company and PHINIA will continue to provide certain services to each other following the Spin-Off.

NOTE 1 SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES

The following paragraphs briefly describe the Company’s significant accounting policies.

Basis of presentation Certain prior period amounts have been reclassified to conform to current period presentation.

As discussed in the Introduction above, as a result of the Spin-Off, the historical results of operations and the financial position of PHINIA for periods prior to the Spin-Off are presented as discontinued operations in these Consolidated Financial Statements. Refer to Note 26, “Discontinued Operations,” to the Consolidated Financial Statements for more information.

The Company’s Consolidated Financial Statements reflect the results of acquisitions following the date of the respective acquisition. Refer to Note 2, “Acquisitions and Dispositions,” to the Consolidated Financial Statements for more information.

Use of estimates The preparation of financial statements in conformity with accounting principles generally accepted in the United States of America (“GAAP”) requires management to make estimates and assumptions. These estimates and assumptions affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities as of the date of the financial statements and the

accompanying notes, as well as the amounts of revenues and expenses reported during the periods covered by these financial statements and accompanying notes. Actual results could differ from those estimates.

Principles of consolidation The Consolidated Financial Statements include all majority-owned subsidiaries in which the Company has a controlling financial interest. All inter-company balances and transactions have been eliminated in consolidation.

Joint ventures and equity securities The Company has investments in two unconsolidated joint ventures, NSK-Warner K.K. and Turbo Energy Private Limited, of which the Company owns 50% and 32.6%, respectively. These joint ventures are non-controlled affiliates in which the Company exercises significant influence but does not have a controlling financial interest and, therefore, are accounted for under the equity method. Generally, under the equity method, the Company's original investments in these joint ventures are recorded at cost and subsequently adjusted by the Company's share of equity in income or losses. The Company monitors its equity method investments for indicators of other-than-temporary declines in fair value on an ongoing basis. If such a decline has occurred, an impairment charge is recorded, which is measured as the difference between the carrying value and the estimated fair value. The Company's investment in these non-controlled affiliates is included within Investments and long-term receivables in the Consolidated Balance Sheets. The Company's share of equity in income or losses is included in Equity in affiliates' earnings, net of tax in the Consolidated Statements of Operations.

The Company also has certain investments for which it does not have the ability to exercise significant influence (generally when ownership interest is less than 20%). The Company's investment in these equity securities is included within Investments and long-term receivables in the Consolidated Balance Sheet. Refer to Note 10, "Other Current and Non-Current Assets," to the Consolidated Financial Statements for more information.

Interests in privately held companies that do not have readily determinable fair values are accounted for using the measurement alternative under ASC Topic 321, "Investments - Equity Securities," which includes monitoring on an ongoing basis for indicators of impairments or upward adjustments. These equity securities are measured at cost less impairments, adjusted for observable price changes in orderly transactions for the identical or similar investment of the same issuer. If the Company determines that an indicator of impairment or upward adjustment is present, an adjustment is recorded, which is measured as the difference between carrying value and estimated fair value. Estimated fair value is generally determined using an income approach on discounted cash flows or negotiated transaction values.

Equity securities that have readily determinable fair values are measured at fair value. Equity securities that do not have a readily determinable fair value and which provide a net asset value ("NAV") or its equivalent, are valued using NAV as a practical expedient. Changes in fair value and NAV are recorded in Realized and unrealized loss on debt and equity securities in the Consolidated Statements of Operations.

Debt securities During 2022, the Company invested $500 million in convertible debt securities of Wolfspeed, Inc. ("Wolfspeed") as part of a strategic partnership with Wolfspeed. The Company elected to classify the debt security as trading and remeasured the investment quarterly using fair value in accordance with ASC Topic 320, "Investments." In 2023, the investment in Wolfspeed was disposed and recorded a loss in Realized and unrealized loss on debt and equity securities in the Consolidated Statements of Operations. Refer to Note 16, "Fair Value Measurements," to the Consolidated Financial Statements for more information regarding the fair value of the convertible debt securities.

The Company's investment in the debt securities was included within Investments and long-term receivables in the Consolidated Balance Sheets. The changes in fair value are recorded in Realized and unrealized loss on debt and equity securities in the Consolidated Statements of Operations.

Business combinations In accordance with ASC Topic 805, "Business Combinations," acquisitions are recorded using the acquisition method of accounting. The Company includes the operating results of

acquired entities from their respective dates of acquisition. The Company recognizes and measures the identifiable assets acquired, liabilities assumed, and any non-controlling interest at the acquisition date fair value. Various valuation techniques are used to determine the fair value of intangible assets, with the primary techniques being forms of the income approach, specifically the relief-from-royalty and multi-period excess earnings valuation methods. Under these valuation approaches, the Company is required to make estimates and assumptions from a market participant perspective that may include revenue growth rates, estimated earnings, royalty rates, obsolescence factors, contributory asset charges, customer attrition and discount rates. The excess, if any, of total consideration transferred in a business combination over the fair value of identifiable assets acquired, liabilities assumed and any non-controlling interest is recognized as goodwill. Costs incurred as a result of a business combination other than costs related to the issuance of debt or equity securities are recorded in the period the costs are incurred. During the measurement period, which may be up to one year from the acquisition date, the Company may record adjustments to assets acquired and liabilities assumed with the corresponding offset to goodwill.

Revenue recognition Revenue is recognized when performance obligations under the terms of a contract are satisfied, which generally occurs with the transfer of control of the products. For most products, transfer of control occurs upon shipment or delivery; however, a limited number of customer arrangements for highly customized products with no alternative use provide the Company with the right to payment during the production process. As a result, for these limited arrangements, revenue is recognized as goods are produced and control transfers to the customer using the input cost-to-cost method. Revenue is measured at the amount of consideration the Company expects to receive in exchange for transferring the goods. Although the Company may enter into long-term supply arrangements with its major customers, the prices and volumes are not fixed over the life of the arrangements, and a contract does not exist for purposes of applying ASC Topic 606, “Revenue from Contracts with Customers,” until volumes are contractually known.

Sales incentives and allowances (including returns) are recognized as a reduction to revenue at the time of the related sale. The Company estimates the allowances based on an analysis of historical experience. Taxes assessed by a governmental authority collected by the Company concurrent with a specific revenue-producing transaction are excluded from net sales. Shipping and handling fees billed to customers are included in sales, while costs of shipping and handling are included in cost of sales. The Company has elected to apply the accounting policy election available under ASC Topic 606 and accounts for shipping and handling activities as a fulfillment cost.

The Company has a limited number of arrangements with customers where the price paid by the customer is dependent on the volume of product purchased over the term of the arrangement. In other arrangements, the Company will provide a rebate to customers based on the volume of products purchased during the course of the arrangement. The Company estimates the volumes to be sold over the term of the arrangement and recognizes revenue based on the estimated amount of consideration to be received from these arrangements.

The Company continually seeks business development opportunities and at times provides customer incentives for new program awards. The Company evaluates the underlying economics of each amount of consideration payable to a customer to determine the proper accounting by understanding the reasons for the payment, the rights and obligations resulting from the payment, the nature of the promise in the contract, and other relevant facts and circumstances. When the Company determines that the payments are incremental and incurred only if the new business is obtained and expects to recover these amounts from the customer over the term of the new business arrangement, the Company capitalizes these amounts. The Company recognizes a reduction to revenue as products that the upfront payments are related to are transferred to the customer, based on the total amount of products expected to be sold over the term of the arrangement (generally three to seven years). The Company evaluates the amounts capitalized each period end for recoverability and expenses any amounts that are no longer expected to be recovered over the term of the business arrangement.

Refer to Note 3, “Revenue from Contracts with Customers,” to the Consolidated Financial Statements for more information.

Cost of sales The Company includes materials, direct labor and manufacturing overhead within cost of sales. Manufacturing overhead is comprised of indirect materials, indirect labor, factory operating costs, warranty costs and other such costs associated with manufacturing products for sale.

Cash and cash equivalents Cash and cash equivalents are valued at fair market value. It is the Company's policy to classify all highly liquid investments with original maturities of three months or less as cash and cash equivalents. Cash and cash equivalents are maintained with several financial institutions. Deposits held with banks may exceed the amount of insurance provided on such deposits. Generally, these deposits may be redeemed upon demand and are maintained with financial institutions of reputable credit and, therefore, bear minimal risk.

Restricted cash Restricted cash includes amounts designated for uses other than current operations and is related to the Company’s commitment to acquire or invest in certain companies. As of December 31, 2023, the Company had no restricted cash. As of December 31, 2022, the Company had restricted cash of $5 million.

Receivables, net and long-term receivables Accounts receivable and long-term receivables are stated at cost less an allowance for credit losses. An allowance for credit losses is recorded for amounts that may become uncollectible in the future. The allowance for credit losses is an estimate based on expected losses, current economic and market conditions, and a review of the current status of each customer’s accounts receivable.

Inventories, net The majority of inventory is measured using first-in, first-out (“FIFO”) or average-cost methods at the lower of cost or net realizable value, with the exception of certain U.S. inventories that are determined using the last-in, first-out (“LIFO”) method at the lower of cost or market. Inventory held by U.S. operations using the LIFO method was $116 million and $184 million at December 31, 2023 and 2022, respectively. Such inventories, if valued at current cost instead of LIFO, would have been greater by $32 million and $25 million at December 31, 2023 and 2022, respectively. Refer to Note 9, “Inventories, Net,” to the Consolidated Financial Statements for more information.

Pre-production costs related to long-term supply arrangements Engineering, research and development and other design and development costs for products sold on long-term supply arrangements are expensed as incurred unless the Company has a contractual guarantee for reimbursement from the customer. Costs for molds, dies and other tools used to make products sold on long-term supply arrangements for which the Company has title to the assets are capitalized in property, plant and equipment and amortized to cost of sales over the shorter of the term of the arrangement or over the estimated useful lives of the assets, typically three to five years. Costs for molds, dies and other tools used to make products sold on long-term supply arrangements for which the Company has a contractual guarantee for lump sum reimbursement from the customer are capitalized in prepayments and other current assets.

Property, plant and equipment, net Property, plant and equipment is valued at cost less accumulated depreciation. Expenditures for maintenance, repairs and renewals of relatively minor items are generally charged to expense as incurred. Renewals of significant items are capitalized. Depreciation is generally computed on a straight-line basis over the estimated useful lives of the assets. Useful lives for buildings range from 15 to 40 years, and useful lives for machinery and equipment range from three to 12 years. For income tax purposes, accelerated methods of depreciation are generally used. Refer to Note 11, “Property, Plant and Equipment, Net,” to the Consolidated Financial Statements for more information.

Impairment of long-lived assets, including definite-lived intangible assets The Company reviews the carrying value of its long-lived assets, whether held for use or disposal, including other amortizing intangible assets, when events and circumstances warrant such a review under ASC Topic 360, “Property, Plant and Equipment.” In assessing long-lived assets for an impairment loss, assets are grouped with other assets and liabilities at the lowest level for which identifiable cash flows are largely independent of the cash flows of other assets and liabilities. In assessing long-lived assets for impairment, management generally considers individual facilities to be the lowest level for which identifiable cash flows are largely independent. A recoverability review is performed using the undiscounted cash flows if there is a triggering event. If the undiscounted cash flow test for recoverability identifies a possible impairment, management will perform a fair value analysis. Management determines fair value under ASC Topic 820, “Fair Value Measurement,” using the appropriate valuation technique of market, income or cost approach. If the carrying value of a long-lived asset is considered impaired, an impairment charge is recorded for the amount by which the carrying value of the long-lived asset exceeds its fair value.

Management believes that the estimates of future cash flows and fair value assumptions are reasonable; however, changes in assumptions underlying these estimates could affect the valuations. Significant judgments and estimates used by management when evaluating long-lived assets for impairment include (i) an assessment as to whether an adverse event or circumstance has triggered the need for an impairment review; (ii) undiscounted future cash flows generated by the asset; and (iii) fair valuation of the asset.

Goodwill and other intangible assets During the fourth quarter of each year, the Company tests goodwill for impairment by either performing a qualitative assessment or a quantitative analysis. The qualitative assessment evaluates various events and circumstances, such as macroeconomic conditions, industry and market conditions, cost factors, relevant events and financial trends, that may impact a reporting unit's fair value. Using this qualitative assessment, the Company determines whether it is more-likely-than-not the reporting unit's fair value exceeds its carrying value. If it is determined that it is not more-likely-than-not the reporting unit's fair value exceeds the carrying value, or upon consideration of other factors, including recent acquisition, restructuring or disposal activity or to refresh the fair values, the Company performs a quantitative goodwill impairment analysis. In addition, the Company may test goodwill in between annual test dates if an event occurs or circumstances change that could more-likely-than-not reduce the fair value of a reporting unit below its carrying value.

The Company has definite-lived intangible assets related to patents and developed technology, customer relationships and trade names. The Company amortizes definite-lived intangible assets over their estimated useful lives. The Company also has intangible assets related to acquired trade names that are classified as indefinite-lived when there are no foreseeable limits on the periods of time over which they are expected to contribute cash flows. Costs to renew or extend the term of acquired intangible assets are recognized as expense as incurred.

Similar to goodwill, the Company can elect to perform the impairment test for indefinite-lived intangibles other than goodwill (primarily trade names) using a qualitative analysis, considering similar factors as outlined in the goodwill discussion, in order to determine if it is more-likely-than-not that the fair value of the trade names is less than the respective carrying values. If the Company elects to perform or is required to perform a quantitative analysis, the test consists of a comparison of the fair value of the indefinite-lived intangible asset to the carrying value of the asset as of the impairment testing date. The Company estimates the fair value of indefinite-lived intangibles using the relief-from-royalty method, which it believes is an appropriate and widely used valuation technique for such assets. The fair value derived from the relief-from-royalty method is measured as the discounted cash flow savings realized from owning such trade names and not being required to pay a royalty for their use.

Refer to Note 12, “Goodwill and Other Intangibles,” to the Consolidated Financial Statements for more information.

Assets and liabilities held for sale The Company classifies assets and liabilities (disposal groups) to be sold as held for sale in the period in which all of the following criteria are met: management, having the authority to approve the action, commits to a plan to sell the disposal group; the disposal group is available for immediate sale in its present condition subject only to terms that are usual and customary for sales of such disposal groups; an active program to locate a buyer and other actions required to complete the plan to sell the disposal group have been initiated; the sale of the disposal group is probable, and transfer of the disposal group is expected to qualify for recognition as a completed sale within one year, except if events or circumstances beyond the Company's control extend the period of time required to sell the disposal group beyond one year; the disposal group is being actively marketed for sale at a price that is reasonable in relation to its current fair value; and actions required to complete the plan indicate that it is unlikely that significant changes to the plan will be made or that the plan will be withdrawn.

The Company initially measures a disposal group that is classified as held for sale at the lower of its carrying value or fair value less any costs to sell. Any loss resulting from this measurement is recognized in the period in which the held for sale criteria are met. Conversely, gains are not recognized on the sale of a disposal group until the date of sale. The Company assesses the fair value of a disposal group, less any costs to sell, each reporting period it remains classified as held for sale and reports any subsequent changes as an adjustment to the carrying value of the disposal group, as long as the new carrying value does not exceed the carrying value of the disposal group at the time it was initially classified as held for sale. Additionally, depreciation is not recorded during the period in which the long-lived assets, included in the disposal group, are classified as held for sale.

Upon determining that a disposal group meets the criteria to be classified as held for sale, the Company reports the assets and liabilities of the disposal group, if material, in the line items assets held for sale and liabilities held for sale in the Consolidated Balance Sheets.

Product warranties The Company provides warranties on some, but not all, of its products. The warranty terms are typically from one to three years. Provisions for estimated expenses related to product warranty are made at the time products are sold. These estimates are established using historical information about the nature, frequency and average cost of warranty claim settlements as well as product manufacturing and industry developments and recoveries from third parties. Management actively studies trends of warranty claims and takes action to improve product quality and minimize warranty claims. Costs of product recalls, which may include the cost of the product being replaced as well as the customer's cost of the recall, including labor to remove and replace the recalled part, are accrued as part of the Company's warranty accrual at the time an obligation becomes probable and can be reasonably estimated. Management believes that the warranty accrual is appropriate; however, actual claims incurred could differ from the original estimates, requiring adjustments to the accrual. The product warranty accrual is allocated to current and non-current liabilities in the Consolidated Balance Sheets.

Refer to Note 13, "Product Warranty," to the Consolidated Financial Statements for more information.

Other loss accruals and valuation allowances The Company has numerous other loss exposures, such as customer claims, workers' compensation claims, litigation and recoverability of certain assets. Establishing loss accruals or valuation allowances for these matters requires the use of estimates and judgment in regard to the risk exposure and ultimate realization. The Company estimates losses using consistent and appropriate methods; however, changes to its assumptions could materially affect the recorded accrued liabilities for loss or asset valuation allowances.

Environmental contingencies The Company accounts for environmental costs in accordance with ASC Topic 450, "Contingencies." Costs related to environmental assessments and remediation efforts at operating facilities are accrued when it is probable that a liability has been incurred and the amount of

that liability can be reasonably estimated. Estimated costs are recorded at undiscounted amounts, based on experience and assessments and are regularly evaluated. The liabilities are recorded in Other current and Other non-current liabilities in the Company's Consolidated Balance Sheets.

Refer to Note 21, "Contingencies," to the Consolidated Financial Statements for more information.

Government grants The Company periodically receives government grants representing assistance provided by a government. These government grants are generally received in cash and typically provide reimbursement related to acquisition of property and equipment, product development or local governmental economic relief. The government grants are generally amortized using a systematic and rational method over the life of the grant. As of December 31, 2023, the Company recorded government grant related liabilities of $3 million in Other current liabilities and $50 million in Other non-current liabilities in the Company's Consolidated Balance Sheet. During the year ended December 31, 2023, the Company recorded $37 million and $12 million of government grant-related credits in Selling, general and administrative expenses and Cost of sales, respectively, in the Company's Consolidated Statement of Operations.

Derivative financial instruments The Company recognizes that certain normal business transactions and foreign currency operations generate risk. Examples of risks include exposure to exchange rate risk related to transactions denominated in currencies other than the functional currency, changes in commodity costs and interest rates. It is the objective of the Company to assess the impact of these transaction risks and offer protection from selected risks through various methods, including financial derivatives. Virtually all derivative instruments held by the Company are designated as hedges, have high correlation with the underlying exposure and are highly effective in offsetting underlying price movements. Accordingly, gains and losses from changes in qualifying hedge fair values are matched with the underlying transactions. Hedge instruments are generally reported gross, with no right to offset, on the Consolidated Balance Sheets at their fair value based on quoted market prices for contracts with similar maturities. The Company does not engage in any derivative transactions for purposes other than hedging specific operational risks.

Refer to Note 17, "Financial Instruments," to the Consolidated Financial Statements for more information.

Foreign currency The financial statements of foreign subsidiaries are translated to U.S. Dollars using the period-end exchange rate for assets and liabilities and an average exchange rate for each period for revenues, expenses and capital expenditures. The local currency is the functional currency for substantially all of the Company's foreign subsidiaries. Translation adjustments for foreign subsidiaries are recorded as a component of accumulated other comprehensive income (loss) in equity. The Company recognizes transaction gains and losses arising from fluctuations in currency exchange rates on transactions denominated in currencies other than the functional currency in earnings as incurred.

Refer to Note 20, "Accumulated Other Comprehensive Loss," to the Consolidated Financial Statements for more information.

Postretirement defined benefits The Company's defined benefit pension and other postemployment benefit plans are accounted for in accordance with ASC Topic 715, "Compensation - Retirement Benefits." Disability, early retirement and other postemployment benefits are accounted for in accordance with ASC Topic 712, "Compensation - Nonretirement Postemployment Benefits."

Pensions and other postemployment benefit costs and related liabilities and assets are dependent upon assumptions used in calculating such amounts. These assumptions include discount rates, expected returns on plan assets, health care cost trends, compensation and other factors. In accordance with GAAP, actual results that differ from the assumptions used are accumulated and amortized over future periods, and accordingly, generally affect recognized expense in future periods.

Refer to Note 18, “Retirement Benefit Plans,” to the Consolidated Financial Statements for more information.

Restructuring Restructuring costs may occur when the Company takes action to exit or significantly curtail a part of its operations or implements a reorganization that affects the nature and focus of operations. A restructuring charge can consist of severance costs associated with reductions to the workforce, costs to terminate an operating lease or contract, professional fees and other costs incurred related to the implementation of restructuring activities.

The Company generally records costs associated with voluntary separations at the time of employee acceptance. Costs for involuntary separation programs are recorded when management has approved the plan for separation, the employees are identified and aware of the benefits they are entitled to and it is unlikely that the plan will change significantly. When a plan of separation requires approval by or consultation with the relevant labor organization or government, the costs are recorded upon agreement. Costs associated with benefits that are contingent on the employee continuing to provide service are expensed over the required service period.

Refer to Note 4, “Restructuring,” to the Consolidated Financial Statements for more information.

Income taxes In accordance with ASC Topic 740, “Income Taxes,” the Company’s income tax expense is calculated based on expected income and statutory tax rates in the various jurisdictions in which the Company operates and requires the use of management’s estimates and judgments. Accounting for income taxes is complex, in part because the Company conducts business globally and, therefore, files income tax returns in numerous tax jurisdictions. Management judgment is required in determining the Company’s worldwide provision for income taxes and recording the related assets and liabilities, including accruals for unrecognized tax benefits and assessing the need for valuation allowances.

The determination of accruals for unrecognized tax benefits includes the application of complex tax laws in a multitude of jurisdictions across the Company’s global operations. Management judgment is required in determining the gross unrecognized tax benefits’ related liabilities. In the ordinary course of the Company’s business, there are many transactions and calculations where the ultimate tax determination is less than certain. Accruals for unrecognized tax benefits are established when, despite the belief that tax positions are supportable, there remain certain positions that do not meet the minimum probability threshold, which is a tax position that is more-likely-than-not to be sustained upon examination by the applicable taxing authority.

The Company records valuation allowances to reduce the carrying value of deferred tax assets to amounts that it expects are more-likely-than-not to be realized. The Company assesses existing deferred tax assets, net operating losses and tax credit carryforwards by jurisdiction and expectations of its ability to utilize these tax attributes through a review of past, current and estimated future taxable income and tax planning strategies.

Refer to Note 7, “Income Taxes,” to the Consolidated Financial Statements for more information.

New Accounting Pronouncements

Recently Adopted Accounting Standards

In October 2021, the FASB issued ASU No. 2021-08, “Business Combinations (Topic 805): Accounting for Contract Assets and Contract Liabilities from Contracts with Customers.” It requires entities to apply Topic 606 to recognize and measure contract assets and contract liabilities in a business combination. The amendments improve comparability after the business combination by providing consistent recognition and measurement guidance for revenue contracts with customers acquired in a business combination and revenue contracts with customers not acquired in a business combination. This guidance is effective for interim and annual reporting periods beginning after December 15, 2022. The Company adopted this guidance prospectively as of January 1, 2023, and there was no impact related to the 2023 acquisitions in the Consolidated Financial Statements.

Accounting Standards Not Yet Adopted

In November 2023, the FASB issued ASU No. 2023-07, “Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures.” It requires incremental disclosures related to an entity’s reportable segments, including (i) significant segment expense categories and amounts for each reportable segment that are provided to the chief operating decision maker (“CODM”), (ii) an aggregate amount and description of other segment items included in each reported measure, (iii) all annual disclosures about a reportable segment’s profit or loss and assets required by Topic 280, to be disclosed in interim periods, (iv) title and position of the individual or the name of the group identified as the CODM and (v) explanation of how the CODM uses the reported measures of segment profit or loss to assess performance and allocate resources to the segment. The standard improves transparency by providing disaggregated expense information about an entity’s reportable segments. The standard does not change the definition of a segment, the method for determining segments or the criteria for aggregating operating segments into reportable segments. This guidance is effective for annual reporting periods beginning after December 15, 2023, and interim reporting periods beginning after December 15, 2024. The Company does not expect this guidance to have a material impact on its Consolidated Financial Statements.

In December 2023, the FASB issued ASU No. 2023-09, “Income Taxes (Topic 740): Improvements to Income Tax Disclosures.” It requires entities to disaggregate information related to the effective tax rate reconciliation and income taxes paid. The standard improve transparency by providing more detailed income tax disclosures that would be useful in making capital allocation decisions. This guidance is effective for annual reporting periods beginning after December 15, 2024. The Company does not expect this guidance to have a material impact on its Consolidated Financial Statements.

NOTE 2 ACQUISITIONS AND DISPOSITIONS

Acquisitions

In accordance with ASC Topic 805, “Business Combinations,” acquisitions are recorded using the acquisition method of accounting. The Company recognizes and measures the acquisition date fair value of the identifiable assets acquired, liabilities assumed, and any non-controlling interest using a range of methodologies as indicated by generally accepted valuation practices. Various valuation techniques are used to determine the fair value of intangible assets, with the primary techniques being forms of the income approach, specifically the relief-from-royalty and multi-period excess earnings valuation methods. Under these valuation approaches, the Company is required to make estimates and assumptions from a market participant perspective and may include revenue growth rates, estimated earnings, royalty rates, obsolescence factors, contributory asset charges, customer attrition and discount rates. For each acquisition disclosed below, management used a third-party valuation firm to assist in the determination of the provisional purchase accounting fair values; however, management ultimately oversees the third-party valuation firm to ensure that the transaction-specific assumptions are appropriate for the Company.

Due to the insignificant size of the 2023, 2022 and 2021 acquisitions, both individually and in the aggregate, relative to the Company, supplemental pro forma financial information for the current and prior reporting periods is not provided.

Eldor Corporation’s Electric Hybrid Systems Business

On December 1, 2023, the Company completed its acquisition of the electric hybrid systems business segment of Eldor Corporation (“Eldor”), which is headquartered in Italy. The Company expects the acquisition to complement its existing ePropulsion product portfolio by enhancing the Company’s engineering capabilities in power electronics. The Company paid €72 million ($78 million) at closing. Pursuant to the Agreement, the Company is obligated to remit up to €175 million ($191 million) of earn-out payments contingent upon booked business for future periods from new customer awards. The Company’s current estimates indicate that the minimum threshold for the earn-out target will not be achieved, thus no amount of the earn-out payment has been included in the purchase consideration or in the Company’s Consolidated Balance Sheet.

The purchase price was allocated on a provisional basis as of December 1, 2023. Assets acquired and liabilities assumed were recorded at estimated fair values based on management’s estimates, available information, and supportable assumptions that management considered reasonable. Certain estimated values for the acquisition, including goodwill, tangible, and intangible assets and deferred taxes, are not yet finalized, and the provisional purchase price allocations are subject to change as the Company completes its analysis of the fair value at the date of acquisition. The final valuation of assets acquired and liabilities assumed may be materially different than the estimated values shown below.

The estimated fair values of assets acquired and liabilities assumed as of December 1, 2023 were assets of $36 million, including goodwill and intangibles of $25 million, and liabilities of $8 million.

Any excess of the purchase price over the estimated fair value of net assets was recognized as goodwill. Goodwill of $14 million was recorded within the Company’s ePropulsion segment. Goodwill consists of the Company’s expected future economic benefits that will be realized from expanding the Company’s electric vehicle portfolio as electric vehicle production continues to increase. Goodwill is not expected to be deductible for tax purposes in Italy.

In connection with the acquisition, the Company preliminarily recorded $11 million for intangible assets, primarily for customer relationships and developed technology. The provisional fair values of goodwill and identifiable intangible assets were valued using the market approach.

The impact of the Eldor acquisition on net sales and net earnings was immaterial for the year ended December 31, 2023.

Hubei Surpass Sun Electric Charging Business

On March 1, 2023, the Company completed its acquisition of 100% of the electric vehicle solution, smart grid and smart energy businesses ("SSE") of Hubei Surpass Sun Electric, pursuant to an Equity Transfer Agreement. The acquisition is expected to complement the Company's existing European and North American charging footprint by adding a presence in China. The total consideration was ¥288 million ($42 million), including ¥268 million ($39 million) of base purchase price and ¥20 million ($3 million) of estimated earn-out payments. The Company paid ¥217 million ($31 million) of base purchase price in the year ended December 31, 2023. Of the remaining ¥51 million ($8 million) of base purchase price, ¥31 million ($5 million) is payable by April 30, 2024 and is recorded in Other current liabilities in the Company's Consolidated Balance Sheet as of December 31, 2023. The remaining ¥20 million ($3 million) of base purchase price is payable before April 30, 2025 and is recorded in Other non-current liabilities in the Company's Consolidated Balance Sheet as of December 31, 2023. Pursuant to the agreement, the Company's obligation to remit up to ¥103 million ($15 million) of earn-out payments is contingent upon the achievement of certain revenue and pre-tax profit margin targets in 2023 and 2024 as well as the retention of key employees during the same time period. As of December 31, 2023, the Company's estimate of the earn-out payments was approximately ¥20 million ($3 million), of which half is recorded in Other current liabilities and half is recorded in Other non-current liabilities in the Company's Consolidated Balance Sheet.

The purchase price was allocated on a provisional basis as of March 1, 2023, and all assets acquired and liabilities assumed were recorded at estimated fair values based on management's estimates, available information, and supportable assumptions that management considered reasonable. The Company subsequently finalized its valuation of the assets and liabilities of the SSE acquisition during the third quarter of 2023, and the estimated fair values of assets acquired and liabilities assumed amounted to total assets of $50 million, including goodwill and intangibles of $5 million, and total liabilities of $8 million. These final amounts were not materially different than the estimated values recorded on March 1, 2023.

Any excess of the purchase price over the estimated fair value of net assets was recognized as goodwill. Goodwill of $2 million was recorded within the Company's Air Management segment. Goodwill consists of the Company's expected future economic benefits that will be realized from expanding the Company's electric vehicle portfolio as electric vehicle production continues to increase. Goodwill is not deductible for tax purposes in China.

The following table summarizes the other intangible assets acquired:

(in millions)	Estimated Life	Estimated Fair Value
Developed technology	5 years	$ 2
Customer relationships	6 years	1
Total other intangible assets		$ 3

Identifiable intangible assets were valued using the income approach.

The impact of the SSE acquisition on net sales and net earnings was immaterial for the year ended December 31, 2023.

Drivetek AG

On December 1, 2022, the Company completed its acquisition of 100% of Drivetek AG ("Drivetek"), an engineering and product development company located in Switzerland. This acquisition is expected to strengthen the Company's power electronics capabilities in auxiliary inverters to accelerate the growth of the High Voltage eFan business. The Company paid ₣27 million ($29 million) at closing, and up to ₣10 million ($10 million) could be paid in the form of contingent earn-out payments over the three years following closing. The earn-out payments are contingent upon achievement of estimated future sales targets associated with newly awarded business and future turnover rate targets. As of December 31, 2023, the Company's estimate of the earn-out payments was approximately ₣10 million ($12 million), which is recorded in Other non-current liabilities in the Company's Consolidated Balance Sheet.

The purchase price was allocated on a preliminary basis as of December 1, 2022. Assets acquired and liabilities assumed were recorded at estimated fair values based on management's estimates, available information, and supportable assumptions that management considered reasonable. The Company subsequently finalized its valuation of the assets and liabilities of the Drivetek acquisition during the third quarter of 2023, and the estimated fair values of assets acquired and liabilities assumed amounted to total assets of $49 million, including goodwill and intangibles of $40 million, and liabilities of $10 million. These final amounts were not materially different than the estimated values recorded on December 1, 2022.

Any excess of the purchase price over the estimated fair value of net assets was recognized as goodwill. Goodwill of $22 million was recorded within the Company's Air Management segment. Goodwill consists of the Company's expected future economic benefits that will be realized from expanding the Company's electric vehicle portfolio as electric vehicle production continues to increase. Goodwill is not expected to be deductible for tax purposes.

The following table summarizes the other intangible assets acquired:

(in millions)	Estimated Life	Estimated Fair Value
Developed technology	8 years	$ 11
Customer relationships	12 years	7
Total other intangible assets		$ 18

Identifiable intangible assets were valued using the market approach.

The impact of the Drivetek acquisition on net sales and net earnings was immaterial for the year ended December 31, 2022.

Rhombus Energy Solutions

On July 29, 2022, the Company completed its acquisition of 100% of Rhombus Energy Solutions ("Rhombus"), a provider of charging solutions in the North American market, pursuant to the terms of an Agreement and Plan of Merger (the "Agreement"). The acquisition is expected to complement the Company's existing European charging footprint to accelerate organic growth and adds North American regional presence to its charging business.

The Company paid $131 million at closing. Pursuant to the Agreement, the Company is obligated to remit up to $30 million of earn-out payments, payable in 2025, contingent upon achievement of certain sales dollars, sales volume, and gross margin targets. The Company's current estimates indicate that the minimum thresholds for these earn-out targets will not be achieved, thus no amount for the earn-out payments has been included in the purchase consideration or in the Company's Consolidated Balance Sheet. Additionally, pursuant to the Agreement, the Company is obligated to remit up to $25 million over

the three years following closing in key employee retention related payments, which include certain performance targets. The amounts will be accounted for as post-combination expense.

The Company finalized its valuation of the assets and liabilities for the Rhombus acquisition during the second quarter of 2023. Any excess of the purchase price over the estimated fair value of net assets was recognized as goodwill. Goodwill of $104 million was recorded within the Company's Air Management segment. Goodwill consists of the Company's expected future economic benefits that will be realized from expanding the Company's electric vehicle portfolio as electric vehicle production continues to increase. Goodwill is not expected to be deductible for tax purposes.

The following table summarizes the other intangible assets acquired:

(in millions)	Estimated Life	Estimated Fair Value
Developed technology	13 years	$ 22
Customer relationships	8 years	5
Total other intangible assets		$ 27

Identifiable intangible assets were valued using the income approach.

The impact of the Rhombus acquisition on net sales and net earnings was immaterial for the year ended December 31, 2022.

Santroll Automotive Components

On March 31, 2022, the Company completed its acquisition of 100% of Santroll Automotive Components ("Santroll"), a carve-out of Santroll Electric Auto's eMotor business, pursuant to the terms of an Equity Transfer Agreement ("ETA"). The acquisition is expected to strengthen the Company's vertical integration, scale and portfolio breadth in light vehicle eMotors while allowing for increased speed to market.

The total final consideration was $192 million, including approximately ¥1.0 billion ($152 million) of base purchase price and ¥0.25 billion ($40 million) of originally estimated earn-out payments. The Company paid approximately ¥1.0 billion ($157 million) of base purchase price in the year ended December 31, 2022 and no longer expects to recapture a previously anticipated $5 million of post-closing adjustments, which has been recorded in Other operating expense, net. Pursuant to the ETA, the obligation of the Company to remit up to ¥0.3 billion (approximately $47 million) of earn-out payments was contingent upon achievement of certain sales volume targets and certain estimated future volume targets associated with newly awarded business. As of December 31, 2023, the Company paid ¥0.2 billion ($24 million) to settle the remaining earn-out liability and related adjustments.

The Company finalized its valuation of the assets and liabilities of the Santroll acquisition during the first quarter of 2023. Any excess of the purchase price over the estimated fair value of net assets was recognized as goodwill. Goodwill of $112 million was recorded within the Company's ePropulsion segment. Goodwill consists of the Company's expected future economic benefits that will arise from future product sales and the added capabilities from vertical integration of eMotors. Goodwill is not expected to be deductible for tax purposes in China.

The following table summarizes the other intangible assets acquired:

(in millions)	Estimated Life	Estimated Fair Value
Customer relationships	12 years	$ 62
Manufacturing processes (know-how)	10 years	25
Total other intangible assets		$ 87

Identifiable intangible assets were valued using the income approach.

The impact of the Santroll acquisition on net sales and net earnings was immaterial for the year ended December 31, 2022.

AKASOL AG

On June 4, 2021, the Company completed its voluntary public takeover offer for shares of AKASOL AG ("AKASOL"), resulting in ownership of 89% of AKASOL's outstanding shares. The Company paid approximately €648 million ($788 million) to settle the offer. During 2021, the Company purchased additional shares of AKASOL for €28 million ($33 million) increasing its ownership to 93% as of December 31, 2021.

At December 31, 2021, the noncontrolling interest in AKASOL of approximately €51 million ($58 million), which was to be acquired through a merger squeeze-out process (the "Squeeze Out"). On February 10, 2022, the Company completed the registration of the Squeeze Out resulting in 100% ownership. The Company settled the Squeeze Out with AKASOL minority shareholders in the first quarter of 2022.

The acquisition is expected to further strengthen BorgWarner's commercial vehicle and industrial electrification capabilities, which positions the Company to capitalize on what it believes to be a fast-growing battery module and pack market. Following the June 4, 2021 acquisition date, AKASOL's operations had net sales of $67 million for the year ended December 31, 2021. The impact on net earnings was immaterial for the year ended December 31, 2021.

Romeo Power, Inc.

In May 2019, the Company invested $50 million in exchange for a 20% equity interest in Romeo Systems, Inc. ("Romeo"), a technology-leading battery module and pack supplier that was then privately held. On December 29, 2020, through the business combination of Romeo Systems, Inc. and special purpose acquisition company RMG Acquisition Corporation, a new entity, Romeo Power, Inc., became a publicly listed company. The Company's ownership in Romeo was reduced to 14%, and the investment was recorded at fair value on an ongoing basis with changes in fair value being recognized in Realized and unrealized loss on debt and equity securities in the Consolidated Statements of Operations. During the years ended December 31, 2021 and 2020, the Company recorded a loss of $362 million and a gain of $382 million, respectively, to adjust the carrying value of the Company's investment to fair value. During the year ended December 31, 2022, the Company recorded a loss of $39 million and liquidated its investment in Romeo shares at a fair value of $31 million. As of March 17, 2022, the Company no longer held any investment in Romeo.

In September 2019, the Company and Romeo contributed total equity of $10 million and formed a new joint venture, BorgWarner Romeo Power LLC ("Romeo JV"), in which the Company owned a 60% interest. Romeo JV was a variable interest entity focusing on producing battery module and pack technology. The Company was the primary beneficiary of Romeo JV and had consolidated Romeo JV in its consolidated financial statements. On October 25, 2021, the Company delivered written notice to Romeo that the Company was electing to exercise its right to put its ownership stake in Romeo JV to Romeo. Based on an independent appraisal, the Company's interest in Romeo JV was valued at $30 million. In February 2022, the Company completed the sale of its 60% interest in the Romeo JV for $29 million, the fair value of $30 million reduced by a 5% discount pursuant to the joint venture agreement. During the year ended December 31, 2022, the Company recorded a gain of $22 million in Other operating expense, net, which represented the difference between the Company's book value of its interest in Romeo JV compared to the fair value of consideration received. As a result of the sale, the Company has no further rights in or involvement with Romeo JV.

Dispositions

PHINIA Inc.

On July 3, 2023, BorgWarner completed the previously announced Spin-Off of its Fuel Systems and Aftermarket segments by the distribution of 100% of the outstanding common stock of PHINIA to holders of record of common stock of the Company on a pro-rata basis. Each holder of record of common stock of the Company received one share of PHINIA common stock for every five shares of common stock of the Company held on June 23, 2023, the record date for the distribution ("Distribution Date"). In lieu of fractional shares of PHINIA, shareholders of the Company received cash. PHINIA is an independent public company trading under the symbol "PHIN" on the New York Stock Exchange.

In connection with the Spin-Off, the Company entered into several agreements with PHINIA on or prior to the Distribution Date that, among other things, provide a framework for the Company's relationship with PHINIA after the Spin-Off, including a separation and distribution agreement, an employee matters agreement, a tax matters agreement, an intellectual property cross-license agreement and a transition services agreement through which the Company and PHINIA will continue to provide certain services to each other following the Spin-Off.

Refer to Note 26, "Discontinued Operations" to the Consolidated Financial Statements for more information.

Water Valley

In 2021, the Company announced its strategy to aggressively grow its electrification product portfolio over time through organic investments and technology-focused acquisitions. Additionally, the Company announced a plan to dispose of certain internal combustion assets in support of that strategy. In December 2021, the Company entered into a definitive agreement to sell its Water Valley, Mississippi manufacturing facility ("Water Valley") and the associated solenoid, transmission control module and stop/start accumulator system business for an estimated $57 million. The consideration consisted of $39 million in cash and promissory notes and up to $30 million in potential earn-out payments. The Company included $18 million as contingent consideration in the proceeds, which reflected its original estimate of the payout pursuant to the earn-out. During the year ended December 31, 2022, the Company changed its estimate of the expected earn-out and recorded a pre-tax loss of $9 million in Other operating expense, net. During the year ended December 31, 2023, the Company and the buyer agreed on a final settlement of $14 million for the earn-out and recorded a pre-tax gain of $5 million in Other operating expense, net. As of December 31, 2023, the unpaid portion of $5 million of the contingent consideration and the $10 million remaining on the promissory notes were included in Receivables, net and Investments and long-term receivables, respectively, on the Consolidated Balance Sheet.

Water Valley had net sales of $177 million during the year ended December 31, 2021 and was included in the Company's Drivetrain & Battery Systems segment. On December 31, 2021, upon the closing of the transaction, based upon the final transaction price agreed to in the fourth quarter of 2021, the Company recorded a loss on divestiture of $22 million.

NOTE 3 REVENUE FROM CONTRACTS WITH CUSTOMERS

The Company manufactures and sells products, primarily to OEMs of light vehicles and, to a lesser extent, to other OEMs of commercial vehicles and off-highway vehicles, to certain tier one vehicle systems suppliers and into the aftermarket. The Company's payment terms are based on customary business practices and vary by customer type and products offered. The Company has evaluated the terms of its arrangements and determined that they do not contain significant financing components.

Generally, revenue is recognized upon shipment or delivery; however, a limited number of the Company's customer arrangements for its highly customized products with no alternative use provide the Company with the right to payment during the production process. As a result, for these limited arrangements, revenue is recognized as goods are produced and control transfers to the customer using the input cost-to-cost method. The Company recorded a contract asset of $18 million and $14 million at December 31, 2023 and 2022, respectively, for these arrangements. These amounts are reflected in Prepayments and other current assets in the Company's Consolidated Balance Sheets.

In limited instances, certain customers have provided payments in advance of receiving related products, typically at the onset of an arrangement prior to the beginning of production. These contract liabilities are reflected as Other current liabilities in the Consolidated Balance Sheets and were $18 million at December 31, 2023 and $14 million at December 31, 2022, respectively. These amounts are reflected as revenue over the term of the arrangement (typically three to seven years) as the underlying products are shipped and represent the Company's remaining performance obligations as of the end of the period.

The Company continually seeks business development opportunities and at times provides customer incentives for new program awards. When the Company determines that the payments are incremental and incurred only if the new business is obtained and expects to recover these amounts from the customer over the term of the new business arrangement, the Company capitalizes these amounts. As of December 31, 2023 and 2022, the Company recorded customer incentive payments of $27 million and $34 million, respectively, in Prepayments and other current assets, and $58 million and $99 million, respectively, in Other non-current assets in the Consolidated Balance Sheets.

The Company's products can be disaggregated by two types: eProducts and Foundational products. eProducts include all products utilized on or for electric vehicles ("EVs") plus those same products and components that are included in hybrid powertrains whose underlying technologies are adaptable or applicable to those used in or for EVs. Foundational products include all products utilized on internal combustion engines plus those same products and components that are also included in hybrid powertrains. The following table represents a disaggregation of revenue from contracts with customers by Foundational products and eProducts for the years ended December 31, 2023, 2022, and 2021.

	Year Ended December 31,		
(in millions)	2023	2022	2021
Foundational products	$ 12,161	$ 11,163	$ 11,061
eProducts	2,037	1,472	742
Total	$ 14,198	$ 12,635	$ 11,803

The following table represents a disaggregation of revenue from contracts with customers by reportable segment and region for the years ended December 31, 2023, 2022, and 2021. Refer to Note 24, "Reportable Segments and Related Information" to the Consolidated Financial Statements for more information.

	Year ended December 31, 2023			
(in millions)	Air Management	Drivetrain & Battery Systems	ePropulsion	Total
North America	$ 2,103	$ 1,532	$ 458	$ 4,093
Europe	3,377	1,332	372	5,081
Asia	2,068	1,474	1,268	4,810
Other	202	8	4	214
Total	$ 7,750	$ 4,346	$ 2,102	$ 14,198

	Year ended December 31, 2022			
(in millions)	Air Management	Drivetrain & Battery Systems	ePropulsion	Total
North America	$ 1,998	$ 1,371	$ 500	$ 3,869
Europe	2,872	1,003	198	4,073
Asia	2,024	1,379	1,099	4,502
Other	182	—	9	191
Total	$ 7,076	$ 3,753	$ 1,806	$ 12,635

	Year ended December 31, 2021			
(in millions)	Air Management	Drivetrain & Battery Systems	ePropulsion	Total
North America	$ 1,700	$ 1,309	$ 324	$ 3,333
Europe	2,808	773	210	3,791
Asia	2,105	1,577	849	4,531
Other	148	—	—	148
Total	$ 6,761	$ 3,659	$ 1,383	$ 11,803

NOTE 4 RESTRUCTURING

The Company's restructuring activities are undertaken as necessary to execute management's strategy and streamline operations, consolidate and take advantage of available capacity and resources, and ultimately achieve net cost reductions. Restructuring activities include efforts to integrate and rationalize the Company's business and to relocate operations to best-cost locations.

The Company's restructuring expenses consist primarily of employee termination benefits (principally severance and/or other termination benefits) and other costs, which are primarily professional fees and costs related to facility closures and exits.

(in millions)	Air Management	Drivetrain & Battery Systems	ePropulsion	Corporate	Total
Year ended December 31, 2023					
Employee termination benefits	$ 67	$ 2	$ 1	$ —	$ 70
Other	7	2	—	—	9
Total restructuring expense	$ 74	$ 4	$ 1	$ —	$ 79
Year ended December 31, 2022					
Employee termination benefits	$ 24	$ 14	$ —	$ —	$ 38
Other	—	9	1	—	10
Total restructuring expense	$ 24	$ 23	$ 1	$ —	$ 48
Year ended December 31, 2021					
Employee termination benefits	$ 34	$ 11	$ —	$ —	$ 45
Other	18	36	7	2	63
Total restructuring expense	$ 52	$ 47	$ 7	$ 2	$ 108

The following table displays a roll forward of the restructuring liability recorded within the Company's Consolidated Balance Sheets and the related cash flow activity:

(in millions)	Employee termination benefits	Other	Total
Balance at January 1, 2022	$ 66	$ 13	$ 79
Restructuring expense, net	38	10	48
Cash payments	(58)	(19)	(77)
Foreign currency translation adjustment and other	(7)	5	(2)
Balance at December 31, 2022	39	9	48
Restructuring expense, net	70	9	79
Cash payments	(43)	(13)	(56)
Foreign currency translation adjustment and other	2	2	4
Balance at December 31, 2023	$ 68	$ 7	$ 75
Less: Non-current restructuring liability	7	—	7
Current restructuring liability at December 31, 2023	$ 61	$ 7	$ 68

2023 Structural Costs Plan In 2023, the Company announced a $130 million to $150 million restructuring plan to address structural costs in its Foundational products businesses. During the year ended December 31, 2023, the Company recorded $79 million of restructuring costs related to this plan.

2020 Structural Costs Plan In February 2020, the Company announced a $300 million restructuring plan to address existing structural costs. The actions under this plan are complete.

2019 Legacy Delphi Technologies Plan In 2019, legacy Delphi Technologies announced a restructuring plan to reshape and realign its global technical center footprint and reduce salaried and contract staff. The actions under this plan are complete.

The following provides details of restructuring expense incurred by the Company's reportable segments during the years ended December 31, 2023, 2022 and 2021, related to the plans discussed above:

Air Management

2023 Structural Costs Plan

- During the year ended December 31, 2023, the segment recorded $74 million of restructuring costs under this plan, primarily related to employee termination benefits associated with the announced closure of a facility in Europe affecting approximately 200 employees.

2020 Structural Costs Plan

- During the year ended December 31, 2022, the segment recorded $24 million of restructuring costs under this plan. This primarily related to $18 million for two voluntary termination programs pursuant to which approximately 74 employees accepted termination packages in 2022.
- During the year ended December 31, 2021, the segment recorded $48 million of restructuring costs, of which $23 million related to a voluntary termination program where approximately 140 employees accepted termination packages in 2021, and $25 million related to severance costs and professional fees for specific actions to reduce structural costs.

2019 Legacy Delphi Technologies Plan

- During the year ended December 31, 2021, the segment recorded $4 million of restructuring costs, primarily related to severance costs.

Drivetrain & Battery Systems

2023 Structural Costs Plan

- During the year ended December 31, 2023, the segment recorded $4 million of restructuring costs under this plan, primarily related to employee termination benefits and equipment moves.

2020 Structural Costs Plan

- During the year ended December 31, 2022, the segment recorded $9 million of restructuring costs primarily related to contractual settlements and professional fees.
- During the year ended December 31, 2021, the segment recorded $47 million of restructuring costs, of which $36 million primarily related to severance costs, equipment relocation and professional fees to reduce existing structural costs, and $11 million related

to contractual settlements, professional fees and other costs associated with the announced closure of a facility in Europe.

- During the year ended December 31, 2022, the segment recorded $14 million of restructuring costs, primarily related to severance costs associated with the announced closure of a technical center in Europe affecting approximately 80 employees.

ePropulsion

2023 Structural Costs Plan

- During the year ended December 31, 2023, the segment recorded $1 million of restructuring costs under this plan, primarily related to employee termination benefits.

2020 Structural Costs Plan

- During the year ended December 31, 2022, the segment recorded $1 million of restructuring costs, primarily related to equipment relocation costs.
- During the year ended December 31, 2021, the segment recorded $7 million of restructuring costs, primarily related to legal, professional fees, and equipment relocation costs.

Corporate

- During the year ended December 31, 2021, $2 million of net restructuring costs were recorded for various corporate restructuring actions.

Estimates of restructuring expense are based on information available at the time such charges are recorded. Due to the inherent uncertainty involved in estimating restructuring expenses, actual amounts paid for such activities may differ from amounts initially recorded. Accordingly, the Company may record revisions of previous estimates by adjusting previously established accruals.

The Company continues to evaluate different options across its operations to reduce existing structural costs over the next few years. The Company will recognize restructuring expense associated with any future actions at the time they are approved and become probable or are incurred. Any future actions could result in significant restructuring expense.

NOTE 5 RESEARCH AND DEVELOPMENT COSTS

The Company's net Research & Development ("R&D") expenditures are primarily included in Selling, general and administrative expenses of the Consolidated Statements of Operations. Customer reimbursements are netted against gross R&D expenditures as they are considered a recovery of cost. Customer reimbursements for prototypes are recorded net of prototype costs based on customer contracts, typically either when the prototype is shipped or when it is accepted by the customer. Customer reimbursements for engineering services are recorded when performance obligations are satisfied in accordance with the contract. Financial risks and rewards transfer upon shipment, acceptance of a prototype component by the customer or upon completion of the performance obligation as stated in the respective customer agreement. The Company has various customer arrangements relating to R&D activities that it performs at its various R&D locations.

The following table presents the Company's gross and net expenditures on R&D activities:

	Year Ended December 31,		
(in millions)	2023	2022	2021
Gross R&D expenditures	$ 856	$ 787	$ 694
Customer reimbursements	(139)	(86)	(108)
Net R&D expenditures	$ 717	$ 701	$ 586

Net R&D expenditures as a percentage of net sales were 5.1%, 5.5% and 5.0% for the years ended December 31, 2023, 2022 and 2021, respectively.

NOTE 6 OTHER OPERATING EXPENSE, NET

Items included in Other operating expense, net consist of:

	Year Ended December 31,		
(in millions)	2023	2022	2021
Merger and acquisition expense, net	$ 23	$ 9	$ 48
Asset impairments and lease modifications	29	30	—
Gain on sale of assets	(13)	—	—
Spin-Off transition services income, net	(10)	—	—
(Gain) loss on sale of business	(5)	(13)	29
Other income, net	(11)	(4)	(11)
Other operating expense, net	$ 13	$ 22	$ 66

Merger and acquisition expense, net: During the year ended December 31, 2023, the Company recorded merger and acquisition expense, net of $23 million, primarily related to professional fees for specific acquisition initiatives. This merger and acquisition expense, net included an $8 million loss related to a change in estimate of the expected earn-out estimate associated with the Santroll acquisition. During the year ended December 31, 2022, the Company recorded merger and acquisition expense, net of $9 million, primarily related to professional fees for specific acquisition initiatives. During the year ended December 31, 2021, the Company recorded merger and acquisition expense of $48 million, primarily for professional fees associated with the Company's acquisition of AKASOL.

Asset impairments and lease modifications: During the year ended December 31, 2023, the Company recorded charges of $29 million, primarily related to the write down of a customer incentive asset, a service and lease agreement termination and impairment of certain property, plant and equipment. During the year ended December 31, 2022, the Company recorded an impairment charge of $30 million to remove the AKASOL indefinite-lived trade name as the Company no longer plans to utilize this trade name in the business. Refer to Note 12 "Goodwill and Other Intangibles," to the Consolidated Financial Statement for more information.

Gain on sale of assets: During the year ended December 31, 2023, the Company recorded a $13 million gain on sale of fixed assets, primarily attributed to the sale of a European manufacturing facility and other fixed assets. The sale of the facility was pursuant to a formal restructuring plan.

Spin-Off transition services income, net: During the year ended December 31, 2023, the Company recorded other income in the amount of $10 million for net service reimbursements related to the Spin-Off. These services provided include information technology, human resources, finance, facilities, procurement, sales, intellectual properties and engineering. Refer to Note 26 - "Discontinued Operations," to the Consolidated Financial Statements for more information.

(Gain) loss on sale of business: During the year ended December 31, 2023, the Company recorded a net gain on sales of businesses of $5 million. During the year ended December 31, 2022, the Company recorded a net gain on sales of businesses of $13 million, which included a $22 million gain related to the sale of its interest in BorgWarner Romeo Power LLC and a $9 million loss related to a change in estimate of the expected earn-out related to the divestiture of the Company's Water Valley facility. During the year ended December 31, 2021, the Company recorded a pre-tax loss of $29 million, which included a $22 million loss in connection with the divestiture of the Company's Water Valley facility and a $7 million loss on the sale of an ePropulsion technical center in Europe. Refer to Note 2 "Acquisitions and Dispositions," to the Consolidated Financial Statements for more information.

NOTE 7 INCOME TAXES

Earnings before income taxes and the provision for income taxes are presented in the following table.

	Year Ended December 31,		
(in millions)	2023	2022	2021
Earnings (loss) before income taxes:			
U.S.[1]	$ (316)	$ (40)	$ (523)
Non-U.S.[1]	1,307	953	1,033
Total	$ 991	$ 913	$ 510
Provision for income taxes:			
Current:			
Federal	$ 42	$ 49	$ 15
State	8	7	5
Foreign	299	233	232
Total current expense	349	289	252
Deferred:			
Federal	(85)	(54)	(90)
State	—	(8)	(10)
Foreign	25	(32)	(87)
Total deferred (benefit) expense	(60)	(94)	(187)
Total provision for income taxes	$ 289	$ 195	$ 65

[1] In 2023, the U.S. loss before income taxes was primarily due to the realized and unrealized loss on debt and equity securities of $74 million that was primarily related to the Company's investment in Wolfspeed convertible debt securities that was sold during the year. In 2021, the U.S. loss before income taxes was primarily related to the $362 million unrealized loss related to the Company's investment in Romeo Power, Inc.

The provision for income taxes resulted in an effective tax rate of approximately 29%, 21% and 13% for the years ended December 31, 2023, 2022 and 2021, respectively.

The following table provides a reconciliation of tax expense based on the U.S. statutory tax rate to final tax expense.

	Year Ended December 31,		
(in millions)	2023	2022	2021
Income taxes at U.S. statutory rate of 21%	$ 209	$ 192	$ 107
Increases (decreases) resulting from:			
Valuation allowance adjustments, net	186	21	(13)
Net tax on remittance of foreign earnings	45	36	35
Foreign rate differentials	20	57	66
U.S. tax on foreign earnings	18	10	10
State taxes, net of federal benefit	6	(3)	(6)
Other permanent differences	6	5	14
Impact of tax law and rate change	(1)	1	(19)
Affiliates' earnings	(7)	(6)	(9)
Reserve adjustments, settlements and claims	(15)	13	(20)
Changes in accounting methods and filing positions	(18)	(15)	(7)
Tax credits	(21)	(15)	(5)
Foreign derived intangible income deduction	(23)	(23)	(8)
Enhanced research and development deductions	(35)	(33)	(22)
Tax holidays	(35)	(38)	(81)
Deferred tax impact of intra-group transactions	(70)	—	—
Other, net	24	(7)	23
Provision for income taxes, as reported	$ 289	$ 195	$ 65

The Company’s tax rate is affected by the tax laws and rates of the U.S. and other jurisdictions in which the Company operates, the relative amount of income earned by jurisdiction and the relative amount of losses or income for which no tax benefit or expense was recognized due to a valuation allowance.

The Company’s effective tax rate was impacted beneficially by tax incentives obtained in various non-U.S. countries, primarily those arising in China related to the High and New Technology Enterprise (“HNTE”) status of various subsidiaries ($35 million, $38 million and $81 million for the years ended December 31, 2023, 2022 and 2021, respectively). HNTE status is granted for three-year periods, and the Company seeks to renew such status on a regular basis. In addition, beneficial impacts were recognized related to tax deductions for qualifying research and development expenditures ($35 million, $33 million and $22 million for the years ended December 31, 2023, 2022 and 2021, respectively).

The Company’s effective tax rate is also impacted by net changes to valuation allowances, where the Company has determined that it is more-likely-than-not that certain deferred tax assets would not be realized. For the years ended December 31, 2023 and 2022, the Company recorded net expense related to valuation allowances of $186 million and $21 million, respectively. For the year ended December 31, 2021, the Company recorded a net benefit of $13 million.

In 2023, the Company recognized a discrete tax benefit of approximately $19 million related to the resolution of tax audits and reductions in certain unrecognized tax benefits and accrued interest related to matters for which the statute of limitation had lapsed. In addition, the Company recognized a discrete tax benefit of approximately $50 million in relation to the Spin-Off, a discrete tax benefit of approximately $30 million in relation to various changes in filling positions for prior years, and a discrete tax expense of approximately $79 million in relation to changes in judgment related to the recovery of deferred tax assets, primarily due to the impact of the Spin-Off on the allocation of the Company’s profits across jurisdictions for tax purposes as well as various tax structuring actions and strategies.

In 2022, the Company recognized discrete tax benefits of $23 million, primarily related to a reduction in certain unrecognized tax benefits and accrued interest related to matters for which the statute of limitations had lapsed and favorable provision-to-return adjustments.

In 2021, the Company recognized a $55 million tax benefit related to a reduction in certain unrecognized tax benefits and accrued interest related to matters for which the statute of limitations had lapsed. In addition, the Company recognized a discrete tax benefit of $20 million related to an increase in its deferred tax assets as a result of an increase in the United Kingdom ("U.K.") tax rate from 19% to 25%. This rate change was enacted in June 2021 and becomes effective April 2023. Further, a net discrete tax benefit of $36 million was recognized, primarily related to changes to certain withholding rates applied to unremitted earnings. In the fourth quarter of 2021, the Company received approval for tax holiday status reducing the statutory tax rate for two of its legal entities, resulting in a reduction in tax expense of $28 million in 2021.

A roll forward of the Company's total gross unrecognized tax benefits is presented below:

(in millions)	2023	2022	2021
Balance, January 1	$ 172	$ 181	$ 198
Additions based on tax positions related to current year	15	19	21
Reductions for tax positions of prior years	(11)	(7)	—
Reductions for lapse in statute of limitations	(10)	—	(36)
Reductions for closure of tax audits and settlements	(5)	(9)	—
Translation adjustment	—	(12)	(2)
Balance, December 31	$ 161	$ 172	$ 181

The Company recognizes interest and penalties related to unrecognized tax benefits in income tax expense and accrued approximately $65 million and $55 million for the payment of interest and penalties at December 31, 2023 and 2022, respectively. For the years ended December 31, 2023, 2022 and 2021, the Company recognized expense related to interest and penalties of $10 million, $6 million and $16 million, respectively.

During the year ended December 31, 2021, the Company also recorded a reduction in tax expense of $34 million for previously recorded interest related to matters for which the statute of limitations lapsed.

As of December 31, 2023, approximately $198 million represents the amount that, if recognized, would affect the Company's effective income tax rate in future periods. This amount includes a decrease in U.S. federal income taxes that would occur upon recognition of the state tax benefits and U.S. foreign tax credits included therein.

The Company estimates that it is reasonably possible there could be a decrease of approximately $108 million in unrecognized tax benefits and interest in the next 12 months related to the closure of an audit and the lapse in statute of limitations subsequent to the reporting period from certain taxing jurisdictions.

The Company and/or one of its subsidiaries files income tax returns in the U.S. federal, various state jurisdictions and various foreign jurisdictions. In certain tax jurisdictions, the Company may have more than one taxpayer. The Company is no longer subject to income tax examinations by tax authorities in its major tax jurisdictions as follows:

Tax jurisdiction	Years no longer subject to audit	Tax jurisdiction	Years no longer subject to audit
U.S. Federal	2015 and prior	Japan	2018 and prior
Barbados	2016 and prior	Luxembourg	2019 and prior
China	2015 and prior	Mexico	2015 and prior
France	2015 and prior	Poland	2016 and prior
Germany	2015 and prior	South Korea	2016 and prior
Hungary	2017 and prior	United Kingdom	2018 and prior

In the U.S., certain tax attributes created in years prior to 2017 were subsequently utilized. Even though the U.S. federal statute of limitations may have expired for years prior to 2017, the years in which these tax attributes were created could still be subject to examination, limited to only the examination of the creation of the tax attribute.

The components of deferred tax assets and liabilities consist of the following:

	December 31,	
(in millions)	2023	2022
Deferred tax assets:		
Net operating loss and capital loss carryforwards	$ 359	$ 256
Research and development capitalization	211	143
Employee compensation	43	41
State tax credits	23	29
Unrecognized tax benefits	21	24
Warranty	20	20
Pension and other postemployment benefits	4	10
Other comprehensive loss	3	—
Unrealized loss on equity securities	—	8
Foreign tax credits	—	6
Other	103	97
Total deferred tax assets	$ 787	$ 634
Valuation allowance	(310)	(136)
Net deferred tax asset	$ 477	$ 498
Deferred tax liabilities:		
Goodwill and intangible assets	(135)	(194)
Fixed assets	(121)	(113)
Unremitted foreign earnings	(115)	(125)
Unrealized gain on equity securities	(2)	—
Other comprehensive income	—	(14)
Other	(7)	(16)
Total deferred tax liabilities	$ (380)	$ (462)
Net deferred taxes	$ 97	$ 36

As of December 31, 2023, the Company had gross deferred tax assets for certain non-U.S. net operating loss ("NOL") carryforwards of $277 million, $43 million of which expire at various dates from 2024 through 2043 and $234 million of which have an indefinite life. The Company has a valuation allowance recorded of $213 million with regards to these deferred tax assets.

As of December 31, 2023, certain U.S. subsidiaries had gross deferred tax assets of approximately $39 million for federal and state NOL carryforwards, $21 million of which expire at various dates from 2024 through 2043 and $18 million of which have an indefinite life. The Company has recorded a valuation allowance of $17 million with regards to these deferred tax assets. In addition, certain U.S. subsidiaries also have state tax credit carryforwards of $23 million, which are offset by a valuation allowance of $23 million.

On a quarterly basis, the Company reviews the likelihood that the benefit of its deferred tax assets will be realized and, therefore, the need for valuation allowances. The Company assesses existing deferred tax assets, net operating loss carryforwards and tax credit carryforwards by jurisdiction and expectations of its ability to utilize these tax attributes through a review of past, current, and estimated future taxable income and tax planning strategies. If, based upon the weight of available evidence, it is more-likely-than-not the deferred tax assets will not be realized, a valuation allowance is recorded. Due to recent restructurings, the Company concluded that the weight of the negative evidence outweighs the positive evidence in certain foreign jurisdictions. As a result, the Company believes it is more-likely-than-not that the net deferred tax assets in certain foreign jurisdictions that include entities in Barbados, Hungary, France, Spain, Ireland and the U.K. will not be realized in the future.

As of December 31, 2023, the Company recorded deferred tax liabilities of $115 million with respect to foreign unremitted earnings. The Company did not provide deferred tax liabilities with respect to certain book versus tax basis differences not represented by undistributed earnings of approximately $449 million as of December 31, 2023, because the Company continues to assert indefinite reinvestment of these basis differences. These basis differences would become taxable upon the sale or liquidation of the foreign subsidiaries. The Company's best estimate of the unrecognized deferred tax liability on these basis differences is approximately $18 million as of December 31, 2023.

NOTE 8 RECEIVABLES, NET

The table below provides details of receivables as of December 31, 2023 and 2022:

	December 31,	
(in millions)	2023	2022
Receivables, net:		
Customers	$ 2,519	$ 2,133
Indirect taxes	238	193
Other	364	156
Gross receivables	3,121	2,482
Allowance for credit losses	(12)	(11)
Total receivables, net	$ 3,109	$ 2,471

The table below summarizes the activity in the allowance for credit losses for the years ended December 31, 2023, 2022 and 2021:

	Year Ended December 31,		
(in millions)	2023	2022	2021
Beginning balance, January 1	$ (11)	$ (9)	$ (12)
Provision	(4)	(2)	—
Write-offs	2	—	2
Translation adjustment and other	1	—	1
Ending balance, December 31	$ (12)	$ (11)	$ (9)

NOTE 9 INVENTORIES, NET

A summary of Inventories, net is presented below:

	December 31,	
(in millions)	2023	2022
Raw material and supplies	$ 991	$ 919
Work-in-progress	160	136
Finished goods	194	187
FIFO inventories	1,345	1,242
LIFO reserve	(32)	(25)
Inventories, net	$ 1,313	$ 1,217

NOTE 10 OTHER CURRENT AND NON-CURRENT ASSETS

Additional detail related to assets is presented below:

(in millions)	December 31, 2023	2022
Prepayments and other current assets:		
Prepaid tooling	$ 89	$ 77
Prepaid taxes	38	33
Derivative instruments	32	12
Customer incentive payments (Note 3)	27	34
Contract assets (Note 3)	18	14
Prepaid insurance	10	10
Other	47	50
Total prepayments and other current assets	$ 261	$ 230
Investments and long-term receivables:		
Investment in equity affiliates	$ 237	$ 235
Investment in equity securities	71	73
Long-term receivables	56	56
Investment in debt securities	—	455
Total investments and long-term receivables	$ 364	$ 819
Other non-current assets:		
Deferred income taxes (Note 7)	$ 257	$ 179
Operating leases (Note 22)	143	106
Customer incentive payments (Note 3)	58	99
Derivative instruments	15	68
Other	39	37
Total other non-current assets	$ 512	$ 489

NOTE 11 PROPERTY, PLANT AND EQUIPMENT, NET

Property, plant and equipment, net is stated at cost less accumulated depreciation and amortization, and consisted of:

(in millions)	December 31, 2023	2022
Land, land use rights and buildings	$ 1,319	$ 1,213
Machinery and equipment	5,327	4,963
Finance lease assets	32	13
Construction in progress	671	529
Total property, plant and equipment, gross	7,349	6,718
Less: accumulated depreciation	3,816	3,531
Property, plant and equipment, net, excluding tooling	3,533	3,187
Tooling, net of amortization	250	239
Property, plant and equipment, net	$ 3,783	$ 3,426

Interest costs capitalized for the years ended December 31, 2023, 2022 and 2021 were $17 million, $17 million and $9 million, respectively.

NOTE 12 GOODWILL AND OTHER INTANGIBLES

During the fourth quarter of 2023, the Company performed a quantitative analysis on each reporting unit to refresh its respective fair value. For 2023 and 2022, the estimated fair value was determined using a combined income and market approach. The market approach is based on market multiples (revenue and "EBITDA", defined as earnings before interest, taxes, depreciation and amortization) and requires an estimate of appropriate multiples based on market data for comparable companies. The market valuation models and other financial ratios used by the Company require certain assumptions and estimates regarding the applicability of those models to the Company's facts and circumstances.

The Company believes the assumptions and estimates used to determine the estimated fair value are reasonable. Different assumptions could materially affect the estimated fair value. The primary assumptions affecting the Company's 2023 goodwill quantitative impairment review are as follows:

- **Discount rates:** The Company used a range of 12.5% to 14.5% weighted average cost of capital ("WACC") as the discount rates for future cash flows. The WACC is intended to represent a rate of return that would be expected by a market participant.

- **Operating income margin:** The Company used historical and expected operating income margins, which may vary based on the projections of the reporting unit being evaluated.

- **Revenue growth rates:** The Company used a global automotive market industry growth rate forecast adjusted to estimate its own market participation for product lines.

In addition to the above primary assumptions, the Company notes the following risks to volume and operating income assumptions that could have an impact on the discounted cash flow models:

- The automotive industry is cyclical, and the Company's results of operations could be adversely affected by industry downturns.
- The automotive industry is evolving, and if the Company does not respond appropriately, its results of operations could be adversely affected.
- The Company is dependent on market segments that use its key products and could be affected by decreasing demand in those segments.
- The Company is subject to risks related to international operations.

Based on the assumptions outlined above, the impairment testing conducted in the fourth quarter of 2023 indicated the Company's goodwill assigned to the respective reporting units was not impaired. Future changes in the judgments, assumptions and estimates from those used in acquisition-related valuations and goodwill impairment testing, including discount rates or future operating results and related cash flow projections, could result in significantly different estimates of the fair values in the future. Due to the Company's recent acquisitions, there is less headroom (the difference between the carrying value and the fair value) associated with certain of the Company's reporting units. Based on the impairment testing conducted in 2023, the amounts by which the estimated fair values of the Company's goodwill reporting units exceeded their carrying values ranged from 22% to 139%. An increase in discount rates, a reduction in projected cash flows or a combination of the two could lead to a reduction in the estimated fair values, which may result in impairment charges that could materially affect the Company's financial statements in any given year.

A summary of the changes in the carrying amount of goodwill is presented in the following tables. The prior period balances have been recast for inter-segment transitions of certain businesses that were completed during 2022. Refer to Note 24, “Reportable Segments and Related Information” for more information.

	2023			
(in millions)	Air Management	Drivetrain & Battery Systems	ePropulsion	Total
Gross goodwill balance, January 1	$ 1,566	$ 1,434	$ 480	$ 3,480
Accumulated impairment losses, January 1	(502)	—	—	(502)
Net goodwill balance, January 1	$ 1,064	$ 1,434	$ 480	$ 2,978
Goodwill during the year:				
Acquisitions[1] (Note 2)	8	—	14	22
Measurement period adjustments (Note 2)	(6)	—	—	(6)
Other, primarily translation adjustment	11	21	(13)	19
Net goodwill balance, December 31	$ 1,077	$ 1,455	$ 481	$ 3,013

	2022			
(in millions)	Air Management	Drivetrain & Battery Systems	ePropulsion	Total
Gross goodwill balance, January 1	$ 1,466	$ 1,364	$ 526	$ 3,356
Accumulated impairment losses, January 1	(502)	—	—	(502)
Net goodwill balance, January 1	$ 964	$ 1,364	$ 526	$ 2,854
Goodwill during the year:				
Acquisitions[1] (Note 2)	126	132	—	258
Measurement period adjustments[2]	—	(20)	—	(20)
Other, primarily translation adjustment	(26)	(42)	(46)	(114)
Net goodwill balance, December 31	$ 1,064	$ 1,434	$ 480	$ 2,978

[1] Acquisitions relate to the Company’s 2023 purchases of SSE, Eldor, and 2022 purchases of Drivetek, Rhombus and Santroll.
[2] Measurement period adjustments primarily relate to the 2023 acquisition of SSE and 2022 acquisition of Santroll.

The Company’s other intangible assets, primarily from acquisitions, consist of the following:

		December 31, 2023			December 31, 2022		
(in millions)	Estimated useful lives (years)	Gross carrying amount	Accumulated amortization	Net carrying amount	Gross carrying amount	Accumulated amortization	Net carrying amount
Amortized intangible assets:							
Patented and unpatented technology	5 - 15	$ 364	$ 145	$ 219	$ 349	$ 111	$ 238
Customer relationships	6 - 15	641	305	336	639	267	372
Miscellaneous	2 - 5	9	6	3	9	6	3
Total amortized intangible assets		1,014	456	558	997	384	613
Unamortized trade names		6	—	6	6	—	6
Total other intangible assets		$ 1,020	$ 456	$ 564	$ 1,003	$ 384	$ 619

Amortization of other intangible assets was $67 million, $69 million and $59 million for the years ended December 31, 2023, 2022 and 2021, respectively. The Company utilizes the straight-line method of amortization recognized over the estimated useful lives of the assets. The estimated future annual amortization expense, primarily for acquired intangible assets, is as follows: $70 million in 2024, $69 million in 2025, $61 million in 2026, $54 million in 2027, $54 million in 2028 and $250 million thereafter.

A roll forward of the gross carrying amounts and related accumulated amortization of the Company's other intangible assets is presented below:

	Gross carrying amounts		Accumulated amortization	
(in millions)	2023	2022	2023	2022
Beginning balance, January 1	$ 1,003	$ 957	$ 384	$ 335
Acquisitions[1] (Note 2)	14	132	—	—
Impairment[2]	—	(35)	—	(2)
Amortization	—	—	67	69
Translation adjustment	3	(51)	5	(18)
Ending balance, December 31	$ 1,020	$ 1,003	$ 456	$ 384

[1] Acquisitions relate to the Company's 2023 purchases of SSE, Eldor, 2022 purchases of Drivetek, Rhombus and Santroll.

[2] During the fourth quarter of 2022, the Company recorded an impairment charge of $30 million to remove the AKASOL indefinite-lived trade name as the Company no longer plans to utilize this trade name in the business.

NOTE 13 PRODUCT WARRANTY

The following table summarizes the activity in the product warranty accrual accounts:

(in millions)	2023	2022
Beginning balance, January 1	$ 185	$ 168
Provisions for current period sales	98	81
Adjustments of prior estimates	(3)	(9)
Payments	(84)	(48)
Other, primarily translation adjustment	—	(7)
Ending balance, December 31	$ 196	$ 185

The product warranty liability is classified in the Consolidated Balance Sheets as follows:

	December 31,	
(in millions)	2023	2022
Other current liabilities	$ 91	$ 110
Other non-current liabilities	105	75
Total product warranty liability	$ 196	$ 185

NOTE 14 NOTES PAYABLE AND DEBT

The Company had short-term and long-term debt outstanding as follows:

(in millions)	December 31, 2023	2022
Short-term debt		
Short-term borrowings	$ 70	$ 58
Long-term debt		
3.375% Senior notes due 03/15/25 ($384 million par value)	384	499
5.000% Senior notes due 10/01/25 ($453 million par value)[1]	477	840
2.650% Senior notes due 07/01/27 ($1,100 million par value)	1,093	1,092
7.125% Senior notes due 02/15/29 ($121 million par value)	120	120
1.000% Senior notes due 05/19/31 (€1,000 million par value)	1,088	1,051
4.375% Senior notes due 03/15/45 ($500 million par value)	495	495
Term loan facilities, finance leases and other	53	45
Total long-term debt	3,710	4,142
Less: current portion	3	2
Long-term debt, net of current portion	$ 3,707	$ 4,140

[1] These notes include the fair value step up of $24 million and $65 million as of December 31, 2023 and 2022, respectively, related to the Delphi Technologies acquisition in 2020. The fair value step up was calculated based on observable market data and is amortized as a reduction to interest expense over the remaining life of the instrument using the effective interest method.

In September 2023, the Company purchased and extinguished $438 million of Senior notes due in 2025, comprised of $115 million and $323 million face value of its 3.375% and 5.000% Senior notes, respectively. Total cash consideration paid was $430 million. The Company recorded a gain of approximately $28 million during the year ended December 31, 2023, consisting of an $8 million gain related to a cash settlement below the face value of the 2025 notes and $20 million related to the write-off of a portion of the unamortized fair value step up on the 5.000% Senior notes due in 2025 from the Delphi Technologies acquisition in 2020 and a portion of the unamortized discount on the 3.375% Senior notes due in 2025 that was recorded at the time of that note issuance. The gain on extinguishment was recorded to Interest expense, net, in the Consolidated Statement of Operations.

On May 19, 2021, in anticipation of the acquisition of AKASOL and to refinance the Company's €500 million 1.800% Senior notes due in November 2022, the Company issued €1.0 billion in 1.000% Senior notes due May 2031. Interest is payable annually in arrears on May 19 of each year. On June 18, 2021, the Company repaid its €500 million 1.80% Senior notes due November 2022 and incurred a loss on debt extinguishment of $20 million, which is reflected in Interest expense, net in the Consolidated Statement of Operations.

The Company may utilize uncommitted lines of credit for short-term working capital requirements. As of December 31, 2023 and 2022, the Company had $70 million and $58 million, respectively, in borrowings under these facilities, which are classified in Notes payable and other short-term debt in the Consolidated Balance Sheets. The short-term borrowings primarily relate to a European money market loan with an interest rate of Euribor plus 1.75% that is callable upon immediate notice by either party.

The weighted average interest rate on short-term borrowings outstanding as of December 31, 2023 and 2022 was3.5% and 0.9%, respectively. The weighted average interest rate on all borrowings outstanding, including the effects of outstanding swaps, as of December 31, 2023 and 2022 was 2.3% and 2.5%, respectively. The following table provides details on Interest expense, net included in the Consolidated Statements of Operations. Interest expense primarily relates to interest on the Company's fixed rate

Senior notes, net of any amortization of premium or discount. Interest income primarily relates to interest received on cash and investments and interest received on the Company's net investment hedges. Interest income has been favorably impacted by rising interest rates.

	Year Ended December 31,		
(in millions)	2023	2022	2021
Interest expense	$ 73	$ 71	$ 82
(Gain) loss on debt extinguishment	(28)	—	20
Interest income	(35)	(20)	(11)
Interest expense, net	$ 10	$ 51	$ 91

Annual principal payments required as of December 31, 2023 are as follows:

(in millions)	
2024	$ 73
2025	846
2026	8
2027	1,107
2028	8
After 2028	1,743
Total payments	$ 3,785
Less: unamortized premiums, net of discount	(5)
Total short and long-term debt	$ 3,780

The Company's long-term debt includes various covenants, none of which are expected to restrict future operations.

The Company has a $2 billion multi-currency revolving credit facility that allows the Company to increase the facility by $1 billion with bank group approval. This facility was renewed in September 2023 and now matures in September 2028. The credit agreement contains customary events of default and one key financial covenant, which is a debt-to-EBITDA (Earnings Before Interest, Taxes, Depreciation and Amortization) ratio. The Company was in compliance with the financial covenant at December 31, 2023. At December 31, 2023 and 2022, the Company had no outstanding borrowings under this facility.

The Company's commercial paper program allows the Company to issue $2 billion of short-term, unsecured commercial paper notes under the limits of its multi-currency revolving credit facility. Under this program, the Company may issue notes from time to time and use the proceeds for general corporate purposes. The Company had no outstanding borrowings under this program as of December 31, 2023 and 2022.

The total current combined borrowing capacity under the multi-currency revolving credit facility and commercial paper program cannot exceed $2 billion.

As of December 31, 2023 and 2022, the estimated fair values of the Company's senior unsecured notes totaled $3,304 million and $3,530 million, respectively. The estimated fair values were $353 million and $567 million lower than carrying value at December 31, 2023 and 2022, respectively. Fair market values of the senior unsecured notes are developed using observable values for similar debt instruments, which are considered Level 2 inputs as defined by ASC Topic 820. The carrying values of the Company's multi-currency revolving credit facility, commercial paper program and other debt facilities approximate fair value. The fair value estimates do not necessarily reflect the values the Company could realize in the current markets.

The Company had outstanding letters of credit of $37 million and $31 million at December 31, 2023 and 2022, respectively. The letters of credit typically act as guarantees of payment to certain third parties in accordance with specified terms and conditions.

NOTE 15 OTHER CURRENT AND NON-CURRENT LIABILITIES

Additional detail related to liabilities is presented in the table below:

(in millions)	December 31, 2023	December 31, 2022
Other current liabilities:		
Payroll and employee related	$ 329	$ 314
Customer related	124	108
Indirect taxes	121	115
Income taxes payable	103	107
Product warranties (Note 13)	91	110
Employee termination benefits (Note 4)	61	21
Operating leases (Note 22)	37	22
Accrued freight	26	30
Interest	26	22
Contract liabilities (Note 3)	18	14
Supplier related	16	15
Insurance	16	18
Deferred engineering	13	23
Other non-income taxes	12	12
Retirement related (Note 18)	11	11
Dividends payable	—	21
Other	144	121
Total other current liabilities	$ 1,148	$ 1,084
Other non-current liabilities:		
Other income tax liabilities	$ 226	$ 242
Deferred income taxes (Note 7)	160	143
Operating leases (Note 22)	112	85
Product warranties (Note 13)	105	75
Deferred income	83	59
Earn-out liability (Note 2)	13	10
Employee termination benefits (Note 4)	7	18
Other	61	54
Total other non-current liabilities	$ 767	$ 686

NOTE 16 FAIR VALUE MEASUREMENTS

ASC Topic 820 emphasizes that fair value is a market-based measurement, not an entity-specific measurement. Therefore, a fair value measurement should be determined based on assumptions that market participants would use in pricing an asset or liability. As a basis for considering market participant assumptions in fair value measurements, ASC Topic 820 establishes a fair value hierarchy, which prioritizes the inputs used in measuring fair values as follows:

Level 1: Observable inputs such as quoted prices for identical assets or liabilities in active markets;

Level 2: Inputs, other than quoted prices in active markets, that are observable either directly or indirectly; and

Level 3: Unobservable inputs in which there is little or no market data, which require the reporting entity to develop its own assumptions.

Assets and liabilities measured at fair value are based on one or more of the following three valuation techniques noted in ASC Topic 820:

A. **Market approach:** Prices and other relevant information generated by market transactions involving identical or comparable assets, liabilities or a group of assets or liabilities, such as a business.
B. **Cost approach:** Amount that would be required to replace the service capacity of an asset (replacement cost).
C. **Income approach:** Techniques to convert future amounts to a single present amount based upon market expectations (including present value techniques, option-pricing and excess earnings models).

The following tables classify assets and liabilities measured at fair value on a recurring basis as of December 31, 2023 and 2022:

		Basis of fair value measurements				
(in millions)	Balance at December 31, 2023	Quoted prices in active markets for identical items (Level 1)	Significant other observable inputs (Level 2)	Significant unobservable inputs (Level 3)	Valuation technique	Assets measured at NAV[1]
Assets:						
Current earn-out receivables	$ 5	$ —	$ —	$ 5	C	$ —
Investment in equity securities	$ 26	$ —	$ —	$ —	—	$ 26
Foreign currency contracts	$ 33	$ —	$ 33	$ —	A	$ —
Net investment hedge contracts	$ 14	$ —	$ 14	$ —	A	$ —
Liabilities:						
Current earn-out liabilities	$ 2	$ —	$ —	$ 2	C	$ —
Non-current earn-out liabilities	$ 13	$ —	$ —	$ 13	C	$ —
Foreign currency contracts	$ 3	$ —	$ 3	$ —	A	$ —
Net investment hedge contracts	$ 2	$ —	$ 2	$ —	A	$ —

(in millions)	Balance at December 31, 2022	Basis of fair value measurements: Quoted prices in active markets for identical items (Level 1)	Significant other observable inputs (Level 2)	Significant unobservable inputs (Level 3)	Valuation technique	Assets measured at NAV[1]
Assets:						
Current earn-out receivables	$ 9	$ —	$ —	$ 9	C	$ —
Investment in debt securities	$ 455	$ —	$ 455	$ —	A	$ —
Investment in equity securities	$ 29	$ —	$ —	$ —	—	$ 29
Foreign currency contracts	$ 12	$ —	$ 12	$ —	A	$ —
Net investment hedge contracts	$ 68	$ —	$ 68	$ —	A	$ —
Liabilities:						
Current earn-out liabilities	$ 21	$ —	$ —	$ 21	C	$ —
Non-current earn-out liabilities	$ 10	$ —	$ —	$ 10	C	$ —
Foreign currency contracts	$ 9	$ —	$ 9	$ —	A	$ —
Net investment hedge contracts	$ 1	$ —	$ 1	$ —	A	$ —

[1] Certain assets that are measured at fair value using the NAV per share (or its equivalent) practical expedient have not been classified in the fair value hierarchy. These amounts represent investments in commingled and managed funds that have underlying assets in fixed income securities, equity securities, and other assets and the fair values have been estimated using the net asset value of the Company's ownership interest in partners' capital. The Company's redemption of its investments with the funds is governed by the partnership agreements and subject to approval from the general partners. With the exception of annual distributions in connection with the Company's deemed tax liability, distributions from each fund will be received as the underlying investments of the funds are liquidated, the timing of which is unknown.

The following tables provide a reconciliation of the Company's Level 3 earn-out assets and liabilities:

(in millions)	Fair Value Measurements Using Significant Unobservable Inputs (Level 3): Long-Term receivables	Current earn-out receivables	Current earn-out liabilities	Non-current earn-out liabilities
Balance at January 1, 2022	$ 18	$ —	$ —	$ —
Contingent earn-out recognized upon acquisition or disposition	—	—	16	34
Change in fair value of contingent consideration	—	(9)	5	(24)
Classification reclass	(18)	18	—	—
Balance at December 31, 2022	$ —	$ 9	$ 21	$ 10
Change in fair value of contingent consideration	—	5	5	3
Earn-out settlements	—	(9)	(24)	—
Balance at December 31, 2023	$ —	$ 5	$ 2	$ 13

Refer to Note 2, "Acquisitions and Dispositions," to the Consolidated Financial Statements for more detail regarding earn-outs.

The PHINIA-related defined benefits pension assets, liabilities and benefits (costs) are included in the tables below for periods prior to the Spin-Off, as they are not reported as discontinued operations in accordance with ASC Topic 205-20, "Discontinued Operations".

The following tables provide a reconciliation of the Company's defined benefit plans assets measured at fair value on a recurring basis:

(in millions)	Balance at December 31, 2023	Basis of fair value measurements: Quoted prices in active markets for identical items (Level 1)	Significant other observable inputs (Level 2)	Significant unobservable inputs (Level 3)	Valuation technique	Assets measured at NAV[2]
U.S. Plans:						
Fixed income securities	$ 87	$ —	$ —	$ —	—	$ 87
Equity securities	18	—	—	—	—	18
Alternative credit fund	16	—	—	—	—	16
Cash	2	2	—	—	A	—
	$ 123	$ 2	$ —	$ —		$ 121
Non-U.S. Plans:						
Fixed income securities	$ 213	$ 167	$ —	$ —	A	$ 46
Equity securities	40	27	—	—	A	13
Cash	5	5	—	—	A	—
Insurance contract[1]	74	—	—	74	C	—
Real estate and other	60	—	23	—	A,C	37
	$ 392	$ 199	$ 23	$ 74		$ 96

(in millions)	Balance at December 31, 2022	Basis of fair value measurements: Quoted prices in active markets for identical items (Level 1)	Significant other observable inputs (Level 2)	Significant unobservable inputs (Level 3)	Valuation technique	Assets measured at NAV[2]
U.S. Plans:						
Fixed income securities	$ 89	$ —	$ —	$ —	—	$ 89
Equity securities	17	—	—	—	—	17
Alternative credit fund	20	—	—	—	—	20
Cash	3	3	—	—	A	—
	$ 129	$ 3	$ —	$ —		$ 126
Non-U.S. Plans:						
Fixed income securities	$ 525	$ 54	$ —	$ —	A	$ 471
Equity securities	142	113	—	—	A	29
Cash	147	147	—	—	A	—
Insurance contract[1]	69	—	—	69	C	—
Real estate and other	273	—	20	46	A,C	207
	$ 1,156	$ 314	$ 20	$ 115		$ 707

[1] A BorgWarner defined benefit plan in the United Kingdom owns an insurance contract that guarantees payment of specified pension liabilities. The Company measures the fair value of the insurance asset by projecting expected future cash flows from the contract and discounting them to present value based on current market rates, including an assessment for non-performance risk of the insurance company. The assumptions used to project expected future cash flows are based on actuarial estimates and are unobservable; therefore, the contract is categorized within Level 3 of the hierarchy.

[2] Certain assets that are measured at fair value using the NAV per share (or its equivalent) practical expedient have not been classified in the fair value hierarchy. These amounts represent investments in commingled and managed funds that have underlying assets in fixed income securities, equity securities, and other assets.

The following tables provide a reconciliation of the Company's Level 3 defined benefit plans assets:

	Fair Value Measurements Using Significant Unobservable Inputs (Level 3)	
(in millions)	Insurance contract	Real estate trust fund
Balance at January 1, 2022	$ 108	$ 127
Purchases, sales and settlements	—	(93)
Realized gains	—	3
Benefits paid	(5)	—
Unrealized (losses) gains on assets still held at the reporting date	(20)	25
Translation adjustment	(14)	(16)
Balance at December 31, 2022	$ 69	$ 46
Purchases, sales and settlements	—	(46)
Benefits paid	5	—
Unrealized (losses) gains on assets still held at the reporting date	(4)	—
Translation adjustment	4	—
Balance at December 31, 2023	$ 74	$ —

Refer to Note 18, "Retirement Benefit Plans," to the Consolidated Financial Statements for more detail surrounding the defined benefit plan's asset investment policies and strategies, target allocation percentages and expected return on plan asset assumptions.

NOTE 17 FINANCIAL INSTRUMENTS

The Company's financial instruments include cash and cash equivalents, marketable securities and accounts receivable. Due to the short-term nature of these instruments, their book value approximates their fair value. The Company's financial instruments may include long-term debt, investments in equity securities, interest rate and cross-currency swaps, commodity derivative contracts and foreign currency derivative contracts. All derivative contracts are placed with counterparties that have an S&P, or equivalent, investment grade credit rating at the time of the contracts' placement. An adjustment for non-performance risk is considered in the estimate of fair value in derivative assets based on the counterparty credit default swap ("CDS") rate. When the Company is in a net derivative liability position, the non-performance risk adjustment is based on its CDS rate. At December 31, 2023 and 2022, the Company had no derivative contracts that contained credit-risk-related contingent features.

The Company, at times, uses certain commodity derivative contracts to protect against commodity price changes related to forecasted raw material and component purchases. The Company had no outstanding commodity contracts at December 31, 2023 and 2022.

The Company manages its interest rate risk by assessing its exposure to fixed and variable rates while attempting to optimize its interest costs. The Company, at times, selectively uses interest rate swaps to reduce market value risk associated with changes in interest rates (fair value hedges and cash flow hedges). At December 31, 2023 and 2022, the Company had no outstanding interest rate swaps or options.

The Company uses foreign currency forward and option contracts to protect against exchange rate movements for forecasted cash flows, including capital expenditures, purchases, operating expenses or sales transactions designated in currencies other than the functional currency of the operating unit. In addition, the Company uses foreign currency forward contracts to hedge exposure associated with its net investment in certain foreign operations (net investment hedges). Foreign currency derivative contracts require the Company, at a future date, to either buy or sell foreign currency in exchange for the operating units' local currency. The following foreign currency derivative contracts were outstanding and mature through the ending duration noted below:

		Foreign currency derivatives (in millions)[1]		
Functional currency	Traded currency	Notional in traded currency December 31, 2023	Notional in traded currency December 31, 2022	Ending duration
British Pound	Euro	83	10	Dec-25
Chinese Renminbi	U.S. Dollar	209	276	Nov-25
Euro	British Pound	15	9	Jan-24
Euro	Hungarian Forint	8,233	—	Dec-25
Euro	Polish Zloty	573	440	Dec-25
Euro	U.S. Dollar	152	120	Dec-25
Euro	Swiss Franc	24	—	Dec-25
Thailand Baht	U.S. Dollar	30	—	Dec-24
U.S. Dollar	Chinese Renminbi	582	1,402	Jun-24
U.S. Dollar	Euro	9	45	Jan-24
U.S. Dollar	Korean Won	34,209	51,786	Nov-24
U.S. Dollar	Mexican Peso	3,280	2,474	Dec-25
U.S. Dollar	Thai Baht	2,100	—	Jun-24

[1] Table above excludes non-significant traded currency pairings with total notional amounts less than $10 million U.S. Dollar equivalent as of December 31, 2023 or 2022.

The Company selectively uses cross-currency swaps to hedge the foreign currency exposure associated with its net investment in certain foreign operations (net investment hedges). At December 31, 2023 and 2022, the following cross-currency swap contracts were outstanding:

	Cross-currency swaps		
(in millions)	December 31, 2023	December 31, 2022	Ending duration
U.S. Dollar to Euro:			
Fixed receiving notional	$ 1,100	$ 1,100	Jul-27
Fixed paying notional	€ 976	€ 976	Jul-27
U.S. Dollar to Euro:			
Fixed receiving notional	$ 500	$ 500	Mar-25
Fixed paying notional	€ 450	€ 450	Mar-25
U.S. Dollar to Japanese Yen:			
Fixed receiving notional	$ 100	$ 100	Feb-29
Fixed paying notional	¥ 12,724	¥ 12,724	Feb-29

At December 31, 2023 and 2022, the following amounts were recorded in the Consolidated Balance Sheets as being payable to or receivable from counterparties under ASC Topic 815, “Derivatives and Hedging”:

(in millions)	Assets			Liabilities		
Derivatives designated as hedging instruments Under Topic 815:	Balance Sheet Location	December 31, 2023	December 31, 2022	Balance Sheet Location	December 31, 2023	December 31, 2022
Foreign currency	Prepayments and other current assets	$ 30	$ 9	Other current liabilities	$ 2	$ 8
Foreign currency	Other non-current assets	$ 1	$ —	Other non-current liabilities	$ —	$ 1
Net investment hedges	Other non-current assets	$ 14	$ 68	Other non-current liabilities	$ 2	$ 1
Derivatives not designated as hedging instruments:						
Foreign currency	Prepayments and other current assets	$ 2	$ 3	Other current liabilities	$ 1	$ —

Effectiveness for cash flow hedges is assessed at the inception of the hedging relationship and quarterly, thereafter. Gains and losses arising from these contracts that are included in the assessment of effectiveness are deferred into accumulated other comprehensive income (loss) (“AOCI”) and reclassified into income as the underlying operating transactions are recognized. These realized gains or losses offset the hedged transaction and are recorded on the same line in the statement of operations. The initial value of any component excluded from the assessment of effectiveness will be recognized in income using a systematic and rational method over the life of the hedging instrument. Any difference between the change in fair value of the excluded component and amounts recognized in income under that systematic and rational method will be recognized in AOCI.

Effectiveness for net investment hedges is assessed at the inception of the hedging relationship and quarterly, thereafter. Gains and losses arising from these contracts that are included in the assessment of effectiveness are deferred into foreign currency translation adjustments and only released when the subsidiary being hedged is sold or substantially liquidated. The initial value of any component excluded from the assessment of effectiveness will be recognized in income using a systematic and rational method over the life of the hedging instrument. Any difference between the change in fair value of the excluded component and amounts recognized in income under that systematic and rational method will be recognized in AOCI.

The table below shows deferred gains (losses) reported in AOCI as well as the amount expected to be reclassified to income in one year or less for designated net investment hedges. The amount expected to be reclassified to income in one year or less assumes no change in the current relationship of the hedged item at December 31, 2023 market rates.

(in millions)	Deferred gain (loss) in AOCI at		Gain (loss) expected to be reclassified to income in one year or less
Contract Type	December 31, 2023	December 31, 2022	
Net investment hedges:			
Foreign currency	$ —	$ (4)	$ —
Cross-currency swaps	12	67	—
Foreign currency-denominated debt	100	133	—
Total	$ 112	$ 196	$ —

Derivative instruments designated as hedging instruments as defined by ASC Topic 815 held during the period resulted in the following gains and losses recorded in income:

	Year ended December 31, 2023			
(in millions)	Net sales	Cost of sales	Selling, general and administrative expenses	Other comprehensive income
Total amounts of earnings and other comprehensive income line items in which the effects of cash flow hedges are recorded	$ 14,198	$ 11,630	$ 1,316	$ 48
Gain (loss) on cash flow hedging relationships:				
Foreign currency				
Gain (loss) recognized in other comprehensive income				$ 25

	Year ended December 31, 2022			
(in millions)	Net sales	Cost of sales	Selling, general and administrative expenses	Other comprehensive income
Total amounts of earnings and other comprehensive income line items in which the effects of cash flow hedges are recorded	$ 12,635	$ 10,266	$ 1,290	$ (325)
Gain (loss) on cash flow hedging relationships:				
Foreign currency				
Gain (loss) recognized in other comprehensive income				$ 4

	Year ended December 31, 2021			
(in millions)	Net sales	Cost of sales	Selling, general and administrative expenses	Other comprehensive income
Total amounts of earnings and other comprehensive income line items in which the effects of cash flow hedges are recorded	$ 11,803	$ 9,630	$ 1,085	$ 100
Gain (loss) on cash flow hedging relationships:				
Foreign currency				
Gain (loss) recognized in other comprehensive income				$ (4)
Gain (loss) reclassified from AOCI to income	$ 1	$ (4)	$ (1)	$ —

The gains or losses recorded in income related to components excluded from the assessment of effectiveness for derivative instruments designated as cash flow hedges were immaterial for the periods presented.

Gains and losses on derivative instruments designated as net investment hedges were recognized in other comprehensive income (loss) during the periods presented below.

(in millions)	Year Ended December 31,		
Net investment hedges	2023	2022	2021
Foreign currency	$ —	$ 6	$ (9)
Cross-currency swaps	$ (55)	$ 129	$ 115
Foreign currency-denominated debt	$ (33)	$ 67	$ 84

Derivatives designated as net investment hedge instruments as defined by ASC Topic 815 held during the period resulted in the following gains recorded in Interest expense on components excluded from the assessment of effectiveness:

(in millions)	Year Ended December 31,		
Net investment hedges	2023	2022	2021
Cross-currency swaps	$ 25	$ 26	$ 22

There were no gains or losses recorded in income related to components excluded from the assessment of effectiveness for foreign currency-denominated debt designated as net investment hedges. There were no gains or losses reclassified from AOCI for net investment hedges during the periods presented.

Derivatives not designated as hedging instruments are used to hedge remeasurement exposures of monetary assets and liabilities denominated in currencies other than the operating units' functional currency. These derivatives resulted in the following gains recorded in earnings, in the Consolidated Statement of Operations:

(in millions)		Year Ended December 31,		
Contract Type	Location	2023	2022	2021
Foreign Currency	Selling, general and administrative expenses	$ 19	$ 23	$ 9

NOTE 18 RETIREMENT BENEFIT PLANS

The Company sponsors various defined contribution savings plans, primarily in the U.S., that allow employees to contribute a portion of their pre-tax and/or after-tax income in accordance with plan specified guidelines. Under specified conditions, the Company will make contributions to the plans and/or match a percentage of the employee contributions up to certain limits. Total expense related to the defined contribution plans was $45 million, $59 million and $58 million in the years ended December 31, 2023, 2022 and 2021, respectively.

The Company has a number of defined benefit pension plans and other postemployment benefit plans covering eligible salaried and hourly employees and their dependents. The defined pension benefits provided are primarily based on (i) years of service and (ii) average compensation or a monthly retirement benefit amount. The Company provides defined benefit pension plans in France, Germany, India, Italy, Japan, Mexico, South Korea, Sweden, Switzerland, Turkey, U.K. and the U.S. The other postemployment benefit plans, which provide medical benefits, are unfunded plans. The Company’s U.S. and U.K. defined benefit plans are frozen, and no additional service cost is being accrued. All pension and other postemployment benefit plans in the U.S. have been closed to new employees. The measurement date for all plans is December 31.
The PHINIA-related defined benefits pension assets, liabilities and benefits (costs) are included in the tables below for periods prior to the Spin-Off, as they are not reported as discontinued operations in accordance with ASC Topic 205-20, “Discontinued Operations”.

The following table summarizes the expenses for the Company's defined contribution and defined benefit pension plans and the other postemployment benefit plans:

	Year Ended December 31,		
(in millions)	2023	2022	2021
Defined contribution expense	$ 45	$ 59	$ 58
Defined benefit pension (income) expense	26	(10)	(19)
Other postemployment benefit income	—	(1)	(1)
Total	$ 71	$ 48	$ 38

The following provides a roll forward of the plans' benefit obligations, plan assets, funded status and recognition in the Consolidated Balance Sheets:

	Pension benefits				Other postemployment benefits	
	Year Ended December 31,				Year Ended December 31,	
	2023		2022			
(in millions)	U.S.	Non-U.S.	U.S.	Non-U.S.	2023	2022
Change in projected benefit obligation:						
Projected benefit obligation, January 1	$ 136	$ 1,322	$ 183	$ 2,227	$ 37	$ 54
Service cost	—	11	—	20	—	—
Interest cost	7	42	4	37	2	1
Plan amendments	—	—	—	(11)	—	—
Settlement and curtailment	—	3	(6)	(3)	—	—
Actuarial loss (gain)	2	25	(33)	(685)	—	(13)
Currency translation	—	57	—	(200)	—	—
Acquisition	—	—	—	8	—	—
PHINIA spin-off	—	(938)	—	—	(1)	—
Benefits paid	(14)	(44)	(12)	(71)	(5)	(5)
Projected benefit obligation, December 31[1]	$ 131	$ 478	$ 136	$ 1,322	$ 33	$ 37
Change in plan assets:						
Fair value of plan assets, January 1	$ 129	$ 1,156	$ 177	$ 2,049		
Actual return on plan assets	7	50	(33)	(655)		
Employer contribution	1	20	2	20		
Plan participants' contribution	—	1	—	—		
Settlements	—	(3)	(5)	(4)		
Currency translation	—	53	—	(189)		
Acquisition	—	—	—	6		
PHINIA spin-off	—	(841)	—	—		
Benefits paid	(14)	(44)	(12)	(71)		
Fair value of plan assets, December 31	$ 123	$ 392	$ 129	$ 1,156		
Funded status	$ (8)	$ (86)	$ (7)	$ (166)	$ (33)	$ (37)
Amounts in the Consolidated Balance Sheets consist of:						
Non-current assets	$ —	$ 30	$ —	$ 26	$ —	$ —
Current liabilities	(1)	(5)	(1)	(6)	(5)	(6)
Non-current liabilities	(7)	(111)	(6)	(186)	(28)	(31)
Net amount	$ (8)	$ (86)	$ (7)	$ (166)	$ (33)	$ (37)
Amounts in accumulated other comprehensive loss consist of:						
Net actuarial loss	$ 77	$ 112	$ 80	$ 104	$ (3)	$ (3)
Net prior service (credit) cost	(2)	1	(3)	(9)	(8)	(11)
Net amount	$ 75	$ 113	$ 77	$ 95	$ (11)	$ (14)
Total accumulated benefit obligation for all plans	$ 131	$ 450	$ 136	$ 1,279		

[1] The decrease in the projected benefit obligation was primarily due to discontinued operations and actuarial losses during the period. The main driver of these losses was the decrease of 0.70% in the weighted average discount rate for Non-U.S. plans.

The funded status of pension plans with accumulated benefit obligations in excess of plan assets is as follows:

	December 31,	
(in millions)	2023	2022
Accumulated benefit obligation	$ (338)	$ (1,185)
Plan assets	235	1,022
Deficiency	$ (103)	$ (163)
Pension deficiency by country:		
United States	$ (8)	$ (6)
United Kingdom	—	(38)
Germany	(39)	(34)
Other	(56)	(85)
Total pension deficiency	$ (103)	$ (163)

The funded status of pension plans with projected benefit obligations in excess of plan assets is as follows:

	December 31,	
(in millions)	2023	2022
Projected benefit obligation	$ (360)	$ (1,223)
Plan assets	237	1,026
Deficiency	$ (123)	$ (197)
Pension deficiency by country:		
United States	$ (8)	$ (6)
United Kingdom	—	(38)
Germany	(40)	(35)
Other	(75)	(118)
Total pension deficiency	$ (123)	$ (197)

The weighted average asset allocations of the Company's funded pension plans and target allocations by asset category are as follows:

	December 31,		
	2023	2022	Target Allocation
U.S. Plans:			
Alternative credit, real estate, cash and other	15 %	18 %	10% - 20%
Fixed income securities	71 %	69 %	66% - 76%
Equity securities	14 %	13 %	10% - 20%
	100 %	100 %	
Non-U.S. Plans:			
Insurance contract, real estate, cash and other	36 %	42 %	32% - 42%
Fixed income securities	54 %	46 %	49% - 59%
Equity securities	10 %	12 %	4% - 14%
	100 %	100 %	

The Company's investment strategy is to maintain actual asset weightings within a preset range of target allocations. The Company believes these ranges represent an appropriate risk profile for the planned benefit payments of the plans based on the timing of the estimated benefit payments. In each asset category, separate portfolios are maintained for additional diversification. Investment managers are retained in each asset category to manage each portfolio against its benchmark. Each investment

manager has appropriate investment guidelines. In addition, the entire portfolio is evaluated against a relevant peer group. The defined benefit pension plans did not hold any Company securities as investments as of December 31, 2023 and 2022. A portion of pension assets is invested in common and commingled trusts.

The Company expects to contribute a total of $20 million to $30 million into its defined benefit pension plans during 2024. Of the $20 million to $30 million in projected 2024 contributions, $6 million are contractually obligated, while any remaining payments would be discretionary.

Refer to Note 16, “Fair Value Measurements,” to the Consolidated Financial Statements for more detail surrounding the fair value of each major category of plan assets, as well as the inputs and valuation techniques used to develop the fair value measurements of the plans’ assets at December 31, 2023 and 2022.

See the table below for a breakout of net periodic benefit cost between U.S. and non-U.S. pension plans:

	Pension benefits						Other postemployment benefits		
	Year Ended December 31,						Year Ended December 31,		
	2023		2022		2021				
(in millions)	U.S.	Non-U.S.	U.S.	Non-U.S.	U.S.	Non-U.S.	2023	2022	2021
Service cost	$ —	$ 11	$ —	$ 20	$ —	$ 25	$ —	$ —	$ —
Interest cost	7	42	4	37	3	30	2	1	1
Expected return on plan assets	(6)	(37)	(8)	(75)	(10)	(83)	—	—	—
Settlements, curtailments and other	—	4	3	—	2	(2)	—	—	—
Amortization of unrecognized prior service (credit) cost	(1)	—	(1)	—	(1)	—	(2)	(2)	(3)
Amortization of unrecognized loss	3	3	3	7	4	13	—	—	1
Net periodic cost (income)	$ 3	$ 23	$ 1	$ (11)	$ (2)	$ (17)	$ —	$ (1)	$ (1)

The components of net periodic benefit cost other than the service cost component are included in Other postretirement income in the Consolidated Statements of Operations.

The Company’s weighted average assumptions used to determine the benefit obligations for its defined benefit pension and other postemployment benefit plans were as follows:

	December 31,	
(percent)	2023	2022
U.S. pension plans:		
Discount rate	5.14	5.46
Rate of compensation increase	N/A	N/A
U.S. other postemployment benefit plans:		
Discount rate	5.10	5.41
Rate of compensation increase	N/A	N/A
Non-U.S. pension plans:		
Discount rate[1]	4.23	4.94
Rate of compensation increase	3.32	3.76

[1] Includes 4.61% and 4.94% for the U.K. pension plans for December 31, 2023 and 2022, respectively.

The Company's weighted average assumptions used to determine the net periodic benefit cost/(income) for its defined benefit pension and other postemployment benefit plans were as follows:

	Year Ended December 31,		
(percent)	2023	2022	2021
U.S. pension plans:			
Discount rate	5.47	2.73	2.31
Effective interest rate on benefit obligation	5.34	2.18	1.62
Expected long-term rate of return on assets	5.00	4.75	5.75
Average rate of increase in compensation	N/A	N/A	N/A
U.S. other postemployment plans:			
Discount rate	5.41	2.46	1.93
Effective interest rate on benefit obligation	5.29	1.84	1.21
Expected long-term rate of return on assets	N/A	N/A	N/A
Average rate of increase in compensation	N/A	N/A	N/A
Non-U.S. pension plans:			
Discount rate[1]	4.85	1.97	1.44
Effective interest rate on benefit obligation	4.88	1.83	1.24
Expected long-term rate of return on assets[2]	4.90	4.10	4.10
Average rate of increase in compensation	3.58	3.21	3.23

[1] Includes 4.94%, 1.91% and 1.39% for the U.K. pension plans for December 31, 2023, 2022 and 2021, respectively.
[2] Includes 5.30%, 4.12% and 4.00% for the U.K. pension plans for December 31, 2023, 2022 and 2021, respectively.

The Company's approach to establishing the discount rate is based upon the market yields of high-quality corporate bonds, with appropriate consideration of each plan's defined benefit payment terms and duration of the liabilities. In determining the discount rate, the Company utilizes a full-yield approach in the estimation of service and interest components by applying the specific spot rates along the yield curve used in the determination of the benefit obligation to the relevant projected cash flows.

The Company determines its expected return on plan asset assumptions by evaluating estimates of future market returns and the plans' asset allocation. The Company also considers the impact of active management of the plans' invested assets.

The estimated future benefit payments for the pension and other postemployment benefits are as follows:

	Pension benefits		
(in millions) Year	U.S.	Non-U.S.	Other postemployment benefits
2024	$ 15	$ 24	$ 5
2025	13	28	5
2026	12	27	4
2027	12	27	4
2028	11	29	3
2029-2033	48	164	12

The weighted average rate of increase in the per capita cost of covered health care benefits is projected to range from6.25% in 2024 down to an ultimate trend rate of 4.75%.

NOTE 19 STOCK-BASED COMPENSATION

The Company has granted restricted common stock and restricted stock units (collectively, "restricted stock") and performance share units as long-term incentive awards to employees and non-employee directors under the BorgWarner Inc. 2018 Stock Incentive Plan, as amended ("2018 Plan") and the BorgWarner Inc. 2023 Stock Incentive Plan ("2023 Plan"). The Company's Board of Directors adopted the 2023 Plan as a replacement to the 2018 Plan in April 2023, and the Company's stockholders approved the 2023 Plan at the annual meeting of stockholders on April 26, 2023. The 2023 Plan authorizes the issuance of a total of 11.3 million shares and approximately 9.1 million shares were available for future issuance as of December 31, 2023.

Restricted Stock: The value of restricted stock is determined by the market value of the Company's common stock at the date of grant. In 2023, restricted stock in the amount of 1.3 million shares and less than 0.1 million shares were granted to employees and non-employee directors, respectively. The value of the awards is recognized as compensation expense ratably over the restriction periods, generally two or three years. As of December 31, 2023, there was $42 million of unrecognized compensation expense related to restricted stock that will be recognized over a weighted average period of approximately 1.7 years.

Restricted stock compensation expense from continuing operations recorded in the Consolidated Statements of Operations is as follows:

	Year Ended December 31,		
(in millions, except per share data)	2023	2022	2021
Restricted stock compensation expense	$ 37	$ 28	$ 27
Restricted stock compensation expense, net of tax	$ 28	$ 21	$ 20

A summary of the status of the Company's nonvested restricted stock for employees and non-employee directors is as follows:

	Shares subject to restriction (thousands)	Weighted average grant date fair value
Nonvested at January 1, 2021	2,140	$ 39.58
Granted	1,175	$ 43.66
Vested	(845)	$ 43.34
Forfeited	(107)	$ 39.86
Nonvested at December 31, 2021	2,363	$ 40.24
Granted	1,060	$ 44.32
Vested	(862)	$ 38.68
Forfeited	(188)	$ 42.09
Nonvested at December 31, 2022	2,373	$ 42.47
Granted	1,099	$ 48.19
Vested	(917)	$ 39.80
Forfeited	(111)	$ 44.63
PHINIA spin-off awards transferred[1]	(389)	$ 45.80
PHINIA spin-off adjustment[2]	274	$ —
Nonvested at December 31, 2023	2,329	$ 40.57

[1] Represents the cancellation of awards outstanding as of Distribution Date held by PHINIA employees. PHINIA employees were granted PHINIA restricted stock after the spin-off replacing the cancelled awards.
[2] Represents the adjustment of unvested awards using a conversion ratio of 1.13 to 1 to preserve the intrinsic value of the awards prior to spin-off as authorized by the 2023 Plan and 2018 Plan.

Performance share units: The Company grants performance share units to members of senior management that vest at the end of three-year periods based the following metrics:

- ***Total Stockholder Return Units:*** This performance metric is based on the Company's market performance in terms of total shareholder return relative to a peer group of automotive companies. Based on the Company's relative ranking within the performance peer group, it is possible for none of the awards to vest or for a range of up to 200% of the target shares to vest.

  The Company recognizes compensation expense relating to this performance share plan ratably over the performance period regardless of whether the market conditions are expected to be achieved. Compensation expense associated with the performance share plans is calculated using a lattice model (Monte Carlo simulation).

  As of December 31, 2023, there was $10 million of unrecognized compensation expense related to total stockholder return units that will be recognized over a weighted average period of approximately 1.7 years.

- ***Relative Revenue Growth Units:*** This performance metric is based on the Company's performance in terms of revenue growth relative to the vehicle market over three-year performance periods. Based on the Company's relative revenue growth in excess of the industry vehicle production, it is possible for none of the awards to vest or for a range of up to 200% of the target shares to vest.

  The value of this performance share award is determined by the market value of the Company's common stock at the date of grant. The Company recognizes compensation expense relating to this performance share plan over the performance period based on the number of shares expected to vest at the end of each reporting period. The actual performance of the Company is evaluated quarterly and the expense is adjusted according to the new projections.

  As of December 31, 2022, these awards were fully expensed.

- ***Adjusted Earnings Per Share Units:*** Introduced in the first quarter of 2020, this performance metric is based on the Company's earnings per share adjusted for certain one-time items and non-operating gains and losses against a pre-defined target measured in the third year of the performance period.

  The value of this performance share award is determined by the adjusted earnings per share performance. The Company recognizes compensation expense relating to this performance share plan over the performance period based on the number of shares expected to vest at the end of each reporting period. The actual performance of the Company is evaluated quarterly and the expense is adjusted according to the new projections.

  As of December 31, 2022, these awards were fully expensed.

- ***eProduct Revenue Mix***: Introduced in the first quarter of 2021, this performance metric is based on the Company's total revenue derived from eProducts in relation to its total proforma revenue in the third year of the performance period. Based on the Company's eProduct revenue mix, it is possible for none of the awards to vest or for a range of up to 200% of the target shares to vest.

  The value of this performance share award is determined by the market value of the Company's common stock at the date of grant. The Company recognizes compensation expense relating to this performance share plan over the performance period based on the number of shares expected to vest at the end of each reporting period. The actual performance of the Company is evaluated quarterly and the expense is adjusted according to the new projections.

  As of December 31, 2023, there was $7 million of unrecognized compensation expense related to the eProduct revenue mix units that will be recognized over a weighted average period of approximately 1.6 years. The unrecognized amount of compensation expense is based on projected performance as of December 31, 2023.

- ***Cumulative Free Cash Flow***: Introduced in the first quarter of 2021, this performance metric is based on the Company's performance in terms of its operating cash flow less capital expenditures over the three-year performance periods. Based on the Company's cumulative free cash flow, it is possible for none of the awards to vest or for a range of up to 200% of the target shares to vest.

  The value of this performance share award is determined by the market value of the Company's common stock at the date of grant. The Company recognizes compensation expense relating to this performance share plan over the performance period based on the number of shares expected to vest at the end of each reporting period. The actual performance of the Company is evaluated quarterly and the expense is adjusted according to the new projections.

  As of December 31, 2023, there was $7 million of unrecognized compensation expense related to the cumulative free cash flow units that will be recognized over a weighted average period of approximately 1.6 years. The unrecognized amount of compensation expense is based on projected performance as of December 31, 2023.

- ***eProduct Revenue***: Introduced in the first quarter of 2022, this performance metric is based on the amount of the Company's total revenue derived from eProducts in the third year of the performance period. Based on the Company's eProduct revenue, it is possible for none of the awards to vest or for a range of up to 200% of the target shares to vest.

  The value of this performance share award is determined by the market value of the Company's common stock at the date of grant. The Company recognizes compensation expense relating to this performance share plan over the performance period based on the number of shares expected to vest at the end of each reporting period. The actual performance of the Company is evaluated quarterly and the expense is adjusted according to the new projections.

  As of December 31, 2023, there was $6 million of unrecognized compensation expense related to the eProduct revenue units that will be recognized over a weighted average period of approximately 1.7 years. The unrecognized amount of compensation expense is based on projected performance as of December 31, 2023.

The amounts expensed and common stock issued from continuing operations for performance share units for the years ended December 31, 2023, 2022 and 2021 were as follows:

	Year Ended December 31,					
	2023		2022		2021	
	Expense (in millions)	Number of shares issued (in thousands)	Expense (in millions)	Number of shares issued (in thousands)	Expense (in millions)	Number of shares issued (in thousands)
Total Stockholder Return	$ 8	107	$ 6	—	$ 6	—
Other Performance-Based	13	313	30	284	17	138
Total	$ 21	420	$ 36	284	$ 23	138

A summary of the status of the Company's nonvested performance share units for the years ended December 31, 2023, 2022 and 2021 were as follows:

	Total Stockholder Return		Other Performance-Based	
	Number of shares (in thousands)	Weighted average grant date fair value	Number of shares (in thousands)	Weighted average grant date fair value
Nonvested at January 1, 2021	271	$ 45.20	385	$ 38.66
Granted	135	$ 70.39	404	$ 45.30
Vested	(143)	$ 47.93	(143)	$ 41.92
Forfeited	(4)	$ 37.28	(6)	$ 36.79
Nonvested at December 31, 2021	259	$ 56.90	640	$ 42.14
Granted	138	$ 66.96	415	$ 44.33
Vested	(127)	$ 28.55	(234)	$ 34.73
Forfeited	(20)	$ 59.87	(56)	$ 43.35
Nonvested at December 31, 2022	250	$ 76.68	765	$ 45.51
Granted	135	$ 79.39	405	$ 48.06
Vested	(114)	$ 70.39	(341)	$ 45.14
Forfeited	(11)	$ 68.66	(34)	$ 44.96
PHINIA spin-off awards transferred[1]	(22)	$ 72.33	(67)	$ 46.03
PHINIA spin-off adjustment[2]	35	$ —	106	$ —
Nonvested at December 31, 2023	273	$ 71.37	834	$ 41.08

[1] Represents the cancellation of awards outstanding as of Distribution Date held by PHINIA employees. PHINIA employees were granted PHINIA restricted stock after the spin-off replacing the cancelled awards.
[2] Represents the adjustment of unvested awards using a conversion ratio of 1.13 to 1 to preserve the intrinsic value of the awards prior to spin-off as authorized by the 2023 Plan and 2018 Plan.

NOTE 20 ACCUMULATED OTHER COMPREHENSIVE LOSS

The following table summarizes the activity within accumulated other comprehensive loss:

(in millions)	Foreign currency translation adjustments	Hedge instruments	Defined benefit postretirement plans	Total
Beginning Balance, January 1, 2021	$ (321)	$ —	$ (330)	$ (651)
Comprehensive (loss) income before reclassifications[1]	(59)	(4)	255	192
Income taxes associated with comprehensive (loss) income before reclassifications	(43)	—	(64)	(107)
Reclassification from accumulated other comprehensive (loss) income	—	4	14	18
Income taxes reclassified into net earnings	—	—	(3)	(3)
Ending Balance December 31, 2021	$ (423)	$ —	$ (128)	$ (551)
Comprehensive (loss) income before reclassifications	(287)	4	(14)	(297)
Income taxes associated with comprehensive (loss) income before reclassifications	(40)	—	7	(33)
Reclassification from accumulated other comprehensive (loss) income	—	—	7	7
Income taxes reclassified into net earnings	—	—	(2)	(2)
Ending Balance December 31, 2022	$ (750)	$ 4	$ (130)	$ (876)
Comprehensive (loss) income before reclassifications	27	25	(9)	43
Income taxes associated with comprehensive (loss) income before reclassifications	24	—	(5)	19
Reclassification from accumulated other comprehensive (loss) income	—	—	3	3
Income taxes reclassified into net earnings	—	—	(1)	(1)
Spin-off of PHINIA	(20)	(1)	5	(16)
Ending Balance December 31, 2023	$ (719)	$ 28	$ (137)	$ (828)

[1] The increase in the defined benefit postretirement plans comprehensive income before reclassifications is primarily due to actuarial gains during the period.

The change in other comprehensive income for the Company's noncontrolling interest entities is related to foreign currency translation.

NOTE 21 CONTINGENCIES

In the normal course of business, the Company is party to various commercial and legal claims, actions and complaints, including matters involving warranty claims, intellectual property claims, governmental investigations and related proceedings, general liability and other risks. It is not possible to predict with certainty whether or not the Company will ultimately be successful in any of these commercial and legal matters or, if not, what the impact might be. The Company's management does not believe that adverse outcomes in any of these commercial and legal claims, actions and complaints are reasonably likely to have a material adverse effect on the Company's results of operations, financial position or cash flows. An adverse outcome could, nonetheless, be material to the results of operations or cash flows.

Environmental

The Company and certain of its current and former direct and indirect corporate predecessors, subsidiaries and divisions have been identified by the United States Environmental Protection Agency and certain state environmental agencies and private parties as potentially responsible parties ("PRPs") at various hazardous waste disposal sites under the Comprehensive Environmental Response,

Compensation and Liability Act (“Superfund”) and equivalent state laws and, as such, may be presently liable for the cost of clean-up and other remedial activities at 17 and 22 such sites as of December 31, 2023 and 2022, respectively. Responsibility for clean-up and other remedial activities at a Superfund site is typically shared among PRPs based on an allocation formula.

The Company believes that none of these matters, individually or in the aggregate, will have a material adverse effect on its results of operations, financial position or cash flows. Generally, this is because either the estimates of the maximum potential liability at a site are not material or the liability will be shared with other PRPs, although no assurance can be given with respect to the ultimate outcome of any such matter.

The Company has an accrual for environmental liabilities of $6 million as of both December 31, 2023 and 2022, included in Other current and Other non-current liabilities in the Consolidated Balance Sheets. As of December 31, 2023, this accrual, which relates to six of the sites, is based on information available to the Company (which, in most cases, includes an estimate of allocation of liability among PRPs; the probability that other PRPs, many of which are large, solvent public companies, will fully pay the cost apportioned to them; currently available information from PRPs and/or federal or state environmental agencies concerning the scope of contamination and estimated remediation and consulting costs; and remediation alternatives). Clean-up and other remedial activities are complete or nearing completion at the other 11 sites, for which there was no accrual as of December 31, 2023.

NOTE 22 LEASES AND COMMITMENTS

The Company’s lease agreements primarily consist of real estate property, such as manufacturing facilities, warehouses, and office buildings, in addition to personal property, such as vehicles, manufacturing and information technology equipment. The Company determines whether a contract is or contains a lease at contract inception. The majority of the Company’s lease arrangements are comprised of fixed payments, and a limited number of these arrangements include a variable payment component based on certain index fluctuations. As of December 31, 2023, a significant portion of the Company’s leases were classified as operating leases.

Generally, the Company’s operating leases have renewal options that extend the lease terms, and some include options to terminate the agreement or purchase the leased asset. The amortizable life of these assets is the lesser of its useful life or the lease term, including renewal periods reasonably assured of being exercised at lease inception.

All leases with an initial term of 12 months or less without an option to extend or purchase the underlying asset that the Company is reasonably certain to exercise (“short-term leases”) are not recorded on the Consolidated Balance Sheets, and lease expense is recognized on a straight-line basis over the lease term.

The following table presents the lease assets and lease liabilities as of December 31, 2023 and 2022:

(in millions)		December 31, 2023	2022
Assets	Balance Sheet Location		
Operating leases	Other non-current assets	$ 143	$ 106
Finance leases	Property, plant and equipment, net	25	8
Total lease assets		$ 168	$ 114
Liabilities			
Current			
Operating leases	Other current liabilities	$ 37	$ 22
Finance leases	Notes payable and other short-term debt	3	2
Non-current			
Operating leases	Other non-current liabilities	112	85
Finance leases	Long-term debt	23	9
Total lease liabilities		$ 175	$ 118

The following table presents lease obligations arising from obtaining leased assets for the years ended December 31, 2023 and 2022:

(in millions)	December 31, 2023	2022
Operating leases	$ 63	$ 14
Finance leases	17	—
Total lease obligations	$ 80	$ 14

The following table presents the maturity of lease liabilities as of December 31, 2023:

(in millions)	Operating leases	Finance leases
2024	$ 37	$ 3
2025	30	5
2026	23	4
2027	19	4
2028	14	5
After 2028	69	12
Total (undiscounted) lease payments	$ 192	$ 33
Less: Imputed interest	43	7
Present value of lease liabilities	$ 149	$ 26

In the years ended December 31, 2023, 2022 and 2021, the Company recorded operating lease expense of $33 million, $32 million and $28 million, respectively.

In the years ended December 31, 2023, 2022 and 2021, the operating cash flows for operating leases were $33 million, $30 million and $28 million, respectively.

In the years ended December 31, 2023, 2022 and 2021, the Company recorded short-term lease costs of $29 million, $21 million and $17 million, respectively.

Finance lease costs and related cash flows were immaterial for the periods presented.

ASC Topic 842, “Leases”, requires that the rate implicit in the lease be used if readily determinable. Generally, implicit rates are not readily determinable in the Company’s agreements, so the incremental borrowing rate is used instead for such lease arrangements. The incremental borrowing rates are determined using rates specific to the term of the lease, economic environments where lease activity is concentrated, value of lease portfolio, and assuming full collateralization of the loans. The following table presents the terms and discount rates:

	December 31,	
	2023	2022
Weighted average remaining lease term (years)		
Operating leases	8	6
Finance leases	8	5
Weighted average discount rate		
Operating leases	5.4 %	3.1 %
Finance leases	6.2 %	2.5 %

NOTE 23 EARNINGS PER SHARE

The Company presents both basic and diluted earnings per share of common stock (“EPS”) amounts. Basic EPS is calculated by dividing net earnings attributable to BorgWarner Inc. by the weighted average shares of common stock outstanding during the reporting period. Diluted EPS is calculated by dividing net earnings attributable to BorgWarner Inc. by the weighted average shares of common stock and common stock equivalents outstanding during the reporting period.

The dilutive impact of stock-based compensation is calculated using the treasury stock method. The treasury stock method assumes that the Company uses the assumed proceeds from the exercise of awards to repurchase common stock at the average market price during the period. The assumed proceeds under the treasury stock method include the purchase price that the grantee will pay in the future and compensation cost for future service that the Company has not yet recognized. The dilutive effects of performance-based stock awards described in Note 19, “Stock-Based Compensation,” to the Consolidated Financial Statements are included in the computation of diluted earnings per share at the level the related performance criteria are met through the respective balance sheet date. There were 0.7 million, 0.8 million and 0.8 million of performance share units excluded from the computation of the diluted earnings per share for the years ended December 31, 2023, 2022 and 2021, respectively, because the related performance criteria had not been met as of the balance sheet dates.

The following table reconciles the numerators and denominators used to calculate basic and diluted earnings per share of common stock:

(in millions except share and per share amounts)	Year Ended December 31, 2023	2022	2021
Basic earnings per share:			
Net earnings from continuing operations	$ 632	$ 636	$ 343
Weighted average shares of common stock outstanding	232.8	235.5	238.1
Basic earnings per share of common stock	$ 2.71	$ 2.70	$ 1.44
Diluted earnings per share:			
Net earnings from continuing operations	$ 632	$ 636	$ 343
Weighted average shares of common stock outstanding	232.8	235.5	238.1
Effect of stock-based compensation	1.6	1.3	1.4
Weighted average shares of common stock outstanding, including dilutive shares	234.4	236.8	239.5
Diluted earnings per share of common stock	$ 2.70	$ 2.69	$ 1.43

NOTE 24 REPORTABLE SEGMENTS AND RELATED INFORMATION

The Company's business is aggregated into three reportable segments, which are further described below. These segments are strategic business groups, which are managed separately as each represents a specific grouping of related automotive components and systems. In previous years, the Company presented its results under four reportable segments: Air Management, ePropulsion & Drivetrain, Fuel Systems and Aftermarket.

In the first quarter of 2023, the Company elected to disaggregate Air Management and ePropulsion & Drivetrain segments into Air Management, Drivetrain & Battery Systems and ePropulsion and reported its results in a total of five reportable segments: Air Management, Drivetrain & Battery Systems, ePropulsion, Fuel Systems and Aftermarket. As a result of the Spin-Off, Fuel Systems and Aftermarket are no longer reportable segments.

The reportable segment disclosures have been updated accordingly, which included recasting prior period information for the new reporting structure.

- ***Air Management.*** This segment develops and manufactures products to improve fuel economy, reduce emissions and enhance performance. The Air Management segment's technologies include turbochargers, eBoosters, eTurbos, timing systems, emissions systems, thermal systems, gasoline ignition technology, smart remote actuators, powertrain sensors, cabin heaters, battery heaters, battery charging and direct current charging stations.
- ***Drivetrain & Battery Systems.*** This segment's technologies include battery modules and systems, control modules, friction and mechanical clutch products for automatic transmissions, torque-management products and rear-wheel drive ("RWD") and all-wheel drive ("AWD") transfer case systems and coupling systems.
- ***ePropulsion.*** This segment's products and technologies provide industry-leading performance and efficiency with quick-to-market solutions powering current and next-generation electric and hybrid vehicles. The ePropulsion segment's technologies include power electronics such as inverters, onboard chargers, DC/DC converters and combination boxes (multiple combined power electronics components). Rotating electric machines are also part of the ePropulsion portfolio, including eMotors and generators as well as fully integrated drive modules consisting of inverter,

eMotor and gear reducer. Additionally, the segment's products include electronic controls such as engine control units, transmission control units, battery management systems, propulsion controllers and domain controllers.

Segment Adjusted Operating Income (Loss) is the measure of segment income or loss used by the Company. Segment Adjusted Operating Income (Loss) is comprised of operating income adjusted for restructuring, merger, acquisition and divestiture expense, intangible asset amortization expense, impairment charges and other items not reflective of ongoing operating income or loss. The Company believes Segment Adjusted Operating Income (Loss) is most reflective of the operational profitability or loss of its reportable segments.

The following tables show segment information and Segment Adjusted Operating Income (Loss) for the Company's reportable segments:

2023 Segment information

(in millions)	Net sales			Year-end assets	Depreciation/ amortization	Long-lived asset expenditures[1]
	Customers	Inter-segment	Net			
Air Management	$ 7,750	$ 83	$ 7,833	$ 5,505	$ 252	$ 252
Drivetrain & Battery Systems	4,346	2	4,348	4,023	172	162
ePropulsion	2,102	64	2,166	3,185	127	384
Inter-segment eliminations	—	(149)	(149)	—	—	—
Total	14,198	—	14,198	12,713	551	798
Corporate[2]	—	—	—	1,740	31	34
Consolidated	$ 14,198	$ —	$ 14,198	$ 14,453	$ 582	$ 832

2022 Segment information

(in millions)	Net sales			Year-end assets	Depreciation/ amortization	Long-lived asset expenditures[1]
	Customers	Inter-segment	Net			
Air Management	$ 7,076	$ 61	$ 7,137	$ 5,329	$ 243	$ 187
Drivetrain & Battery Systems	3,753	(18)	3,735	3,963	166	166
ePropulsion	1,806	100	1,906	2,349	111	257
Inter-segment eliminations	—	(143)	(143)	—	—	—
Total	12,635	—	12,635	11,641	520	610
Corporate[2]	—	—	—	1,691	32	12
Consolidated	$ 12,635	$ —	$ 12,635	$ 13,332	$ 552	$ 622

2021 Segment information

(in millions)	Net sales			Year-end assets	Depreciation/ amortization	Long-lived asset expenditures[1]
	Customers	Inter-segment	Net			
Air Management	$ 6,761	$ 106	$ 6,867	$ 5,174	$ 281	$ 214
Drivetrain & Battery Systems	3,659	1	3,660	4,229	160	150
ePropulsion	1,383	44	1,427	1,828	102	113
Inter-segment eliminations	—	(151)	(151)	—	—	—
Total	11,803	—	11,803	11,231	543	477
Corporate[2]	—	—	—	1,656	31	37
Consolidated	$ 11,803	$ —	$ 11,803	$ 12,887	$ 574	$ 514

[1] Long-lived asset expenditures include capital expenditures and tooling outlays.

[2] Corporate assets include cash and cash equivalents, investments and long-term receivables, and deferred income taxes.

Segment Adjusted Operating Income (Loss)

(in millions)	Year Ended December 31, 2023	2022	2021
Air Management	$ 1,171	$ 1,073	$ 1,037
Drivetrain & Battery Systems	545	449	524
ePropulsion	(90)	(88)	(48)
Segment Adjusted Operating Income	1,626	1,434	1,513
Corporate, including stock-based compensation	278	282	234
Intangible asset amortization expense	67	69	59
Restructuring expense	79	48	108
Merger and acquisition expense, net	23	9	48
Asset impairments and lease modifications	29	30	—
(Gain) loss on sale of business	(5)	(13)	29
Gain on sale of assets	(13)	—	—
Other non-comparable items	8	—	(3)
Customer warranty settlement	—	—	124
Equity in affiliates' earnings, net of tax	(30)	(28)	(42)
Realized and unrealized loss on debt and equity securities	174	73	362
Interest expense, net	10	51	91
Other postretirement expense (income)	15	—	(7)
Earnings from continuing operations before income taxes and noncontrolling interest	$ 991	$ 913	$ 510

Geographic Information

During the year ended December 31, 2023, approximately 84% of the Company's consolidated net sales were outside the U.S., attributing sales to the location of production rather than the location of the customer. Outside the U.S., China, Mexico, Germany, Poland, Hungary and South Korea exceeded 5% of consolidated net sales during the year ended December 31, 2023. The Company's investments in equity securities are excluded from the definition of long-lived assets, as are goodwill and certain other non-current assets.

(in millions)	Net sales 2023	2022	2021	Long-lived assets 2023	2022	2021
United States	$ 2,334	$ 2,143	$ 2,125	$ 778	$ 469	$ 489
Europe:						
Germany	1,669	1,443	1,338	393	354	401
Poland	1,121	937	961	342	277	276
Hungary	904	636	469	167	170	193
Other Europe	1,386	1,056	1,023	456	327	289
Total Europe	5,080	4,072	3,791	1,358	1,128	1,159
China	2,998	2,864	2,958	876	821	776
Mexico	1,760	1,727	1,208	374	612	561
South Korea	1,163	1,073	1,117	197	205	239
Other foreign	863	756	604	200	191	164
Total	$ 14,198	$ 12,635	$ 11,803	$ 3,783	$ 3,426	$ 3,388

Sales to Major Customers

Consolidated net sales to Ford (including its subsidiaries) were approximately14%, 15% and 13% for the years ended December 31, 2023, 2022 and 2021, respectively. Consolidated net sales to Volkswagen (including its subsidiaries) were approximately 11%, 9% and 9% for the years ended December 31, 2023, 2022 and 2021. Such sales consisted of a variety of products to a variety of customer locations and regions. No other single customer accounted for more than 10% of consolidated net sales in any of the years presented.

Sales by Product Line

Sales of turbochargers for light vehicles represented approximately21%, 26% and 24% of consolidated net sales for the years ended December 31, 2023, 2022 and 2021, respectively. No other single product line accounted for more than 10% of consolidated net sales in any of the years presented.

NOTE 25 OPERATING CASH FLOWS AND OTHER SUPPLEMENTAL FINANCIAL INFORMATION

(in millions)	Year Ended December 31, 2023	2022	2021
OPERATING ACTIVITIES OF CONTINUING OPERATIONS			
Net earnings	$ 695	$ 1,026	$ 639
Net (loss) earnings from discontinued operations	(7)	308	194
Net earnings from continuing operations	702	718	445
Adjustments to reconcile net earnings from continuing operations to net cash provided by operating activities from continuing operations:			
Depreciation and tooling amortization	515	483	515
Intangible asset amortization	67	69	59
Restructuring expense, net of cash paid	66	41	81
Stock-based compensation expense	58	64	50
(Gain) loss on sales of businesses	(5)	(16)	29
(Gain) loss on debt extinguishment	(28)	—	20
Asset impairments	20	30	—
Realized and unrealized loss on debt and equity securities	174	73	362
Deferred income tax benefit	(44)	(76)	(96)
Gain on insurance recovery received for property damages	—	—	(5)
Other non-cash adjustments	(16)	(33)	(38)
Adjustments to reconcile net earnings from continuing operations to net cash provided by operating activities from continuing operations	1,509	1,353	1,422
Retirement plan contributions	(19)	(21)	(28)
Changes in assets and liabilities, excluding effects of acquisitions, divestitures and foreign currency translation adjustments:			
Receivables	(482)	(409)	(55)
Inventories	(72)	(158)	(179)
Prepayments and other current assets	(3)	17	14
Accounts payable and accrued expenses	375	433	(61)
Prepaid taxes and income taxes payable	(20)	25	(10)
Other assets and liabilities	109	(60)	107
Net cash provided by operating activities from continuing operations	$ 1,397	$ 1,180	$ 1,210
SUPPLEMENTAL CASH FLOW INFORMATION			
Cash paid during the year for:			
Interest	$ 126	$ 131	$ 127
Income taxes, net of refunds	$ 362	$ 268	$ 290
Non-cash investing transactions:			
Period end accounts payable related to property, plant and equipment purchases	$ 148	$ 165	$ 95

NOTE 26 DISCONTINUED OPERATIONS

The historical results of operations and the financial position of PHINIA for periods prior to the Spin-Off are presented as discontinued operations in these Consolidated Financial Statements.

The following table summarizes the assets and liabilities from discontinued operations of PHINIA.

(in millions)	December 31, 2022
ASSETS	
Cash, cash equivalents and restricted cash	$ 255
Receivables, net	852
Inventories, net	470
Prepayments and other current assets	39
Total current assets of discontinued operations	$ 1,616
Property, plant and equipment, net	939
Investments and long-term receivables	77
Goodwill	419
Other intangible assets, net	432
Other non-current assets	179
Total non-current assets of discontinued operations	$ 2,046
LIABILITIES	
Notes payable and other short-term debt	$ 2
Accounts payable	538
Other current liabilities	406
Total current liabilities of discontinued operations	$ 946
Long-term debt	26
Retirement-related liabilities	94
Other non-current liabilities	175
Total non-current liabilities of discontinued operations	$ 295

The following table summarizes the financial results from discontinued operations of PHINIA.

(in millions)	Year Ended December 31,		
	2023	2022	2021
Net sales	$ 1,723	$ 3,348	$ 3,227
Cost of sales	1,362	2,616	2,545
Gross profit	361	732	682
Selling, general and administrative expenses	173	320	375
Restructuring expense	7	11	55
Other operating expense, net	132	36	15
Operating (loss) income	49	365	237
Equity in affiliates' earnings, net of tax	(5)	(10)	(6)
Interest expense, net	—	1	2
Other postretirement expense (income)	—	(31)	(38)
Earnings from discontinued operations before income taxes	54	405	279
Provision for income taxes	61	97	85
Net (loss) earnings from discontinued operations attributable to PHINIA	$ (7)	$ 308	$ 194

In connection with the Spin-Off, the Company entered into a transition services agreement through which the Company and PHINIA will continue to provide certain services to each other following the Spin-Off. These services are related to information technology, human resources, finance, facilities, procurement, sales, intellectual properties and engineering. The combined impact of these services is reported in results of continuing operations in the Consolidated Financial Statements. From the date of the Spin-Off through the year ended December 31, 2023, the Company provided services at a cost of $13 million to PHINIA, and PHINIA provided services at a cost of $3 million to the Company .

The Company incurred $134 million and $30 million of costs relating to the Spin-Off during the year ended December 31, 2023 and 2022, respectively, which are reflected within Net (loss) earnings from discontinued operations in the Company's Consolidated Statements of Operations. Spin-Off costs are primarily comprised of professional fees and costs to separate certain operational activities.

NOTE 27 INTERIM FINANCIAL INFORMATION (UNAUDITED)

The following table presents summary quarterly financial information for the years ended December 31, 2023 and 2022.The table presents recast financial information reflecting the presentation of discontinued operations related to the Spin-Off:

(in millions, except per share amounts)	2023					2022				
Quarter ended	Mar-31	Jun-30	Sep-30	Dec-31	Year	Mar-31	Jun-30	Sep-30	Dec-31	Year
Net sales	$ 3,383	$ 3,671	$ 3,622	$ 3,522	$ 14,198	$ 3,079	$ 3,013	$ 3,226	$ 3,317	$ 12,635
Gross profit	578	679	652	659	2,568	575	548	607	639	2,369
Operating income	273	334	272	281	1,160	270	209	265	265	1,009
Amounts attributable to BorgWarner Inc.:										
Net earnings from continuing operations	$ 166	$ 230	$ 87	$ 149	$ 632	$ 130	$ 161	$ 173	$ 172	$ 636
Net earnings (loss) from discontinued operations	51	(26)	(37)	5	(7)	70	54	102	82	308
Net earnings attributable to BorgWarner Inc.	$ 217	$ 204	$ 50	$ 154	$ 625	$ 200	$ 215	$ 275	$ 254	$ 944
Earnings per share from continuing operations — basic	$ 0.72	$ 0.99	$ 0.37	$ 0.64	$ 2.71	$ 0.55	$ 0.68	$ 0.74	$ 0.74	$ 2.70
Earnings per share from discontinued operations — basic	$ 0.21	$ (0.11)	$ (0.16)	$ 0.02	$ (0.03)	$ 0.29	$ 0.23	$ 0.43	$ 0.35	$ 1.31
Earnings per share attributable to BorgWarner Inc. — basic[1]	$ 0.93	$ 0.87	$ 0.21	$ 0.67	$ 2.68	$ 0.84	$ 0.91	$ 1.17	$ 1.09	$ 4.01
Earnings per share from continuing operations — diluted	$ 0.72	$ 0.97	$ 0.37	$ 0.64	$ 2.70	$ 0.54	$ 0.68	$ 0.73	$ 0.73	$ 2.69
Earnings per share from discontinued operations — diluted	$ 0.21	$ (0.10)	$ (0.16)	$ 0.02	$ (0.03)	$ 0.30	$ 0.23	$ 0.42	$ 0.36	$ 1.30
Earnings per share attributable to BorgWarner Inc. — diluted[1]	$ 0.93	$ 0.87	$ 0.21	$ 0.66	$ 2.67	$ 0.84	$ 0.91	$ 1.15	$ 1.09	$ 3.99

[1] Due to the use of quarterly weighted average shares outstanding for computing earnings per share, the sum of the quarterly per share amounts may not equal the per share amount for the year.

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

Not applicable.

Item 9A. Controls and Procedures

Disclosure Controls and Procedures

A control system, no matter how well conceived and operated, can provide only reasonable, not absolute, assurance that the objectives of the control system are met. Further, the design of a control system must reflect the fact that there are resource constraints, and the benefits of controls must be considered relative to costs. Because of the inherent limitations in all control systems, no evaluation of controls can provide absolute assurance that all control issues and instances of fraud, if any, within the Company have

been detected. Because of the inherent limitations in a cost-effective control system, misstatements due to error or fraud may occur and not be detected. However, the Company's disclosure controls and procedures are designed to provide reasonable assurance of achieving their objectives.

The Company has adopted and maintains disclosure controls and procedures that are designed to provide reasonable assurance that information required to be disclosed in the reports filed or submitted under the Exchange Act, such as this Form 10-K, is recorded, processed, summarized and reported within the time periods specified in the rules and forms of the Securities and Exchange Commission. The Company's disclosure controls and procedures are also designed to provide reasonable assurance that such information is accumulated and communicated to management, as appropriate, to allow timely decisions regarding required disclosure. As required under Exchange Act Rule 13a-15, the Company's management, including the Chief Executive Officer and Chief Financial Officer, has conducted an evaluation of the effectiveness of disclosure controls and procedures as of December 31, 2023, the end of the period covered by this report. Based on that evaluation, the Chief Executive Officer and Chief Financial Officer concluded that the disclosure controls and procedures are effective at the reasonable assurance level.

Management's Report on Internal Control Over Financial Reporting

The Company's management is responsible for establishing and maintaining adequate internal control over financial reporting, as defined in Exchange Act Rule 13a-15(f). Management conducted an assessment of the Company's internal control over financial reporting based on the framework and criteria established by the Committee of Sponsoring Organizations of the Treadway Commission in "Internal Control - Integrated Framework" (2013). Management excluded from its assessment of internal control over financial reporting Hubei Surpass Sun Electric's charging business, which was acquired on March 1, 2023, and Eldor Corporation's electric hybrid systems business, which was acquired on December 1, 2023. In aggregate, these acquisitions accounted for less than 1% of consolidated total assets and less than 1% of consolidated net sales, as of and for the year ended December 31, 2023, respectively. Based on the assessment, management concluded that as of December 31, 2023, the Company's internal control over financial reporting was effective based on those criteria.

PricewaterhouseCoopers LLP, an independent registered public accounting firm, has audited the Company's consolidated financial statements and the effectiveness of internal control over financial reporting as of December 31, 2023 as stated in its report included herein.

Changes in Internal Control over Financial Reporting

There have been no changes in internal control over the financial reporting that occurred during the most recent fiscal quarter that have materially affected or are reasonably likely to materially affect the Company's internal control over financial reporting.

Item 9B. Other Information

Not applicable.

Item 9C. Disclosure Regarding Foreign Jurisdictions that Prevent Inspections

Not applicable.

PART III

Item 10. Directors, Executive Officers and Corporate Governance

Information with respect to directors, executive officers and corporate governance that appears in the Company's proxy statement for its 2024 Annual Meeting of Stockholders under the captions "Election of Directors," "Information on Nominees for Directors," "Board Committees," "Compensation Committee Report," and "Code of Ethics" is incorporated herein by this reference and made a part of this report.

Code of Ethics

The Company has long maintained a Code of Ethical Conduct, updated from time to time, which is applicable to all directors, officers, and employees of the Company. In addition, the Company has adopted a Code of Ethics for CEO and Senior Financial Officers, which applies to the Company's CEO, CFO, Treasurer, and Controller. Each of these codes is posted on the Company's website at www.borgwarner.com/investors/corporate-governance. The Company intends to disclose any amendments to, or waivers from, a provision of its Code of Ethical Conduct or Code of Ethics for CEO and Senior Financial Officers on its website within four business days following the date of any amendment or waiver.

Item 11. Executive Compensation

Information with respect to director and executive compensation that will appear in the Company's proxy statement for its 2024 Annual Meeting of Stockholders under the captions "Director Compensation," "Compensation Committee Interlocks and Insider Participation," "Compensation Discussion and Analysis," "Restricted Stock," "Long-Term Equity Incentives," and "Change of Control Agreements and Transitional Income Plan" is incorporated herein by this reference and made a part of this report.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters

Information with respect to security ownership and certain beneficial owners and management and related stockholders matters that will appear in the Company's proxy statement for its 2024 Annual Meeting of Stockholders under the caption "Security Ownership of Certain Beneficial Owners and Management" is incorporated herein by this reference and made a part of this report.

For information regarding the Company's equity compensation plans, see Item 5 "Market for the Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities" in this Annual Report on Form 10-K.

Item 13. Certain Relationships and Related Transactions, and Director Independence

Information with respect to certain relationships and related transactions and director independence that will appear in the Company's proxy statement for its 2024 Annual Meeting of Stockholders under the captions "Certain Relationships and Related Transactions" and "Independence of the Directors" is incorporated herein by this reference and made a part of this report.

Item 14. Principal Accountant Fees and Services

Information with respect to principal accountant fees and services that will appear in the Company's proxy statement for its 2024 Annual Meeting of Stockholders under the caption "Fees Paid to PwC" is incorporated herein by this reference and made a part of this report.

PART IV

Item 15. Exhibits and Financial Statement Schedules

The information required by Section (a)(3) of Item 15 is set forth on the Exhibit Index that precedes the Signatures page of this Form 10-K. The information required by Section (a)(1) of Item 15 is set forth above in Item 8, Financial Statements and Supplementary Data. All financial statement schedules have been omitted, since the required information is not applicable or is not present in amounts sufficient to require submission of the schedule or because the information required is included in the consolidated financial statements and notes thereto included in this Form 10-K.

Item 16. Form 10-K Summary

Not applicable.

EXHIBIT INDEX

Exhibit Number	Description
1.1	Separation and Distribution Agreement, dated July 2, 2023, by and between the Company and PHINIA (incorporated by reference to Exhibit 2.1 to the Company's Current Report on Form 8-K filed on July 7, 2023).
3.1	Composite Restated Certificate of Incorporation of the Company, as amended through July 22, 2022 (incorporated by reference to Exhibit 3.1 to the Company's Quarterly Report on Form 10-Q for the quarter ended June 30, 2022 filed August 3, 2022).
3.2	Amended and Restated By-Laws of the Company, as amended through April 25, 2018 (incorporated by reference to Exhibit 3.2 to the Company's Quarterly Report on Form 10-Q for the quarter ended June 30, 2018 filed July 26, 2018).
4.1	Indenture, dated as of February 15, 1999, between Borg-Warner Automotive, Inc. and The Bank of New York Mellon Trust Company, N.A. (successor in interest to The First National Bank of Chicago), as trustee (incorporated by reference to Exhibit No. 4.5 to the Company's Registration Statement No. 333-172198 filed on February 11, 2011).
4.2	Indenture, dated as of September 23, 1999, between Borg-Warner Automotive, Inc. and The Bank of New York Mellon Trust Company, N.A. (successor in interest to Chase Manhattan Trust Company, National Association), as trustee (incorporated by reference to Exhibit No. 4.6 to the Company's Registration Statement 333-172198 filed on February 11, 2011).
4.3	Fourth Supplemental Indenture dated as of March 16, 2015, between the Company and The Bank of New York Mellon Trust Company, N.A., as the indenture trustee (incorporated by reference to Exhibit 4.2 to the Company's Current Report on Form 8-K filed March 16, 2015).
4.4	Sixth Supplemental Indenture, dated as of June 19, 2020, between the Company and Deutsche Bank Trust Company Americas, as the indenture trustee (incorporated by reference to Exhibit 4.2 to the Company's Current Report on Form 8-K filed June 19, 2020).
4.5	Seventh Supplemental Indenture, dated as of October 5, 2020, between the Company and Deutsche Bank Trust Company Americas, as the indenture trustee (incorporated by reference to Exhibit 4.2 to the Company's Current Report on Form 8-K filed October 5, 2020).
4.6	Eighth Supplemental Indenture, dated as of May 19, 2021, between the Company and Deutsche Bank Trust Company Americas, as the indenture trustee (incorporated by reference to Exhibit 4.2 to the Company's Current Report on Form 8-K filed May 19, 2021)
4.7	Description of Securities (incorporated by reference to Exhibit 4.6 to the Company's Annual Report on Form 10-K for the fiscal year ended December 31, 2019 filed February 13, 2020).
10.1	Transition Services Agreement, dated July 2, 2023, by and between the Company and PHINIA (incorporated by reference to Exhibit 10.1 to the Company's Current Report on Form 8-K filed on July 7, 2023).
10.2	Tax Matters Agreement, dated July 2, 2023, by and between the Company and PHINIA (incorporated by reference to Exhibit 10.2 to the Company's Current Report on Form 8-K filed on July 7, 2023).
10.3	Employee Matters Agreement, dated July 2, 2023, by and between the Company and PHINIA (incorporated by reference to Exhibit 10.3 to the Company's Current Report on Form 8-K filed on July 7, 2023).
10.4	Intellectual Property Cross-License Agreement, dated July 2, 2023, by and between the Company and PHINIA (incorporated by reference to Exhibit 10.4 to the Company's Current Report on Form 8-K filed on July 7, 2023).
10.5	Fifth Amended and Restated Credit Agreement dated as of September 22, 2023, among BorgWarner Inc., Bank of America, N.A., as Administrative Agent for the Lenders and as Swingline Lender and an Issuing Bank, and the Lenders party thereto from time to time (incorporated by reference to Exhibit 10.1 to the Company's Current Report on Form 8-K filed on September 22, 2023).
†10.6	BorgWarner Inc. 2023 Stock Incentive Plan, as Amended and Restated (incorporated by reference to Exhibit 10.2 to the Company's Quarterly report on Form 10-Q for the quarter ended September 30, 2023 filed on November 2, 2023).

Exhibit Number	Description
†10.7	Form of BorgWarner Inc. 2023 Stock Incentive Plan Restricted Stock Agreement for Employees (incorporated by reference to Exhibit 10.5 to the Company's Quarterly report on Form 10-Q for the quarter ended June 30, 2023 filed on August 2, 2023).
†10.8	Form of BorgWarner Inc. 2023 Stock Incentive Plan Stock Units Award Agreement for Non-U.S. Employees (incorporated by reference to Exhibit 10.6 to the Company's Quarterly report on Form 10-Q for the quarter ended June 30, 2023 filed on August 2, 2023).
†10.9	Form of 2023 BorgWarner Inc. 2023 Stock Incentive Plan Performance Stock Units Award Agreement (incorporated by reference to Exhibit 10.7 to the Company's Quarterly report on Form 10-Q for the quarter ended June 30, 2023 filed on August 2, 2023).
†10.10	Form of 2023 BorgWarner Inc. 2023 Stock Incentive Plan Restricted Stock Agreement for Non-Employee Directors (incorporated by reference to Exhibit 10.8 to the Company's Quarterly report on Form 10-Q for the quarter ended June 30, 2023 filed on August 2, 2023).
†10.11	Form of 2022 BorgWarner Inc. 2018 Stock Incentive Plan Restricted Stock Agreement for Employees (incorporated by reference to Exhibit 10.1 to the Company's Quarterly Report on Form 10-Q for the quarter ended March 31, 2022 filed on May 4, 2022).
†10.12	Form of 2022 BorgWarner Inc. 2018 Stock Incentive Plan Stock Units Award Agreement for Non-U.S. Employees (incorporated by reference to Exhibit 10.2 to the Company's Quarterly Report on Form 10-Q for the quarter ended March 31, 2022 filed on May 4, 2022).
†10.13	Form of 2022 BorgWarner Inc. 2018 Stock Incentive Plan Performance Share Award Agreement (incorporated by reference to Exhibit 10.3 to the Company's Quarterly Report on Form 10-Q filed for the quarter ended March 31, 2022 on May 4, 2022).
†10.14	Form of 2022 BorgWarner Inc. 2018 Stock Incentive Plan Restricted Stock Agreement for Non-Employee Directors (incorporated by reference to Exhibit 10.8 to the Company's Annual Report on Form 10-K for the year ended December 31, 2022 filed on February 9, 2023).
†10.15	Form of 2021 BorgWarner Inc. 2018 Stock Incentive Plan Restricted Stock Agreement for Employees (incorporated by reference to Exhibit 10.8 to the Company's Annual Report on Form 10-K for the year ended December 31, 2021 filed on February 15, 2022).
†10.16	Form of 2021 BorgWarner Inc. 2018 Stock Incentive Plan Restricted Stock Units Agreement for Non-U.S. Employees (incorporated by reference to Exhibit 10.9 to the Company's Annual Report on Form 10-K for the year ended December 31, 2021 filed on February 15, 2022).
†10.17	Form of 2021 BorgWarner Inc. 2018 Stock Incentive Plan Performance Share Award Agreement (incorporated by reference to Exhibit 10.10 to the Company's Annual Report on Form 10-K for the year ended December 31, 2021 filed on February 15, 2022).
†10.18	Form of 2021 BorgWarner Inc. 2018 Stock Incentive Plan Restricted Stock Agreement for Non-Employee Directors (incorporated by reference to Exhibit 10.11 to the Company's Annual Report on Form 10-K for the year ended December 31, 2021 filed on February 15, 2022).
†10.19	Amended and Restated BorgWarner Inc. Management Incentive Bonus Plan, effective as of December 31, 2008 (incorporated by reference to Exhibit 10.22 to the Company's Annual Report on Form 10-K for the year ended December 31, 2018 filed on February 19, 2019).
†10.20	BorgWarner Inc. Retirement Savings Excess Benefit Plan, as amended and restated, effective January 1, 2009 (incorporated by reference to Exhibit 10.23 to the Company's Annual Report on Form 10-K for the year ended December 31, 2018 filed on February 19, 2019).
†10.21	BorgWarner Inc. Board of Directors Deferred Compensation Plan, as amended and restated, effective January 1, 2009 (incorporated by reference to Exhibit 10.24 to the Company's Annual Report on Form 10-K for the year ended December 31, 2018 filed on February 19, 2019).
†10.22	First Amendment, dated as of January 1, 2011, to BorgWarner Inc. Board of Directors Deferred Compensation Plan (incorporated by reference to Exhibit 10.25 to the Company's Annual Report on Form 10-K for the year ended December 31, 2018 filed on February 19, 2019).

Exhibit Number	Description
†10.23	Second Amendment, dated as of August 1, 2016, to BorgWarner Inc. Board of Directors Deferred Compensation Plan (incorporated by reference to Exhibit 10.31 to the Company's Annual Report on Form 10-K for the year ended December 31, 2016 filed February 9, 2017).
†10.24	Form of Change of Control Employment Agreement entered into by the Company and each of Tonit Calaway, Stefan Demmerle, Brady Ericson, Joe Fadool, Paul Farrell, Davide Girelli, Frederic Lissalde, Kevin Nowlan, Volker Weng, and Tania Wingfield (incorporated by reference to Exhibit 10.4 to the Company's Quarterly Report on Form 10-K for the quarter ended June 30, 2022 filed on August 3, 2022).
†10.25	BorgWarner Inc. 2004 Deferred Compensation Plan, as amended and restated, effective January 1, 2009 (incorporated by reference to Exhibit 10.29 to the Company's Annual Report on Form 10-K for the year ended December 31, 2018 filed on February 19, 2019).
†10.26	Employment Agreement, dated as of September 9, 2022, between BorgWarner Inc. and Frederic B. Lissalde (incorporated by reference to Exhibit 10.2 on the Company's Quarterly Report on Form 10-Q filed for the quarter ended September 30, 2022 on October 27, 2022).
21.1	Subsidiaries of the Company.*
23.1	Independent Registered Public Accounting Firm's Consent.*
31.1	Rule 13a-14(a)/15d-14(a) Certification by Principal Executive Officer.*
31.2	Rule 13a-14(a)/15d-14(a) Certification by Principal Financial Officer.*
32.1	Section 1350 Certifications.*
†97.1	BorgWarner Inc. Compensation Recovery Policy.*
101.INS	Inline XBRL Instance Document.*
101.SCH	Inline XBRL Taxonomy Extension Schema Document.*
101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document.*
101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document.*
101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document.*
101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document.*
104.1	Cover Page Interactive Data File (formatted as Inline XBRL and contained in Exhibit 101).*

*Filed herewith.
† Indicates a management contract or compensatory plan or arrangement.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

BORGWARNER INC.

By: /s/ Frederic B. Lissalde
Frederic B. Lissalde
President and Chief Executive Officer

Date: February 8, 2024

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following person on behalf of the registrant and in the capacities indicated on the 8th day of February, 2024.

Signature	Title
/s/ Frederic B. Lissalde Frederic B. Lissalde	President and Chief Executive Officer (Principal Executive Officer) and Director
/s/ Kevin A. Nowlan Kevin A. Nowlan	Executive Vice President and Chief Financial Officer (Principal Financial Officer)
/s/ Craig D. Aaron Craig D. Aaron	Vice President and Controller (Principal Accounting Officer)
/s/ Sara A. Greenstein Sara A. Greenstein	Director
/s/ Michael S. Hanley Michael S. Hanley	Director
/s/ Shaun E. McAlmont Shaun E. McAlmont	Director
/s/ Deborah D. McWhinney Deborah D. McWhinney	Director
/s/ Alexis P. Michas Alexis P. Michas	Director and Non-Executive Chairman
/s/ Sailaja K. Shankar Sailaja K. Shankar	Director
/s/ Hau N. Thai-Tang Hau N. Thai-Tang	Director